ACME

Science Overview

Dennis Stocker – Project Scientist

Advanced Combustion via Microgravity Experiments

Glenn Research Center

- Conduct 5+ distinct experiments using a **single modular insert** for the Combustion Integrated Rack (CIR) on the ISS
  - *The only existing insert (MDCA) is for the FLEX droplet combustion experiments*





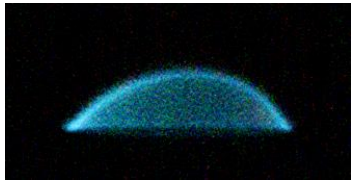
Koichi Wakata with MDCA in CIR (left), ISS (right)

BRE

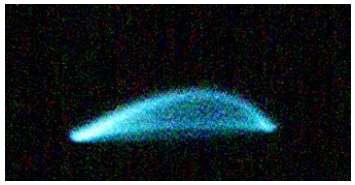
- **Burning Rate Emulator**
  - **Jim Quintiere** & Peter Sunderland (both U. Maryland)
- **Concept**
  - Flat-plate burner with gaseous fuel flow simulating the combustion of condensed fuels
  - Heat flux to the burner is measured, where the flux vaporizes condensed-phase fuels
    - Fuels to be simulated include paper, plastic, alcohols
- **Goals**
  - Quantify the ignition and extinction behavior of condensed fuels in microgravity.
  - Quantify the effects of burning at enhanced oxygen conditions in microgravity.
- **Why?**
  - Spacecraft fire prevention through improved assessment for materials selection

0g C2H4 flames as a function of the heated burner temperature.

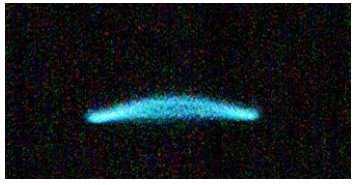
ambient



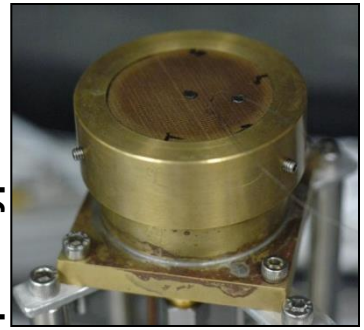
75 C



200 C



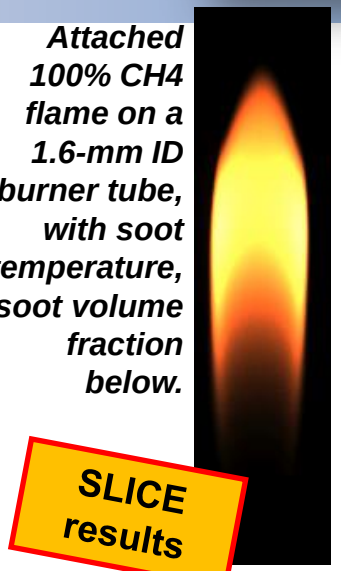
prototype burner



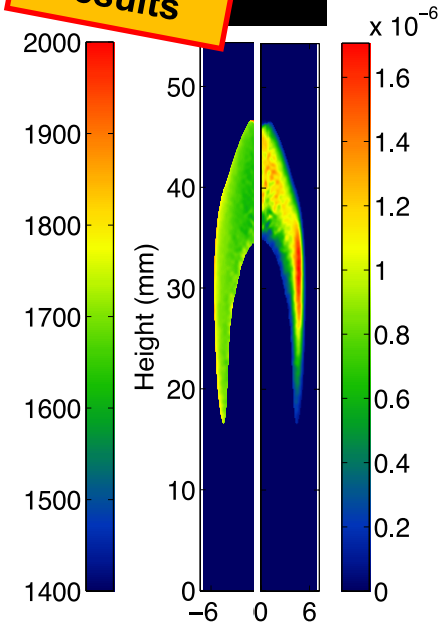
CLD Flame

- **Coflow Laminar Diffusion Flame**
  - **Marshall Long** & Mitchell Smooke (both Yale U.)
- **Concept**
  - Lifted coflow (2D) flame, so little/no heat loss to burner
    - enabling improved modeling of the flame
- **Goals**
  - Generation of modified kinetic mechanisms for hydrocarbon fuels that are able to model effectively diffusion flame structure under a larger parameter range than existing mechanisms.
  - Development of submodels for soot formation that are capable of predicting both high and low soot loading levels in hydrocarbon flames of various fuels.
- **Why?**
  - Improved design capability through validation of combustion models over wider parameter range

Attached 100% CH4 flame on a 1.6-mm ID burner tube, with soot temperature, soot volume fraction below.



SLICE results



E-FIELD Flames

- Electric-Field Effects on Laminar Diffusion Flames
  - Derek Dunn-Rankin (UC Irvine), Felix Weinberg (Imperial College, London), Zeng-Guang Yuan (NCSER at NASA Glenn)
- Concept
  - Gas-jet or coflow (2D) flame with high-voltage mesh (up to ±10 kV) electrode above (downstream of) burner
- Goal
  - Understand chemi-ionization behavior and the resulting ion driven winds sufficiently well so that electrical properties of flames can be used to characterize (by monitoring ion current) and control them (via direct chemical or local convective influences).
- Why?
  - Advanced control capability enabling improved combustion performance



1g, 0 kV, gas-jet



0g, -0.6 kV, gas-jet



0g, 0kv, 24 cm/s coflow

Flame Design

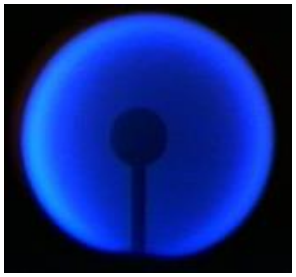
- *A novel approach to clean, efficient diffusion flames*
  - Rich Axelbaum (Washington U., St. Louis), Beei-Huan Chao (U. Hawaii), Peter Sunderland (U. Maryland), David L. Urban (NASA Glenn)
- Concept
  - Spherical (1D) flame with nitrogen exchange
- Goals
  - Evaluate the effects of flame structure on soot inception and flame extinction
  - Obtain a correlation of sooting limits with stoichiometric mixture fraction, *Zst,* and adiabatic temperature, *Tad*
  - Determine the importance of gas-phase oxidation on soot inception for high *Z*st flames
  - Demonstrate the existence of steady spherical flames
  - Identify pseudo-flammability limits for diffusion flames in terms of *Zst* and *Tad*
- Why?
  - To reduce soot and NOx through nitrogen exchange



18% C2H4 into 27% O2



18% C2H4 Into 28% O2



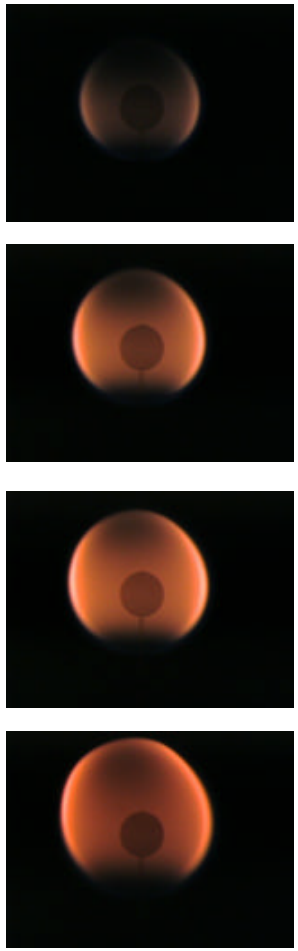
100% O2 into 12% C2H4



100% O2 into 13% C2H4

s-Flame

- Structure and Response of Spherical Diffusion Flames
  - C.K. Law (Princeton U.), Stephen D. Tse (Rutgers U.), Kurt R. Sacksteder (NASA Glenn)
- Concept
  - Spherical (1D) flame stabilized on a porous burner
- Goals
  - Use spherical diffusion flames in ISS environment to:
    - obtain experimental data in simple, well-defined flow fields
    - understand various key elemental diffusion flame processes
  - Compare data with independent computational simulation including detailed chemistry and transport to improve the accuracy and comprehensiveness of these descriptions
- Why?
  - Improvements in the design of practical, high-efficiency, low-emission combustion systems on Earth

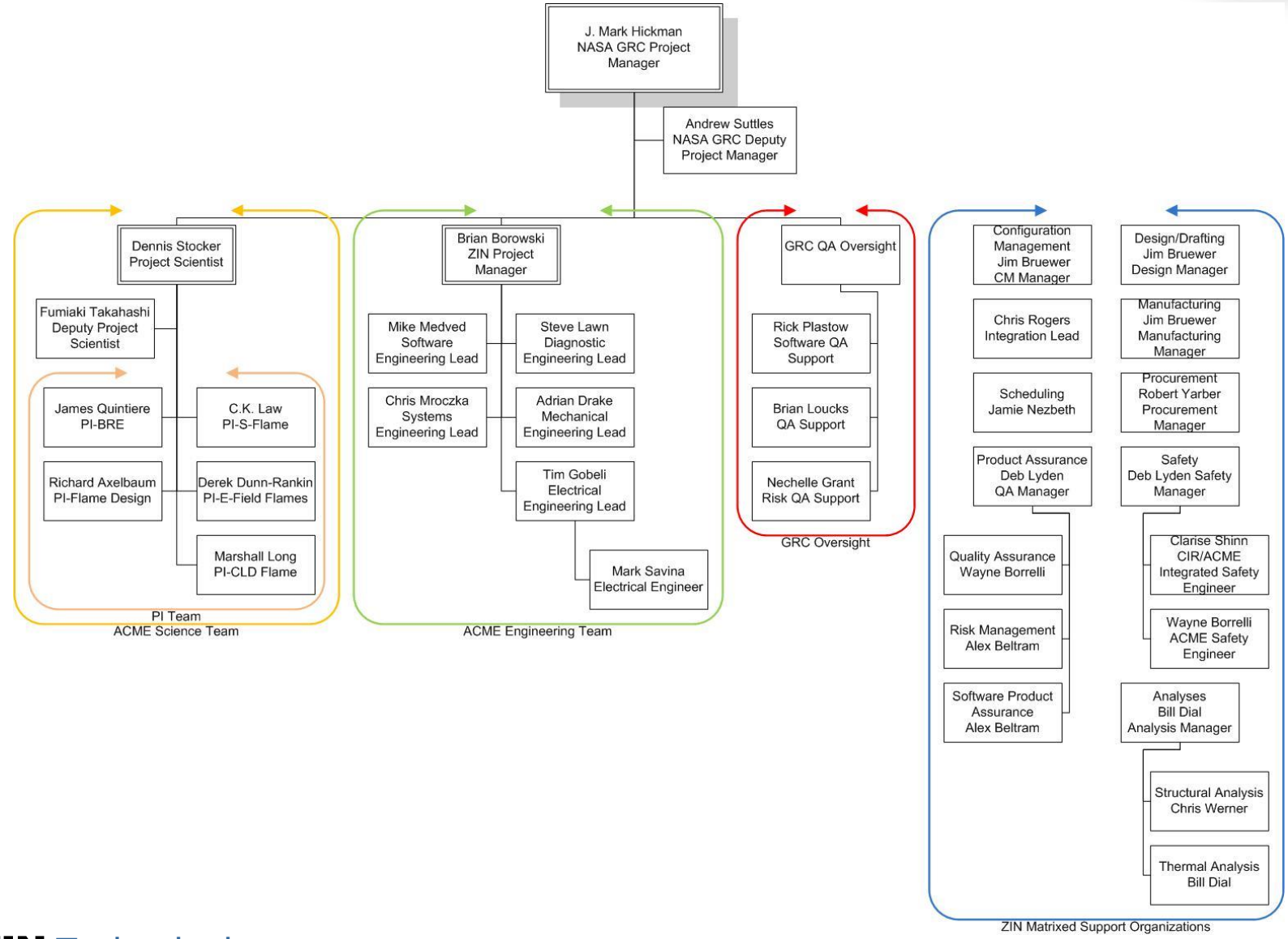


flame evolution with time

Project Overview

Brian Borowski – Project Manager

ACME Project Organization



CIR Sub-rack Payload Relationship

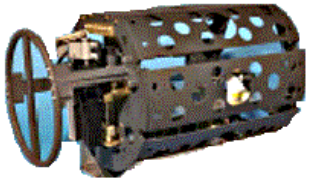
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The Combustion Integrated Rack (CIR) includes support subsystems and combustion diagnostics commonly needed for combustion investigations and supplements the laboratory with unique science hardware developed for each Principal Investigator (PI).

Payload specific and multi-user hardware customizes the CIR in a unique laboratory configuration to perform research effectively.



Advanced Combustion via Microgravity Experiments Insert Assembly





Payload Specific Hardware

- Unique hardware components
- Specific Diagnostics
- Unique PI fuels

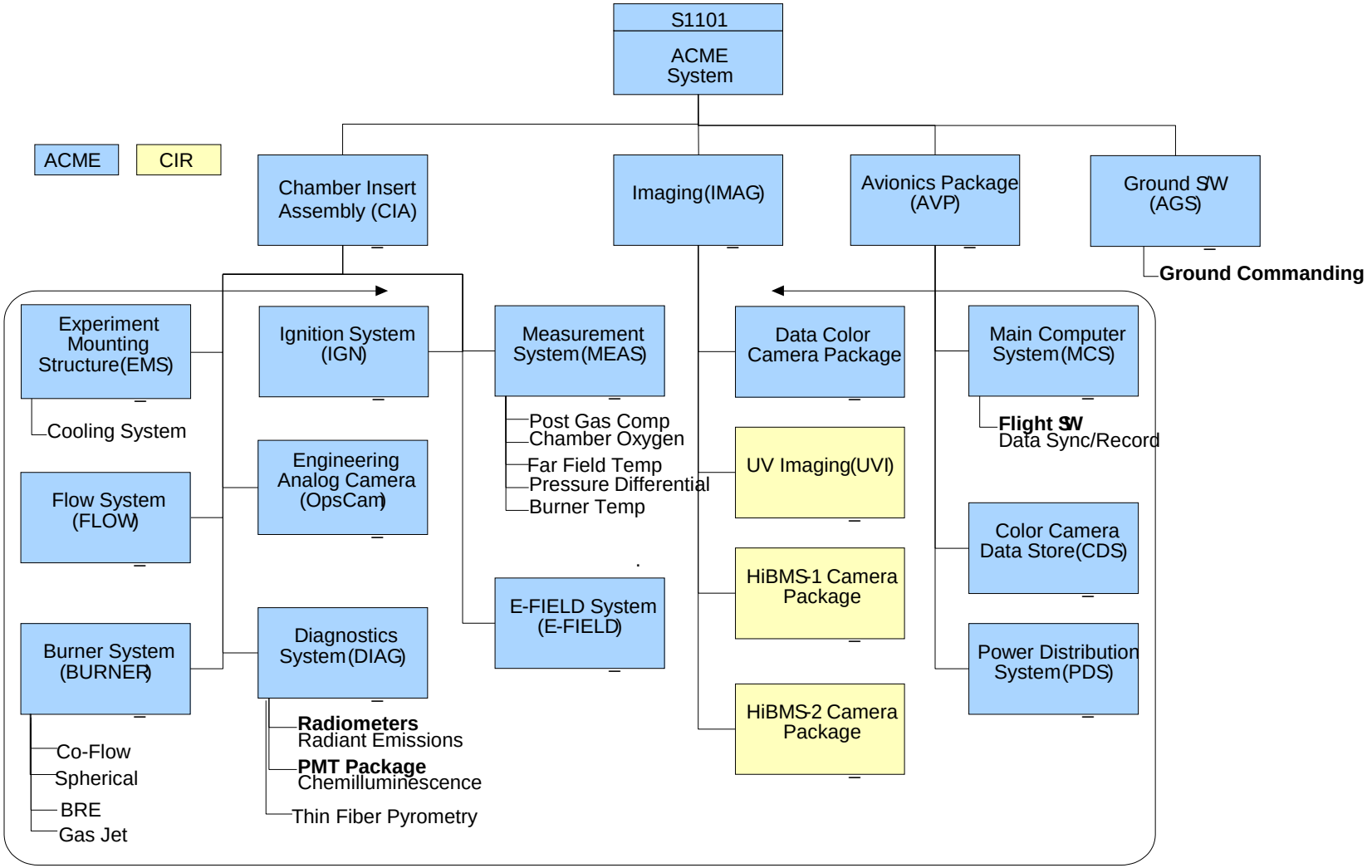
ACME Chamber Insert

- Platform for exchangeable burner:
  - Gas Delivery
  - Retracting ignitor
  - Electric field generator
- Infrastructure that uniquely meets the needs of PI experiments
- Unique science requirements

FCF Combustion Integrated Rack

- Power Supply
- Avionics/Control
- Structural Support & Power
- Common Diagnostics and Illumination
- Environmental Control
- Data Processing/Distribution
- Fluids Control/Distribution
- Combustion Containment

ACME Product Breakdown Structure



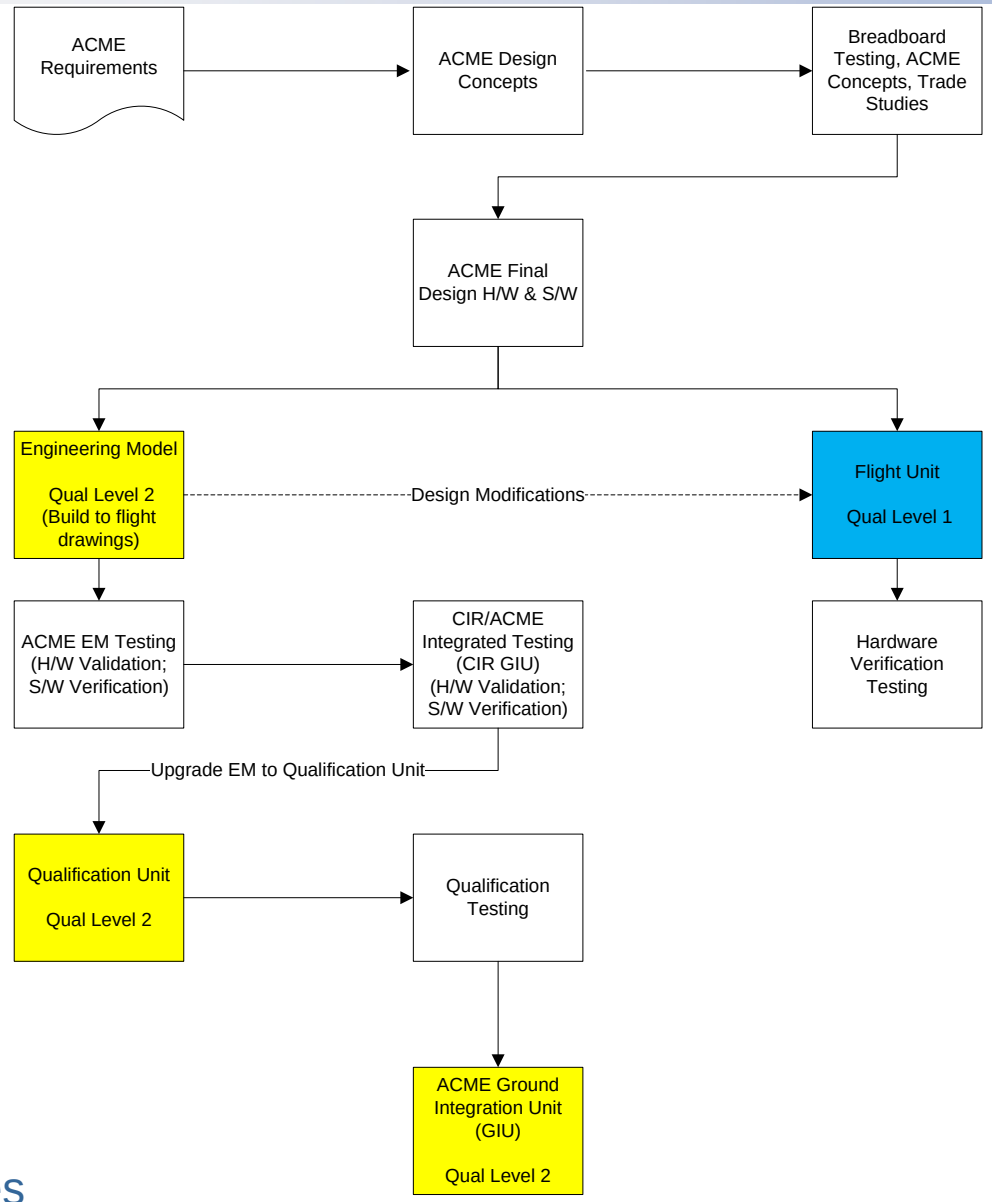
ACME Subsystems

ACME Hardware Development

ZIN Tech. P02001 Quality Planning Procedure
(Explanation of ZIN Qual Level Builds)

Quality Level 1 - Flight Model
Quality Level 2 - GIU Qual. Unit
Quality Level 3 - Engineering Model
Quality Level 4 - Breadboard Model

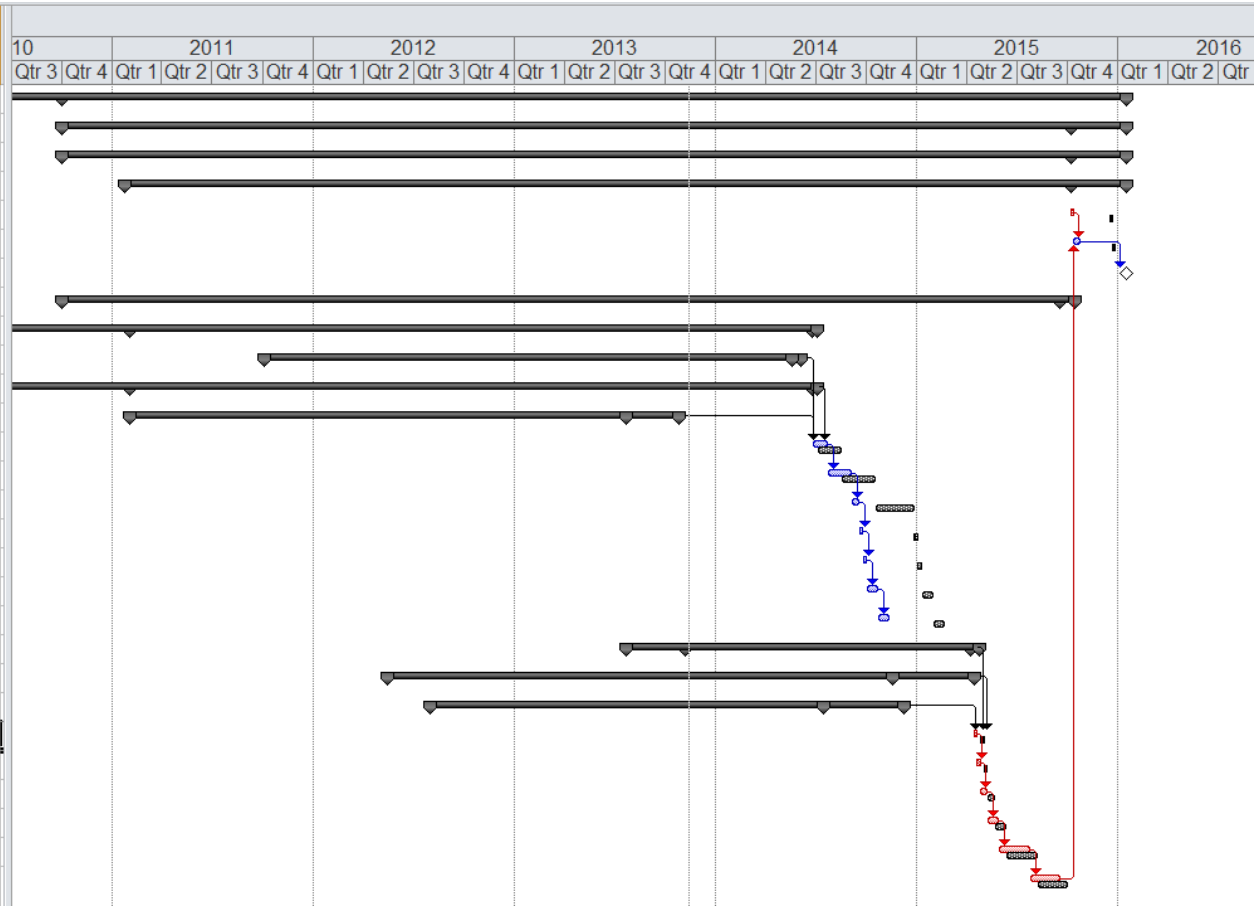
Hardware Development Plan



Hardware Development Schedule

Glenn Research Center

Task Name
Advanced Combustion via Microgravity Experiment (ACME)
1.0 Management
1.1 Project Management
Project Reviews
Phase 3 Flight Safety Review
SAR (FHA)
Launch (SpaceX-9)
3.0 Quality Assurance
7.0 Final Design and Fabrication (EM)
7.3 EM Data Camera Package
7.5 EM CIA
7.6 EM AVP
ACME Internal Engineering Model Testing
CIR/ACME Integrated Testing
Qual Unit Build (upgrade from EM)
Qual Unit Thermal Test
Qual Unit Vibe Test
Qual Unit EMI Test
Verification of Physical Dimensions at an Inpsection House
Flight Data Camera Package
Flight CIA
Flight AVP
Flight Model Launch Vibe
Flight Model Workmanship Vibe
Flight Model EMI Test
Flight Unit Offgas
Verification Testing
Verification Completion



ZIN Technologies

SpaceDOC ACME Contract Status

- ACME SpaceDOC involvement began in 2007
- ACME Science Concept Review was held in February 2008
- ACME Requirements Definition Review was held in May 2010
- ACME Preliminary Design Review was held in January 2011
- ACME Phase 0/1 Flight Safety Review (Integrated with CIR) was held in February 2012
- ACME Interim Design Review was held in June 2012
- BRE Requirements Definition Review and Preliminary Design Review (combined) was held in June 2013
- The ACME SpaceDOC work is assigned as incremental Delivery Orders (DOs) to ZIN Technologies under the SpaceDOC contract
- Current Delivery Order contract began in October 2010 and continues through December 2013

PDR RFA-001 Disposition

RFA # 01

Originator: **Anita Tenteris**

Originator Concurrence: 23 September 2013

Statement of Concern	Recommended Action	Closure Rationale
One of the criteria for PDR is a preliminary limited life items list. Just because other experiments may not have such a list does not mean ACME should not have such a list. ACME presents a unique situation where experiments will occur in the same chamber as the Cube is located as well as tubing, wiring, cooling lines, etc. The project should to start identifying their materials lists from the Computer-Aided Design (CAD) drawings so that compatibility issues/soft goods shelf life/limited life items can be identified early in the design cycle. Maybe testing would also be needed as well for materials with unknown compatibility?	Identify the material compatibility and shelf life limits for the soft goods used in ACME and CIR	It has been determined by the project that a materials analysis would not be done for PDR due to funding and resource limitations. The plan is to have this analysis and MIUL completed at CDR. Although the analysis was not complete at PDR, the ACME project is aware of potential issues with the Data Cube and other electronics and their compatibility with the gaseous fuels. These issues are currently being tracked in the Risk Management Implementation Tool database.

PDR RFA-002 Disposition

RFA # 02

Originator: **Anita Tenteris**

Originator Concurrence: 23 September 2013

Statement of Concern	Recommended Action	Closure Rationale
In many of the presentations, the presenters indicated CIR heritage for a choice of hardware. It seemed as though CIR was treated as a 'black box' that ACME hooks into. ACME fuels need to be compatible with CIR and also cause no issues to CIR. The following items need to be investigated further to assess compatibility: • ACME Venting compatibility with CIR hardware • Are the gases/liquids compatible with CIR hardware, particularly with the soft goods in the solenoid valves? The board was told the solenoid valves contained FKM o-ring. For the ACME gases, NBr is the recommended rubber for most of the gases, if not all. • What are the limited life items for CIR? Do any of the items need be replaced prior to ACME use?		An assessment of the venting compatibility has been addressed and was deemed acceptable. As the project matures, CIR will assess compatibility of materials in their system with ACME gasses, products of combustion and any other applicable interactions. ACME anticipates this will be completed by CDR.

PDR RFA-003 Disposition

RFA # 03

Originator: **Al Linne**

Originator Concurrence: 7 November 2013

Statement of Concern	Recommended Action	Closure Rationale
There are no warnings to eliminate damage to the test hardware	To minimize the likelihood of hardware damage, the flight & ground software designs should incorporate software requirements to identify invalid commands and command parameters that are outside of allowable limits and to alert operations personnel that invalid commands were entered.	A complete software command list for the ACME hardware has been derived from operational requirements for the ACME system. Operational limits have been defined for each command and incorporated into the software requirement set. The software requirements are documented in the Software section of the ACME Engineering Requirements Document (ERD) ACME-REQ-001. This list of signals is also documented in the ACME Data Dictionary Document (ACME-DOC-008). These documents will be available for NASA review and comment during the CDR. The approved operational limitations will be built into the ACME ground command station software at the TSC. Every command sent by an operator will be checked against these established limits. Commands that violate safe ranges will be rejected (on the ground) and the operator will receive an error message. This ground software will be verified, as part of the overall ACME test and verification campaign, to ensure no command can ever reach the ACME hardware that is outside of these safely established limits.

PDR RFA-004 Disposition

RFA # 04

Originator: **Al Linne**

Originator Concurrence: 7 November 2013

Statement of Concern	Recommended Action	Closure Rationale
A detonation analysis has not been completed.	Perform a detonation analysis on the expected ACME gas mixtures. This should be done prior to the ACME Phase 0/1 Flight Safety Review, since it may inform some of the hazard analysis. Also, the results of this analysis could drive the consumables usage and could limit the amount of science performed.	An ACME detonation analysis has been performed for the Inverse Flame Experiment, which is the the worst case hazard. It was chosen because of the possible oxygen interaction with ACME CIA hardware (i.e. materials compatibility). The analysis validated that the present configuration could safely meet requirements. The analysis has not impacted the design. A full detonation analysis of all experiments will be completed prior to the Phase 2 Flight Safety Review.

PDR RFA-005 Disposition

RFA # 05

Originator: **Al Linne**

Originator Concurrence: 7 November 2013

Statement of Concern	Recommended Action	Closure Rationale
A number of different oxygen percentages were discussed & identified in the PDR presentation and documentation. What oxygen percentage was used in the design and does this cover the ranges requested by the PI.	Determine the worst case oxygen percentage (and pressure) that could potentially be in the CIR chamber with ACME installed, based on credible failures to CIR & ACME equipment and use this oxygen percentage as the design point for ACME. This will drive integrated safety/hazard analyses as well as ACME material compatibility. Also, based on the resulting worst case oxygen percentage determine if an oxygen compatibility assessment needs to performed for the ACME Chamber Insert Assembly.	Oxygen compatibility with the ACME CIA has been assessed and is documented in the Draft Materials Identification/ Usage List (MIUL), which is a review item for the CDR. An Integrated Hazard Analysis has been completed and the worst-case oxygen percentages and pressures have been identified and addressed in the Hazard Report (see ACME Hazard Report 5, documented in ACME-SDP-001).

PDR RFA-006 Disposition

RFA # 06

Originator: **Alan Linne, Nechelle Grant, Terry O'Malley**

Originator Concurrence: 25 October 2013

Statement of Concern	Recommended Action	Closure Rationale
CIR hazards mitigations are being used by ACME. It was not clear that the ACME team had reviewed the CIR hazards and confirmed the mitigation steps are applicable to ACME. There were no integrated hazards included in the ACME Phase 0/1 Flight Safety Data Package safety hazard analysis (ACME-SDP-001.	The ACME and CIR teams should provide an integrated CIR/ACME Safety story. There are likely integrated safety hazards that exist in the integrated system and these should be identified and documented. Perform an integrated hazard analysis and prepare integrated hazard reports. The integrated CIR/ACME hazard analysis should be part of the Phase 0/1 Flight Safety Review. Also, a pressure switch, a solenoid valve, and a pressure regulator in the ACME system were included in the design on the N2 input to the chamber. Perform a integrated hazard analysis of this system (CIR & ACME hardware) to ensure system meets the intent of CIR hazard analyses.	The Integrated Hazard Analysis has been completed with the participation of the CIR Team. An Integrated Safety Data Package has been prepared and was included as input to the ACME Phase 0/1 Flight Safety Review.

PDR RFA-007 Disposition

RFA # 07

Originator: **Terry O'Malley**

Originator Concurrence: 25 October 2013

Statement of Concern	Recommended Action	Closure Rationale
The ACME project plan did not provide sufficient detail of the end-to-end project planning. There was no evidence of a bottoms up cost estimate or a bottoms up schedule. Both of these are vital in ensuring the project has been thought through and that sufficient resources are available to successfully complete the project. The only budget shown was the program budget, and it may not be adequate complete the current scope of the ACME project. The ACME Systems Engineering Management Plan also did not provide sufficient detail of the end-to-end project technical management. It provided details through PDR/CDR, but not the project lifecycle.	Update the Project Plan. Complete a bottoms up cost estimate and a bottoms up end-to-end schedule. Consider performing a parametric cost estimate. To be of value, these should be performed within 6 months of the PDR. Update the SEMP to include technical details through launch. Include things as spare required, environmental testing required, Ground/Flight Support Equipment etc.	An updated Project Plan has been baselined, which includes a bottoms-up cost estimate and schedule. An independent cost estimate by the GRC Cost & Economic Analysis Office has been performed and is in-line with the project plan. Updates to the SEMP will be made regularly as more information becomes available to the project team. The SEMP is a living document that will be revisited as necessary.

PDR RFA-008 Disposition

RFA # 08

Originator: **Terry O'Malley**

Originator Concurrence: 25 October 2013

Statement of Concern	Recommended Action	Closure Rationale
The ACME Engineering requirements document is incomplete. While it appears that the Science requirements have been thoroughly addressed, carrier requirements (both CIR & ISS) and safety and mission assurance have not been addressed in the ACME ERD. It is important to thoroughly identify all requirements which are imposed on the system and subsequently must be verified. A related concern is the number of requirements that must be verified. There are a large number of requirements on the system and the verification program could be extensive.	Complete the ERD. This should be done within 6 months of the PDR to ensure the CDR design meets all requirements that are levied on the system. Environmental testing requirements should be included as part of this, waivers for fragile equipment should be identified early. When developing the verification program look for ways to combine & streamline the verification process.	The document FCF-ICD-CIR-ACME addresses carrier requirements (both CIR and ISS). ACME-REQ-001 points there. Safety and Mission Assurance requirements will be detailed in the ACME Hazards and Integrated CIR-ACME hazard reports. Following the Phase 0/1 Flight Safety Review, where these Hazard Reports are approved, the requirements will be incorporated into ACME-REQ-001. The verification process will group requirements by discipline identified to facilitate verification and validation.

PDR RFA-009 Disposition

RFA # 09

Originator: **Lisa Van Der Aar**

Originator Concurrence: 9 September 2013

Statement of Concern	Recommended Action	Closure Rationale
The project has done a good job of writing operational requirements, but they have not yet been vetted with the PIs. This concern stems from RDR when the PIs were assuming operational requirements that the GRC science and engineering teams were not aware of. This concern follows on from the ACME RDR-004 RFA which has not been closed out as of the PDR.	It is imperative that: 1) the currently documented operational requirements are vetted with each of the PIs; 2) the engineering team talks with each of the PIs. This will ensure that both parties fullly understand what is meant by each of the requirements, and also give the engineering team a chance to uncover additional operational requirements. Close out the RFA ACME RDR-004.	The Project Scientist, Dennis Stocker, has vetted the operations requirements documented in ACME-SRD-001 with the respective Pis. Dennis has also discussed the Ops requirements with the ACME engineering team to be sure they are well understood. The ERD revision including the updates with the operations requirements has been completed and released. This information should be sufficient to close RFA ACME-RDR-004.

PDR BRE RFA1 Disposition

BRE RFA #1

Originator: **Roger Schwarz**

Originator Concurrence:

Statement of Concern	Recommended Action	Closure Rationale
The BRE Science Requirements, which will be incorporated in the ACME Integrated Science Document Revision B, were presented in a draft version (non-signed) at the review. While a draft (ready for signature) ISRD is applicable at RDR, by PDR the ISRD should be signed as it baselines the science requirements. Additional changes and additions were discussed at the review, and these changes could affect the BRE burner and radiometer design requirements.	Expedite approval of the ACME Revision B Integrated Science Requirements Document, including any changes from RDR. Any delay could impact the engineering effort to design the burner and radiometer components of the BRE.	The BRE Science Requirements documented in the Integrated Science Requirements Document have been baselined.

PDR BRE RFA2 Disposition

BRE RFA 2

Originator: **Scott Darpel**

Originator Concurrence:

Statement of Concern	Recommended Action	Closure Rationale
Although this is specifically targeted at BRE, the ACME project in general is lacking a completed SAR matrix which is required at PDR. The project SEMP states that the project will not have its own SMAP, but will follow ZIN's P30012. P30012 is being replace with the newer P30020, but in either case, following each document requires the individual projects to complete an applicability matrix which details what S&MA requirements they plan to fully meet, or those they intend to partially meet or feel are not applicable, along with a rationale	The project is reported to be working on complying with P30020. BRE and all of ACME has to get approved and signed their SMAP applicability matrix.	The ACME SAR Checklist has been completed

PDR BRE RFA3 Disposition

BRE RFA 3

Originator: Scott Darpel, Roger Schwarz

Originator Concurrence:

Statement of Concern	Recommended Action	Closure Rationale
The charge for the engineering board, as communicated via the convening memo, included an action to evaluate the project's performance vs. plan on budget and schedule. The presentation package did not include sufficient information that would allow the board to make this evaluation. Specifically, as BRE is a late addition to the ACME project, the board is interested to learn what additional risks to budget and schedule were incurred.	Provide the members of the engineering board with information that communicates the overall performance against plan with respect to budget and schedule. Specifically, the board would like to see what the addition of BRE did to the overall cost and schedule to ACME. The information should make it clear what risks may or may not have been levied upon the ACME project with the late inclusion of the BRE.	Data to be provided by the ACME CDR

PDR BRE RFA4 Disposition

BRE RFA 4

Originator: **Nancy Hall**

Originator Concurrence:

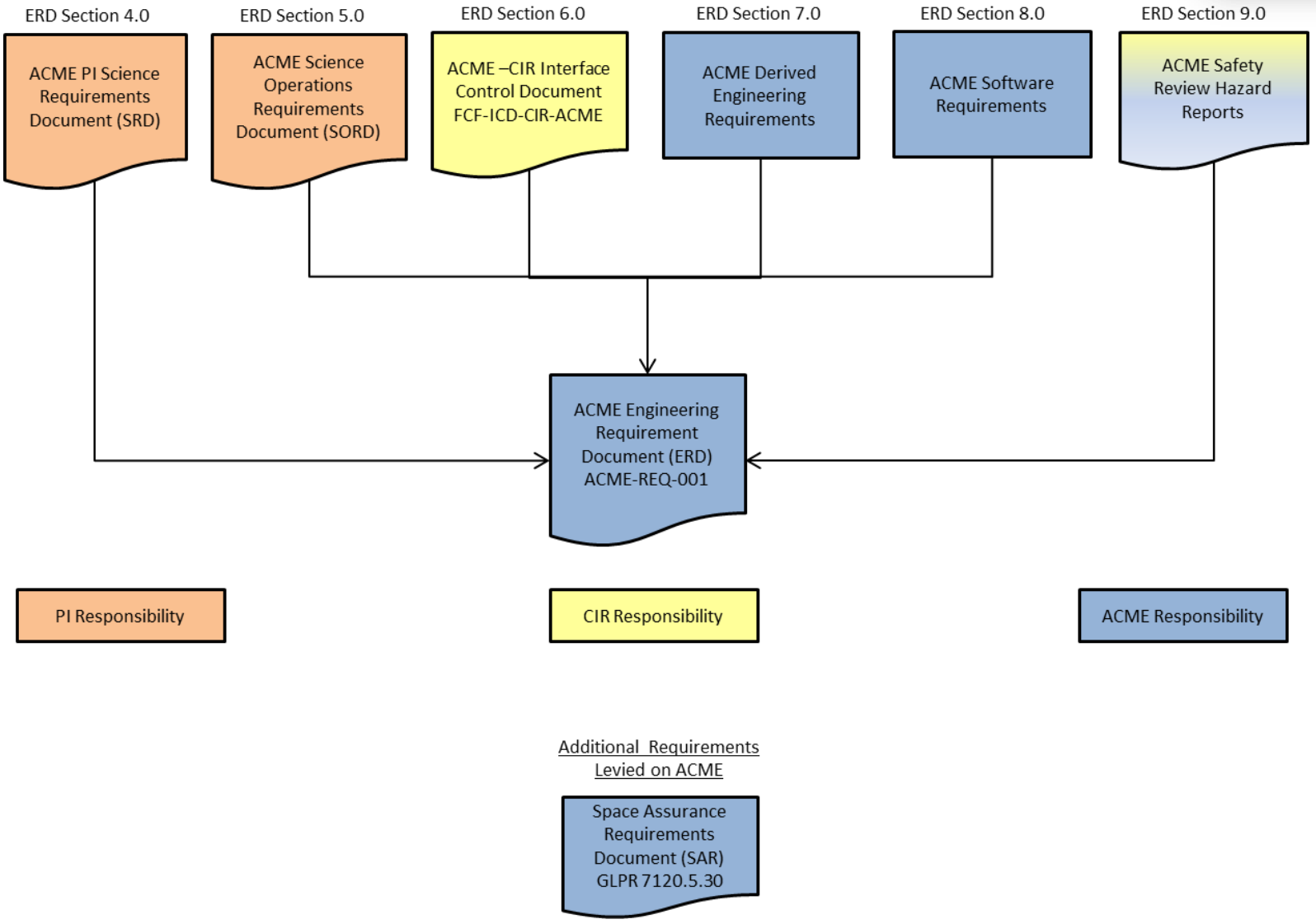
Statement of Concern	Recommended Action	Closure Rationale
ACME project risk due to addition of the BRE experiment was not presented at PDR. The engineering chair was later given access to RMIT and the engineering panel reviewed the ACME risk status. . There is a risk covering the BRE science requirements, but the risk mitigation does not include details of the project's forward plan to meet the BRE requirements at an acceptable level of risk and does not indicate that BRE requirements are not finalized	Update the BRE risk to include mitigation steps to address changes to the BRE science requirements	The risk regarding BRE science requirements was updated to reflect the desired information and has since been closed with the delivery of the BRE science requirements. (Risk ACME-022)

Project Requirements Overview

Chris Mroczka – Systems Engineering Lead

ACME Requirements Flow

Glenn Research Center



ACME Requirements

- ACME ERD – 735 Total Requirements.
  - 407 Engineering build-to requirements 55.37%
  - 107 Operations requirements 14.56%
  - 202 Software requirements 27.48%
  - 19 Derived requirements 2.59%
- Integration requirements are generated by the Mission Integration and Planning Team (MIP) and are documented in FCF-ICD-CIR-ACME. This document encompasses requirements flowed down to ACME form the ISS Program and CIR. All carrier interface requirements are documented here
- Safety requirements are documented in the ACME Safety Verification Tracking Log (SVTL)

ACME Integration Requirements

- Operations
  - Use of chamber fan
  - Use of gas chromatograph (GCIP)
  - Rack acceleration measurement
- Atmosphere and Flow
  - Venting to specific pressures
  - Atmosphere scrubbing
  - Vary gas composition
  - Provide source gases
- Diagnostics
  - LLL-UV
    - OH* 310 nm Filter
  - LLL-UV FOV
    - 30, 50, 80mm lenses
  - HiBMS-2
    - 90mm lens
    - Adjust gain setting
  - HiBMS-1
    - 30, 50mm lenses
  - Illumination Package

Sample Requirements from ACME Requirements Matrix

System	SSC_ID	ISRD Trace	ISRD Description Trace	ERD REQ Title/Description	Comply? (Y/W/N)	Compliance Approach	V-type	V-area	Verification Activities
	SPACE110	1.1		**General Experiment Configuration**	Y				
BURNER	530_0010	1.1.1	1.1.1. The experimental hardware shall accommodate all of the burners described in sections 1.2-1.4, and 1.13, but only a single burner shall be installed at a time within the CIR chamber.*	**Burners – Installation** ACME shall install a single burner one at a time within the CIR chamber.	Y	ACME design includes replaceable burners that will be installed on orbit one at a time in the CIA to support each experiment within the CIR.	D	ME	The verification shall be considered successful when a demonstration of the flight hardware shows that a single burner can be installed in the CIA one at a time.
BURNER	530_0020	1.1.2	1.1.2. All burners shall be Orbital Replacement Units (ORUs) and exchangeable during spaceflight.	**Burners – Exchangeable** ACME shall provide that all burners are exchangeable during spaceflight.	Y	ACME design includes exchangeable burners that can be changed out during spacefilght.	I	ME	The verification shall be considered successful when the inspection of each flight burner mounting plate shows that it is compatible with the flight switching manifold.
BURNER	530_0030	1.1.2		**Burners – ORU** ACME shall provide all burners as ORUs.	Y	ACME design includes burners that are orbital replaceable units.	I	PS	The verification shall be considered successful when each burner is approved by HFIT for ORU classification.
FLOW	520_0010	1.1.4.1	1.1.4. The gas delivery system and all burners shall be compatible with at least the following gases: 1.1.4.1. gaseous fuels: hydrogen, methane, ethane, propane, ethylene, propylene, or mixtures of any of these fuels;	**Flow System H/W Compatibility - Hydrogen** ACME shall provide a gas delivery system that is compatible with the gaseous fuel hydrogen.	Y	ACME design includes a flow system that is compatible with the gaseous fuel hydrogen.	A	MP	The verification shall be considered successful when the material analysis of the flow system hardware shows compatibility with the gaseous fuel hydrogen.

ACME Compliance Summary

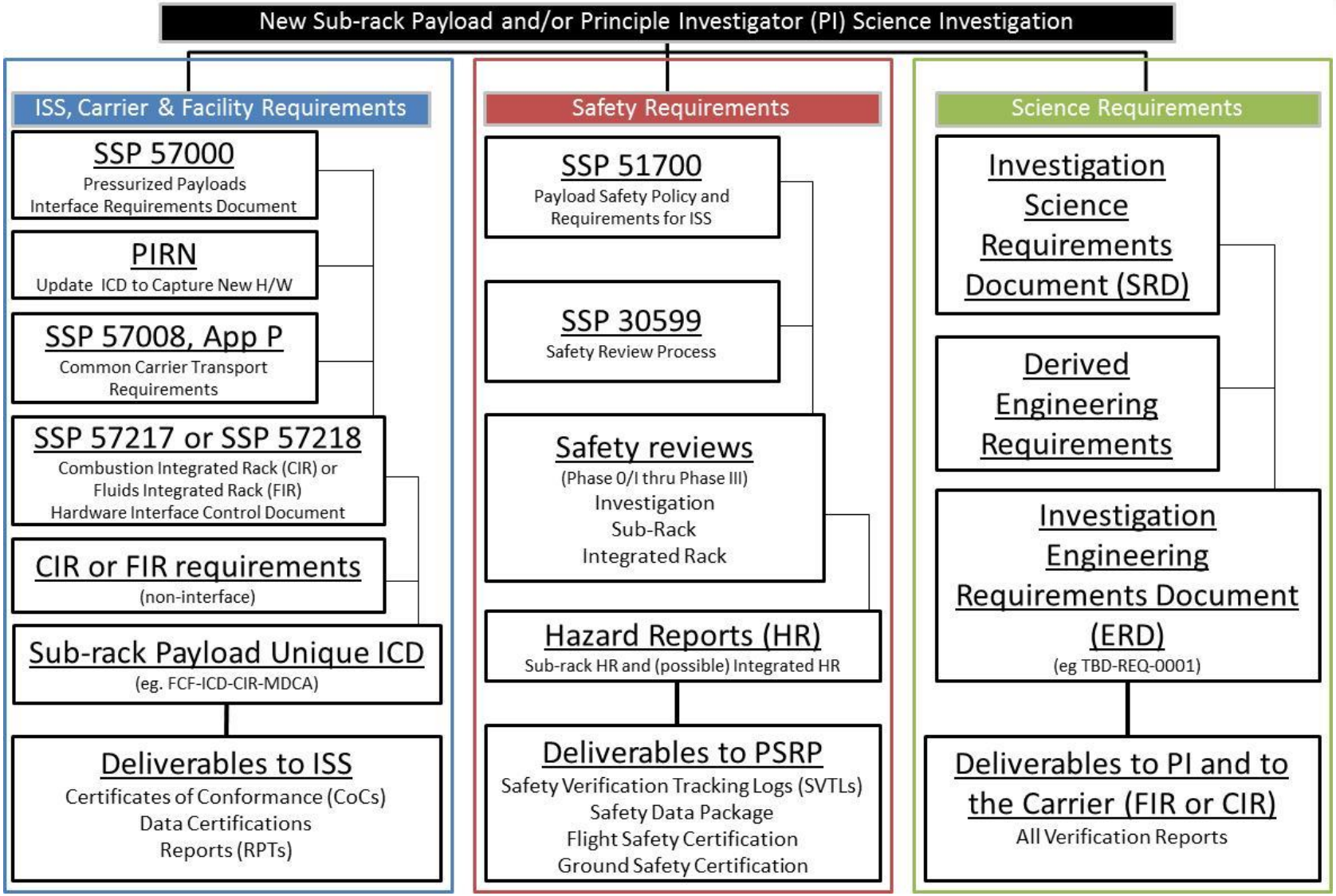
- ACME-DOC-002 Requirements Compliance Matrix
  - Engineering build-to requirements: 100% Compliance
  - Operational requirements: 100% Compliance
  - Software requirements: 100% Compliance
  - Derived requirements: 100% Compliance

Verification Planning

- ACME and CIR will verify system performance
  - Engineering, operational, software, and derived requirements in the ACME ERD are verified in the Science & Engineering Verification Plan; ACME-PLN-003 (Baselined)
    - ACME requirements called out in the ERD are ACME's responsibility to verify or obtain verification from carrier (i.e. CIR cameras, GCIP, etc.)
    - CIR component performance is verified and reported to ACME via the MIP Team.
- ACME will also verify payload requirements passed down from the ISS Program and CIR (FCF-ICD-CIR-ACME)
  - Payload requirements are verified in the Payload Verification Plan (FCF-PVP-CIR-ACME)
    - PVP is generated by the MIP Team
    - ACME verifies payload requirements
    - ACME delivers Payload Verification Report to CIR to be rolled up and reported in the integrated rack verifications

Sub-rack Payload Requirement & Verification Flow

Glenn Research Center



Project Testing Overview

Chris Mroczka – Systems Engineering Lead

ACME Testing Plan

- As Engineering Model sub-systems are built, they will be functionally tested to validate performance at that level
- When the system level Engineering Model is complete we will conduct ACME functional testing between the AVP, Data Camera and CIA to validate performance at the system level
- These 3 units comprising the Engineering Model will then be installed on the CIR GIU for integrated testing with the rack, validating carrier interfaces
- Since the EM is built to flight drawings, it will then undergo Qual testing: Vibration, EMI and thermal
- When the Flight Unit is complete the Flight AVP, CIA and Data Camera will undergo flight level environmental testing: Vibe, EMI and offgas
- The remainder of flight verification testing will then occur after the flight environmental testing

Environmental and Other Significant Testing

Radiation Testing

- May 1-3 at Indiana University Cyclotron facility
- Tested boards with complex electronics
  - CPUs, DSPs, FPGAs, etc
  - Lippert SBC, RTD Dual GigE, RTD Power, Connecttech Serial, PMD Motion Conroller, Crucial SSD
- Added a power cycle capability to camera
- Changed version of PMD board to not include ARM processor

E-Field EMI Testing

- GRC-EMI-RPT-331: *June 2012* - Most tests over the limit
- GRC-EMI-RPT-343: *April 2013* - Better results, still over the limit
- GRC-EMI-RPT-345: *July 2013* - Acceptable results
- Iterated the design and subsystem capabilities with PS

EM Avionics Package Functional Testing

- Completed EM Avionics Package assembly successfully functionally tested

Engineering Testing

Document ID	Document Title	Status
ACME-TST-002	Flow System Model Breadboard Test Report	Breadboard Level, Validated all test points.
ACME-TST-003	Breadboard System Level Test	Breadboard Level, Validated all test points.
ACME-TST-004	E-Field Mesh Assembly Test Plan	EM level functional test (Passed)
ACME-TST-005	E-Field Power Supply Assembly Test Procedure	EM level functional test (Passed)
ACME-TST-006	Hard Drive Tray Assembly/ SATA Cable Test Plan	EM level functional test (Passed)
ACME-TST-007	ACME EMI E-Field HV Assembly Test Plan	EMI test, results in project repository.
ACME-TST-008	ACME Hall Effect Sensor Board	EM level functional test (Passed)
ACME-TST-009	ACME Avionics Package Test Plan	EM level functional test (Passed)
ACME-TST-010	ACME STEPPER MOTOR ASSEMBLY/SUBASSEMBLY TEST PROCEDURE	EM level functional test (Passed)
ACME-TST-011	ACME Power Distribution Board Assembly Test Procedure	EM level functional test (Passed)
ACME-TST-013	ACME Battery Vacuum Test Procedure	EM level functional test (Passed)
ACME-TST-014	ACME TC Cold Junction Compensation Board Test Procedure	Draft, hardware to be built
ACME-TST-015	ACME Analog Camera LED Board Test Procedure	EM level functional test (Passed)

Engineering Testing Cont.

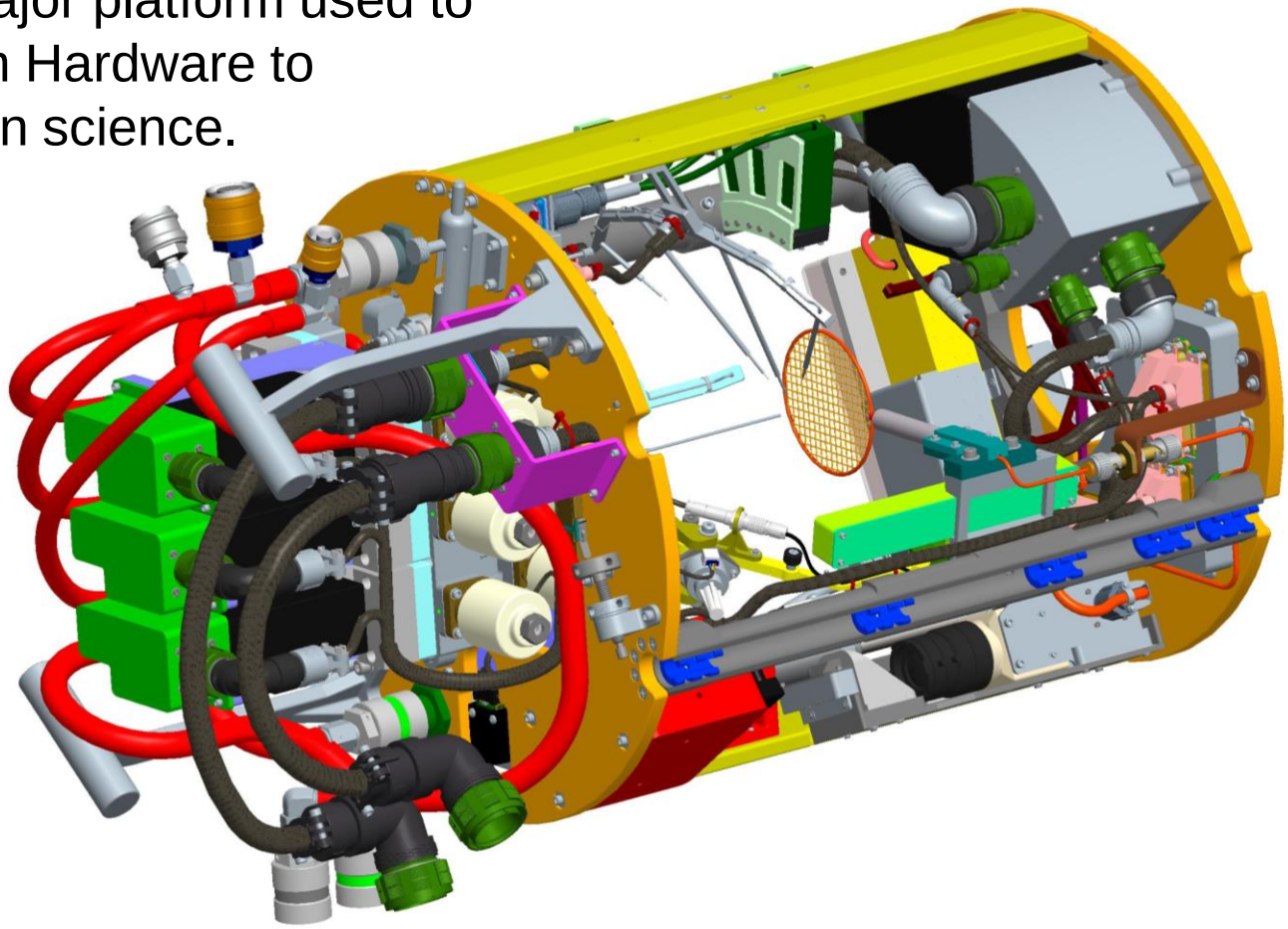
Document ID	Document Title	Status
ACME-TST-016	ACME Filter Board Test Procedure	EM level functional test (Passed)
ACME-TST-017	ACME Camera Stepper Driver Board Test Procedure	Draft, hardware to be built
ACME-TST-018	ACME Cube Assembly Test	EM level functional test (Passed)
ACME-TST-019	ACME Camera Enclosure Test Procedure	Draft
ACME-TST-020	E-Field HV assembly EMI Retest	EMI test, results in project repository.
ACME-TST-021	ACME Camera Control Power Distribution Board Test Procedure	Draft
ACME-TST-022	ACME Radiometer Board Test Procedure	Draft
ACME-TST-023	Motion Control Board Interface Assembly Test Procedure	EM level functional test (Passed)
ACME-TST-024	VGA Interface Board Test Procedure	EM level functional test (Passed)
ACME-TST-025	E-Field HV Assembly Test 3	EMI test, results in project repository.
ACME-TST-026	Down Stream Relay Board Test Procedure	Draft
ACME-TST-027	ACME Camera Control Test Procedure	Draft
ACME-TST-028	Upstream Relay Board Test Procedure	Draft
ACME-TST-029	ACME Flight Cube Board Test	Hardware to be built.

Chamber Insert Assembly Designs

Adrian Drake – Mechanical Engineering Lead

ACME Chamber Insert Assembly

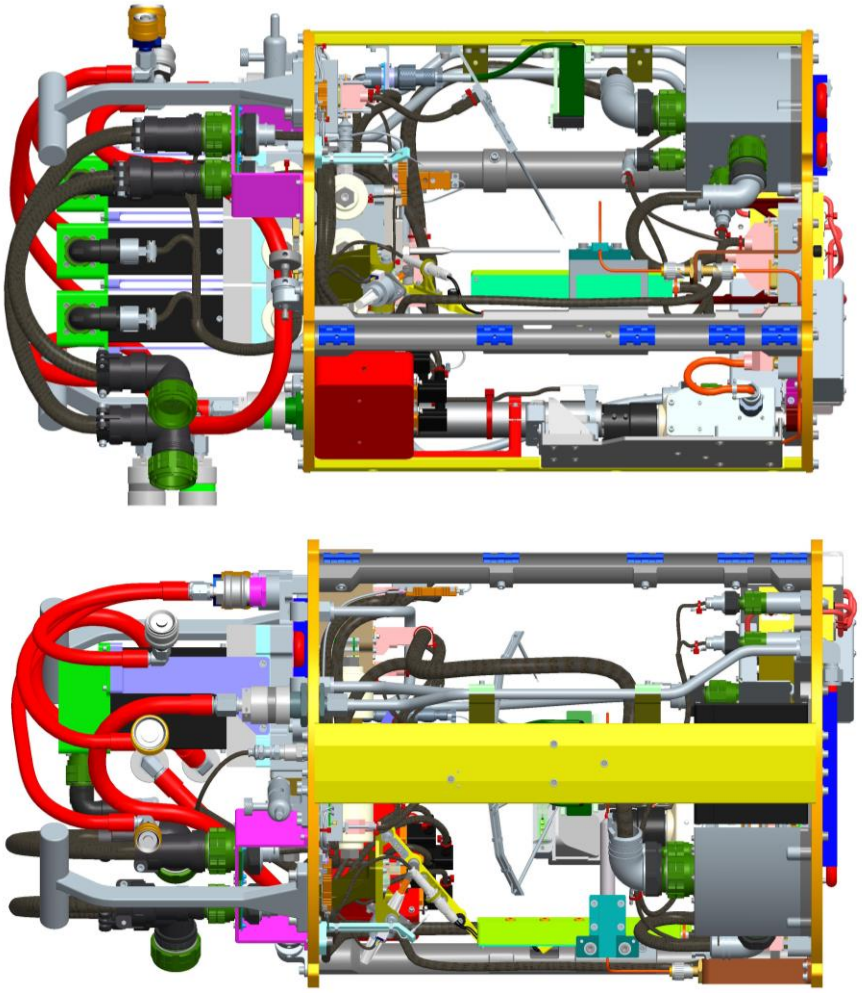
The ACME Chamber Insert Assembly functions as the major platform used to mount Sub-System Hardware to perform combustion science.



ACME Chamber Insert Assembly

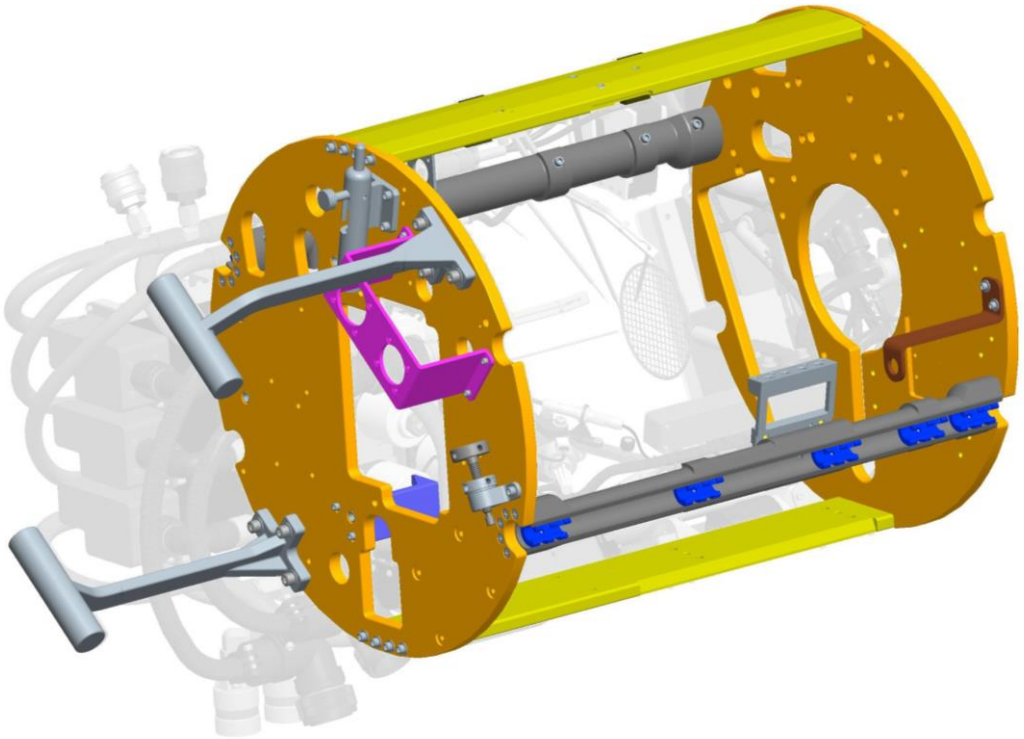
- Component placement maximizes open volume at and downstream of the chamber window interface to satisfy open volume requirements

- Creates mass symmetry where possible per requirements



Experiment Mounting Structure

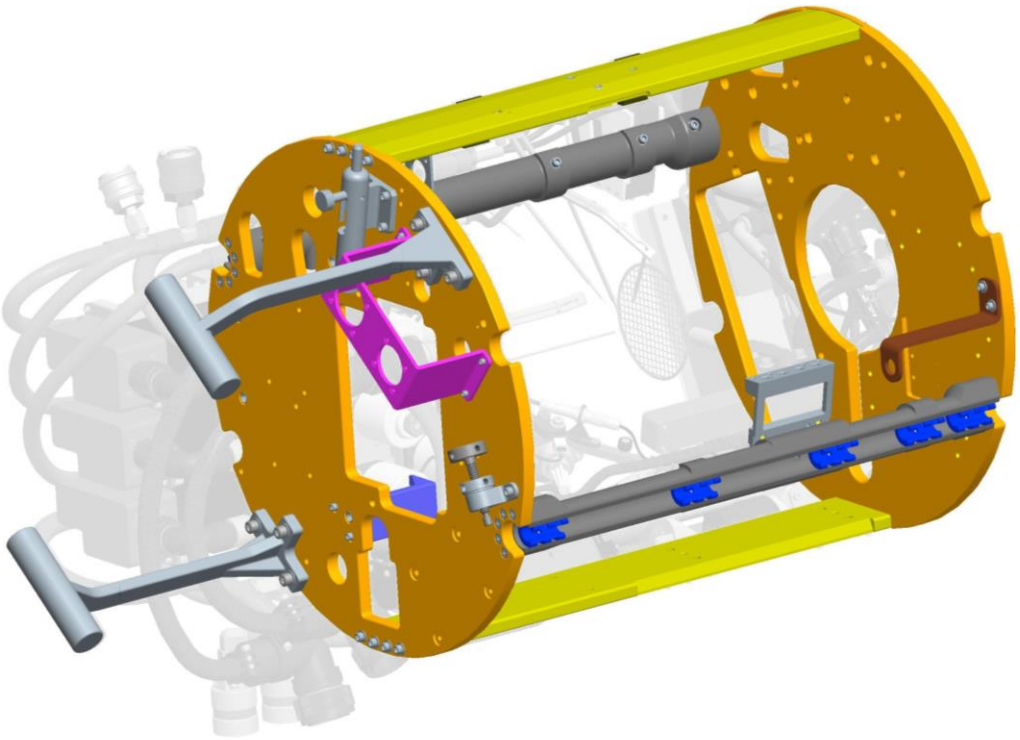
- The Experiment Mounting Structure is the framework to which all other subsystems are attached.
- Aligns the insert within the CIR Chamber



It consists of:
- 2 End Plates
- Frame Rails
- Locking Mechanism and Rail Lock
- Insertion Handles
- Component mounting brackets

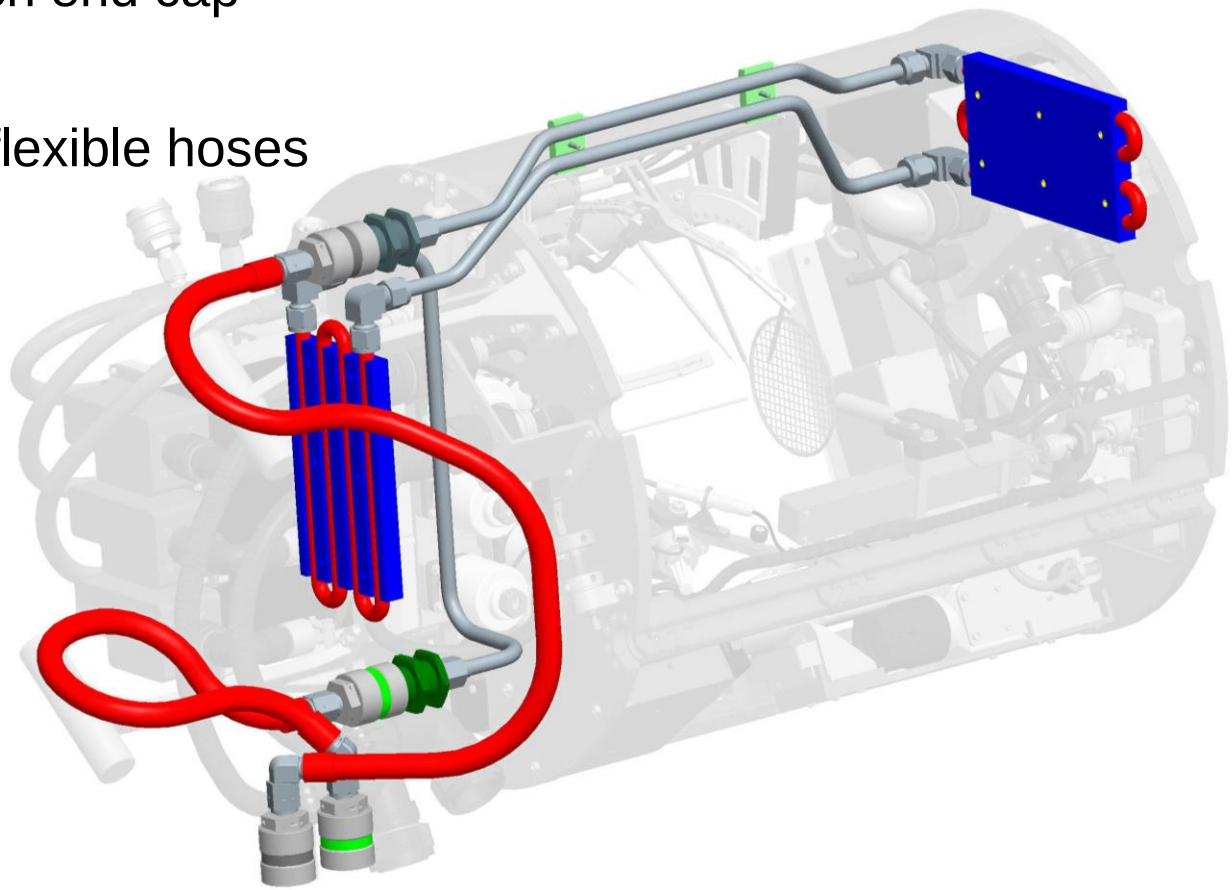
Experiment Mounting Structure

Subsystem Status:
Design Completion: 100%
Drawing Count: 32
Drawing Completion: 50%
EM Build: 28%
Flight Build: 25%
Requirement Compliance: 100%



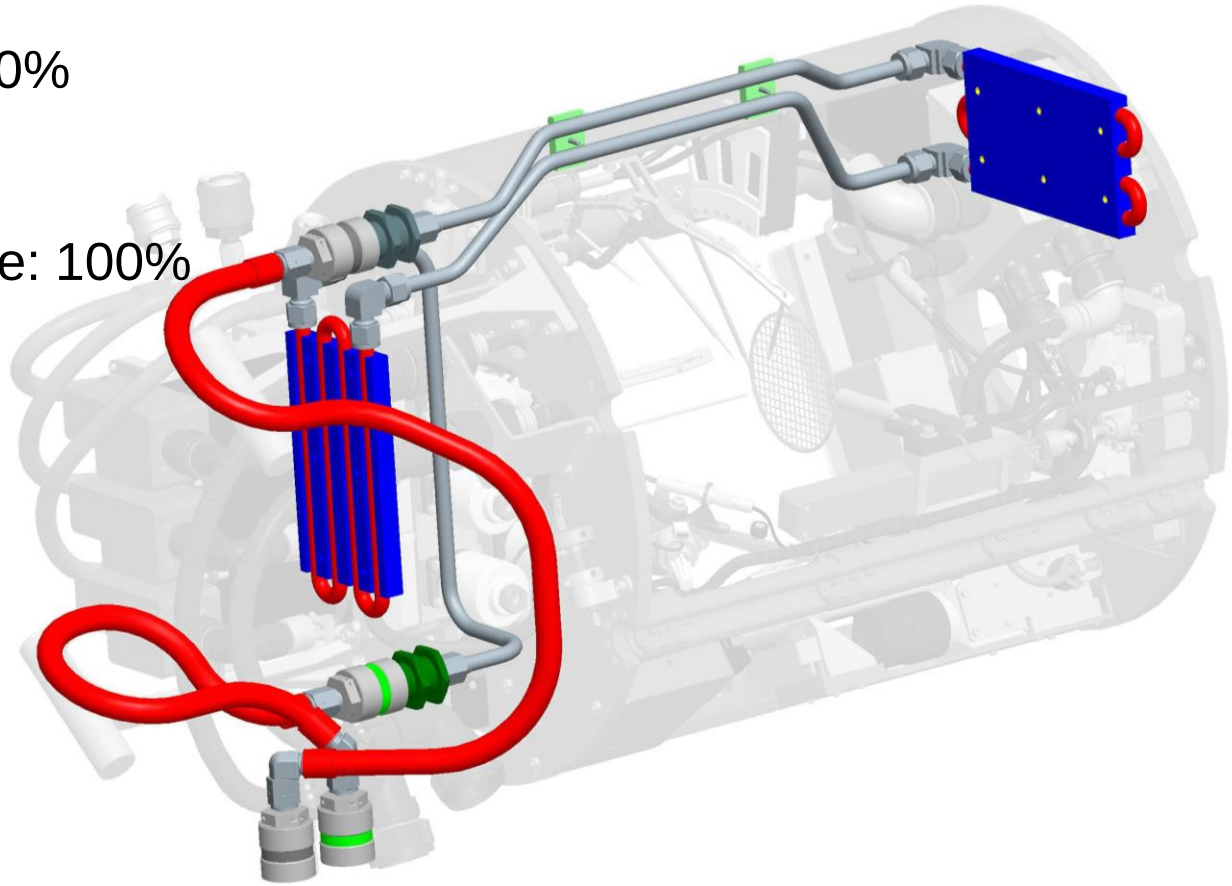
Cooling System

- The Cooling system is made up of 2 cold plates, one on each end cap
- Lytron CP-10
- Attaches to IRR with flexible hoses
- Internal tubing is rigid



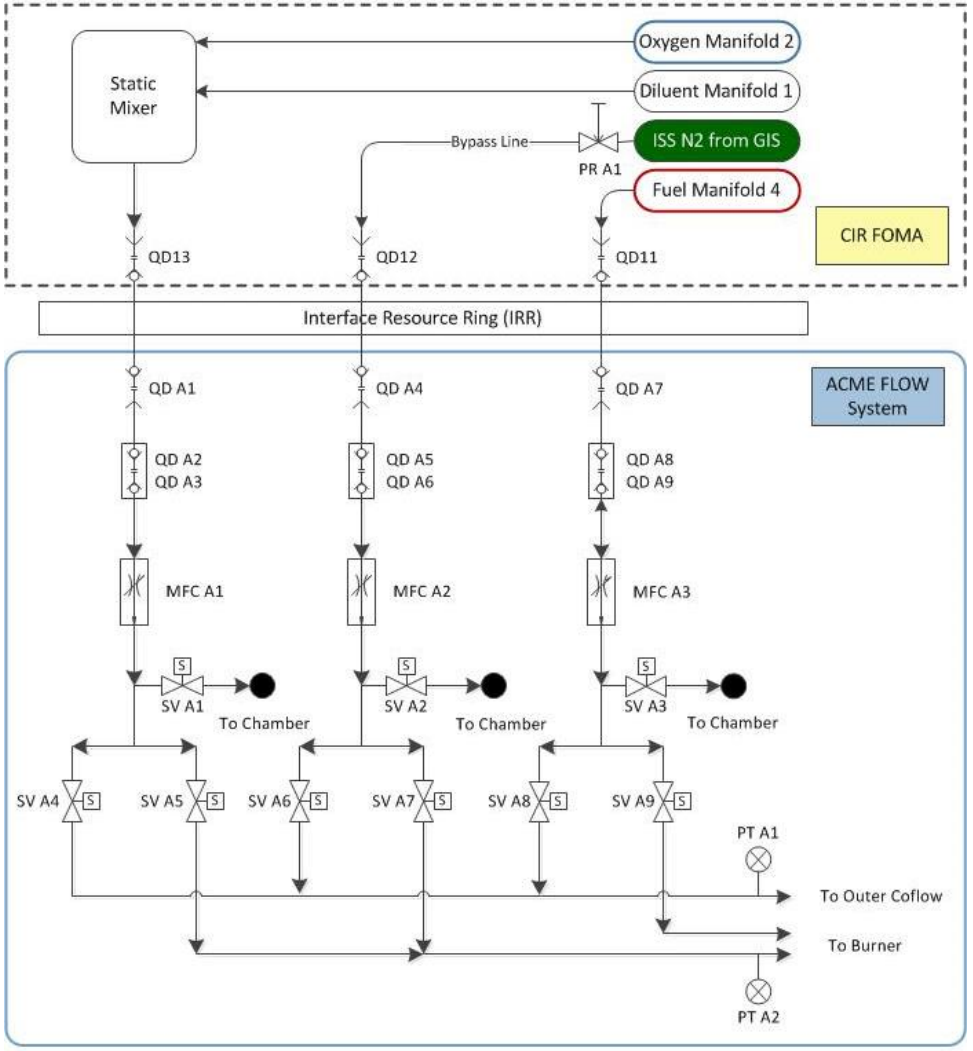
Cooling System

Subsystem Status:
Design Completion: 100%
Drawing Count: 13
Drawing Completion: 100%
EM Build: 15%
Flight Build: 8%
Requirement Compliance: 100%



Gas Flow System

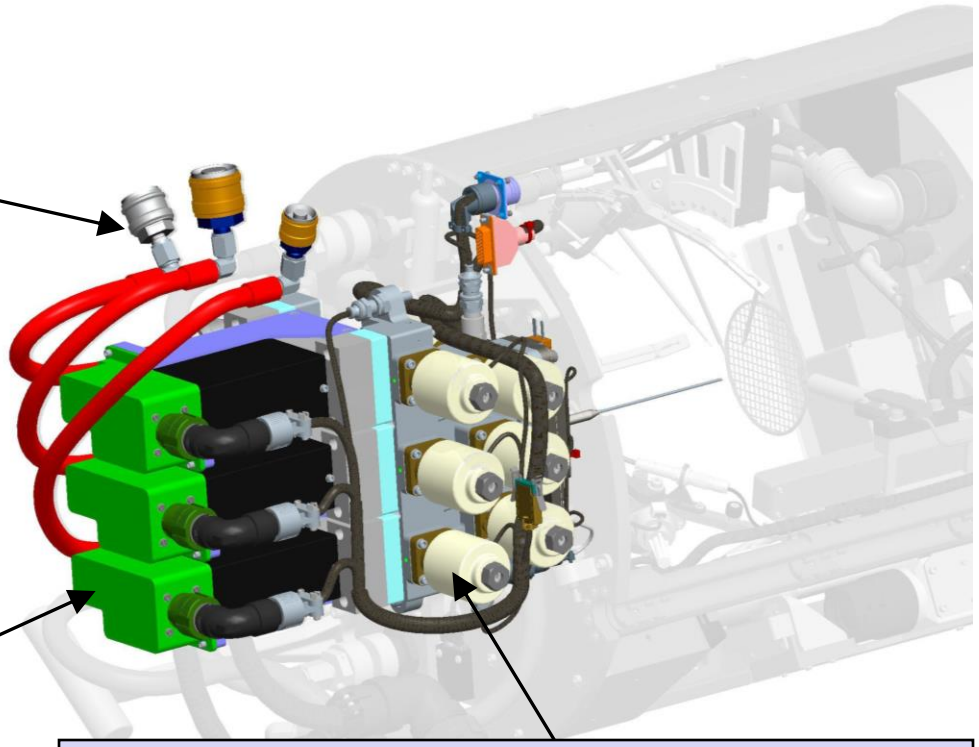
- Receives CIR supplied and regulated gases from FOMA
- Provides pressure regulation of gas from nitrogen bypass line to replace those features bypassed on CIR
- Routes gases to the chamber for chamber fills, bypassing the burner
- Meters gas flow for science compliance via ACME Mass Flow Controllers
- Mixes gases as required
- Delivers gases to the burner



ACME Gas Flow System

Glenn Research Center

IRR QD hookup
• Interface with FOMA



Mass Flow Controllers
• Hastings HFC-D-302 Series
• ORU configuration based on experiment
• Sufficient flexibility to meet all experiment requirements

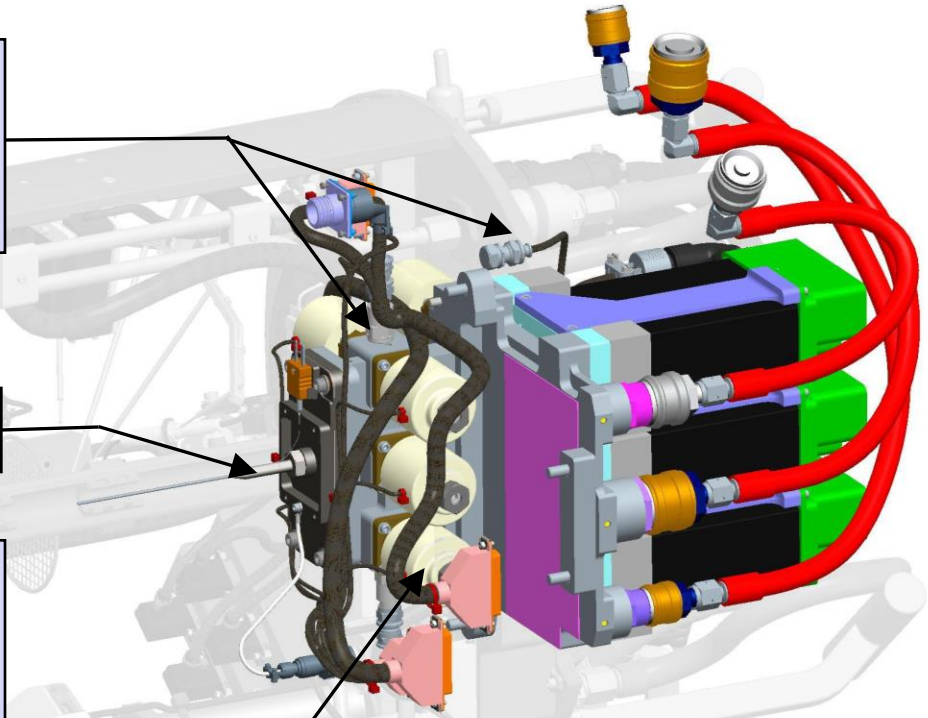
Chamber Vent Valves
• Skinner 7000 Series Low Pressure Solenoid Operated Valves and Lisk L-7040 Solenoid Coil
•Used to bypass the burner for faster chamber filling

ACME Gas Flow System

Entran EPRB-2 Series pressure transducers for pressure drop calculation

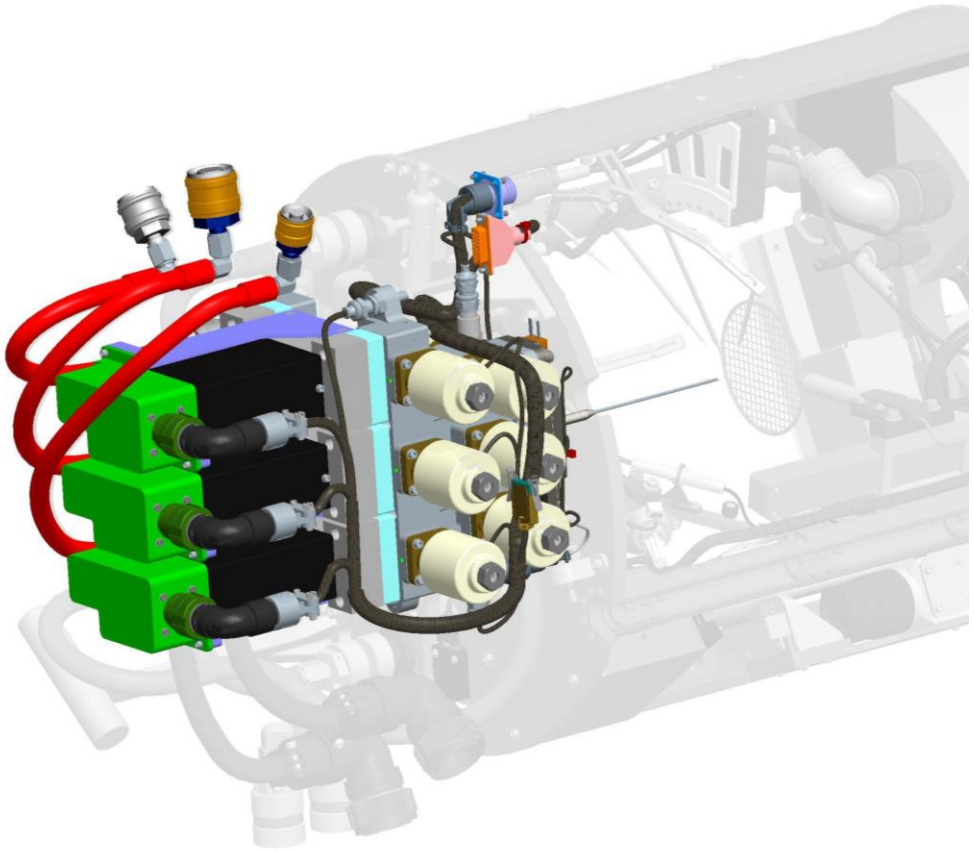
Replaceable burner

Flow Switching Valves
- Skinner 7000 Series Valves with Lisk L-7040 Solenoid Coils
- Controls gas flow to burners
- Requires no crew interface to modify gas flow paths to the burners for each experiment

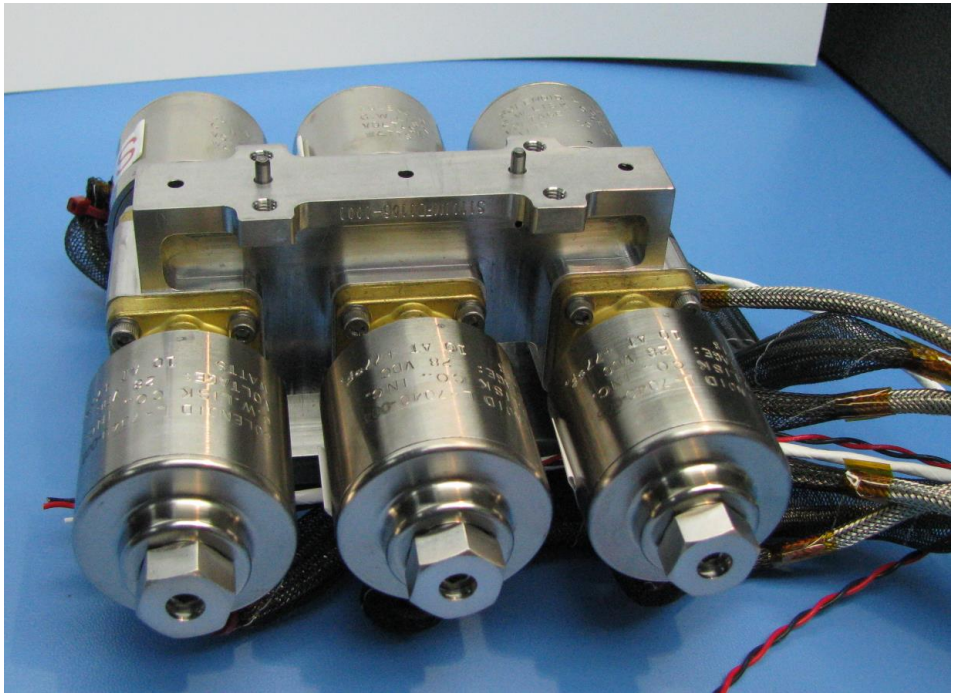


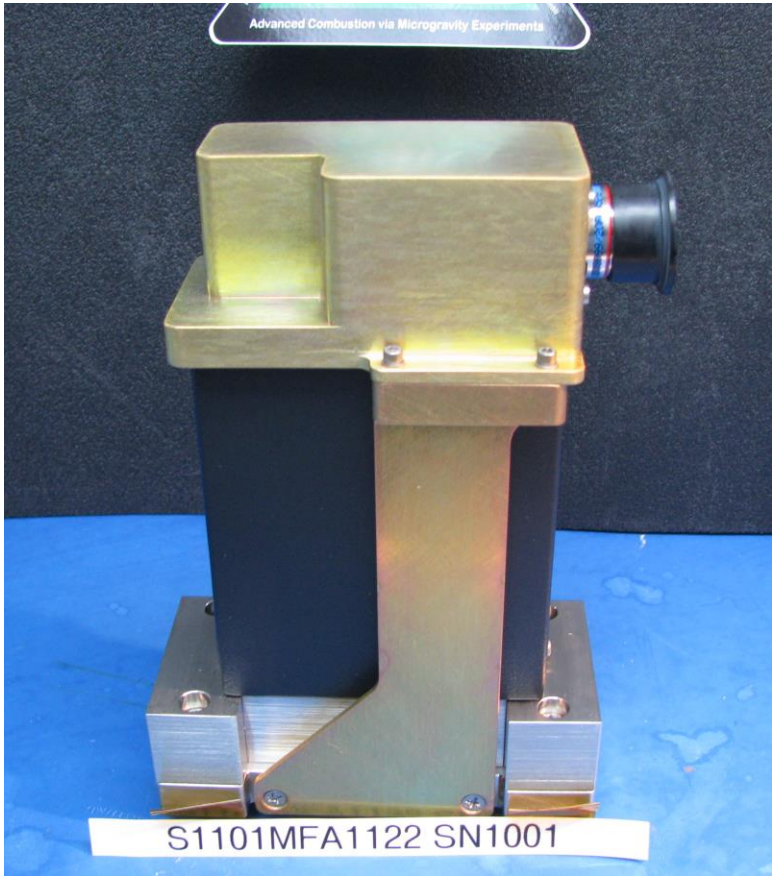
ACME Gas Flow System

Subsystem Status:
Design Completion: 100%
Drawing Count: 54
Drawing Completion: 63%
EM Build: 54%
Flight Build: 20%
Requirement Compliance: 100%



ACME Gas Flow System Hardware





Burner Assembly

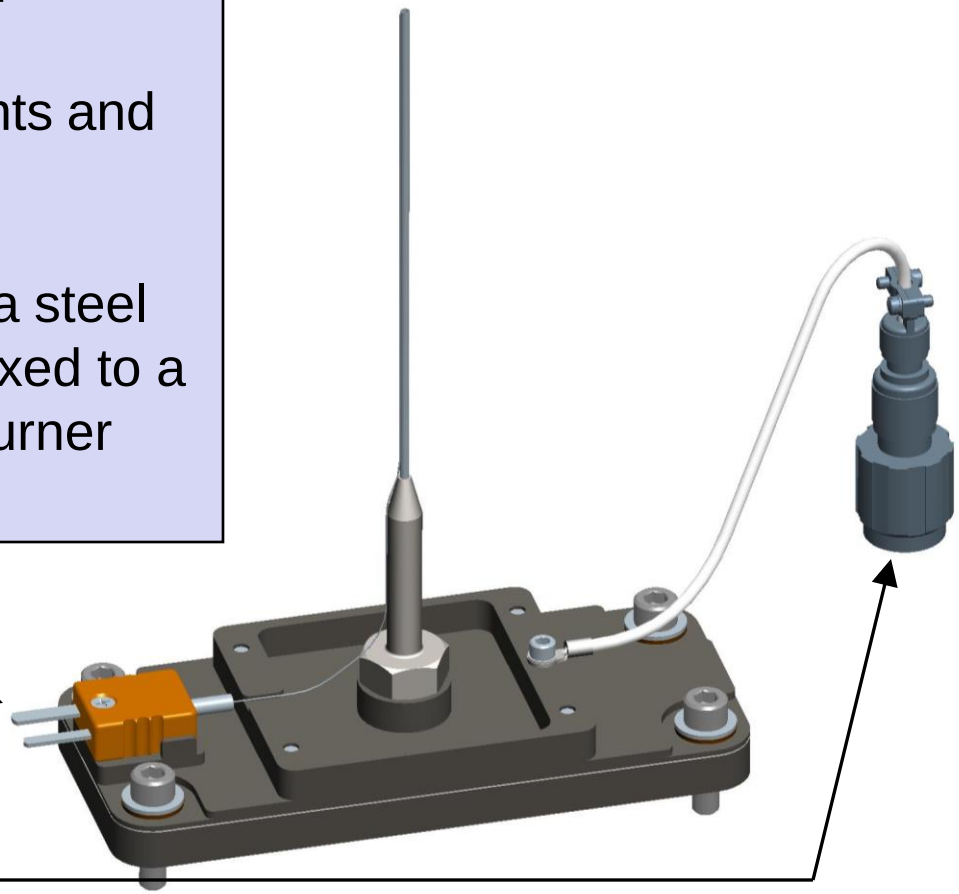
Four Burner types:
Gas Jet, Spherical, Co-flow, BRE

Burners are ORU per requirements and experiment setup

Construction is a burner fixed to a steel mounting plate, which is in turn fixed to a ceramic insulator to isolate the burner electrically

Burner temperature thermocouple connection

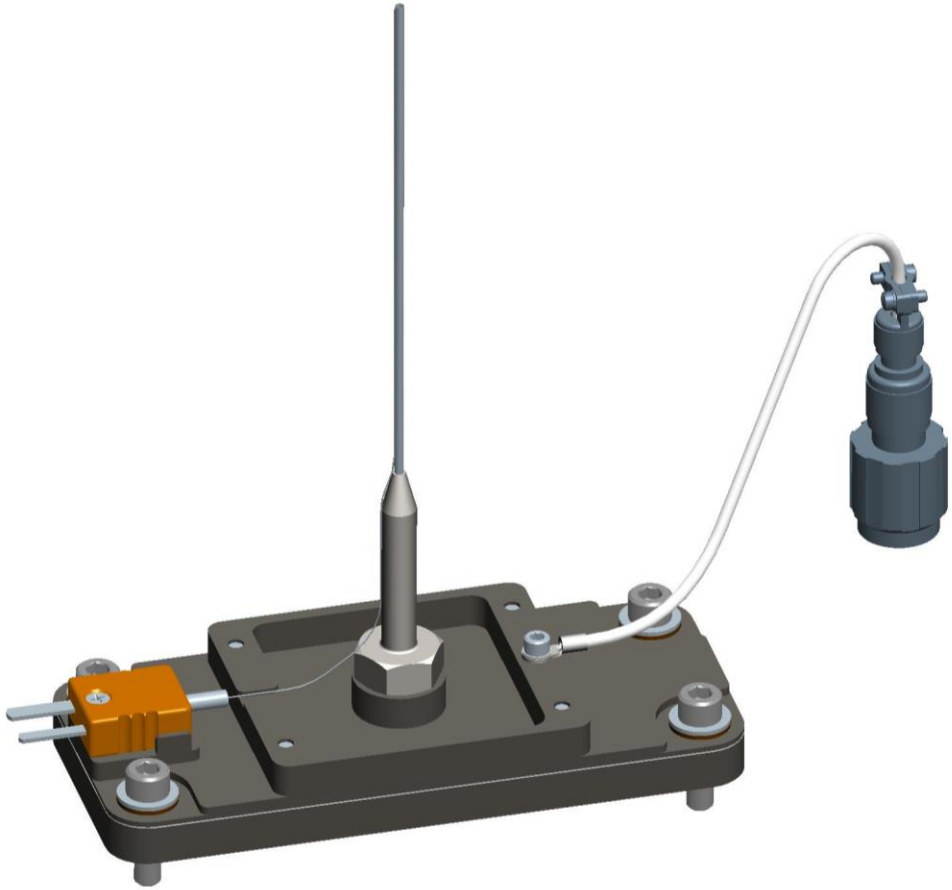
E-field measurement attachment



Gas Jet Burner Status

Gas Jet Design Completion: 100%
Gas Jet Drawing Count: 16
Gas Jet Drawing Completion: 100%

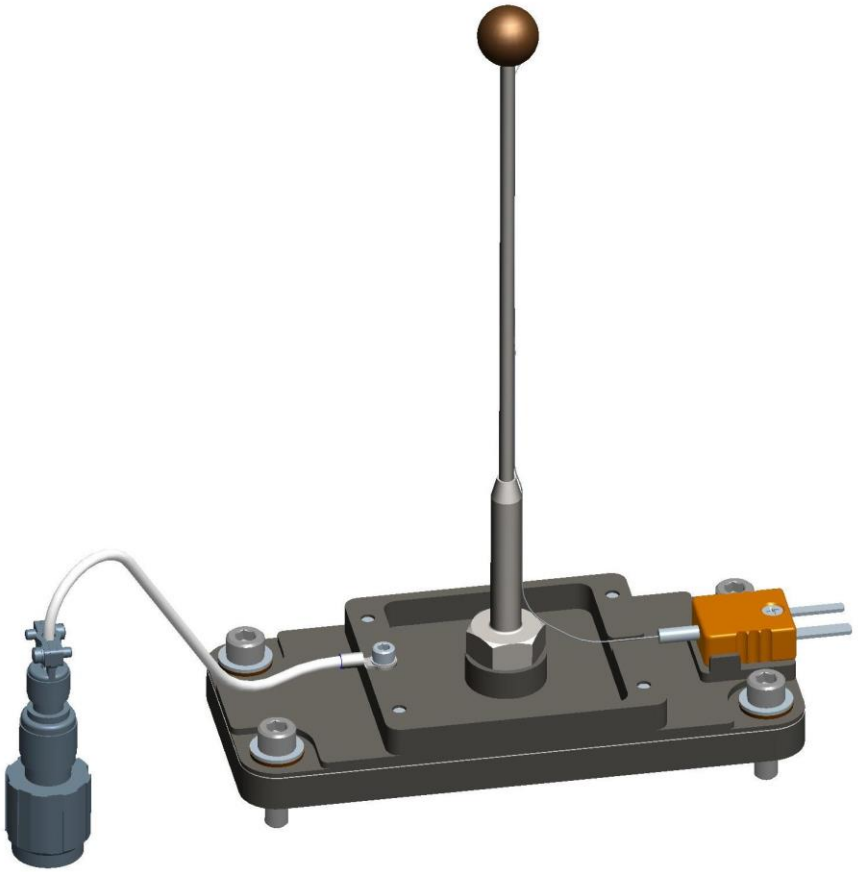
EM Build: 0%
Flight Build: 0%
Requirement Compliance: 100%



Spherical Burner Status

Spherical Design Completion: 100%
Spherical Drawing Count: 14
Spherical Drawing Completion: 100%

EM Build: 0%
Flight Build: 0%
Requirement Compliance: 100%



Spherical Burner

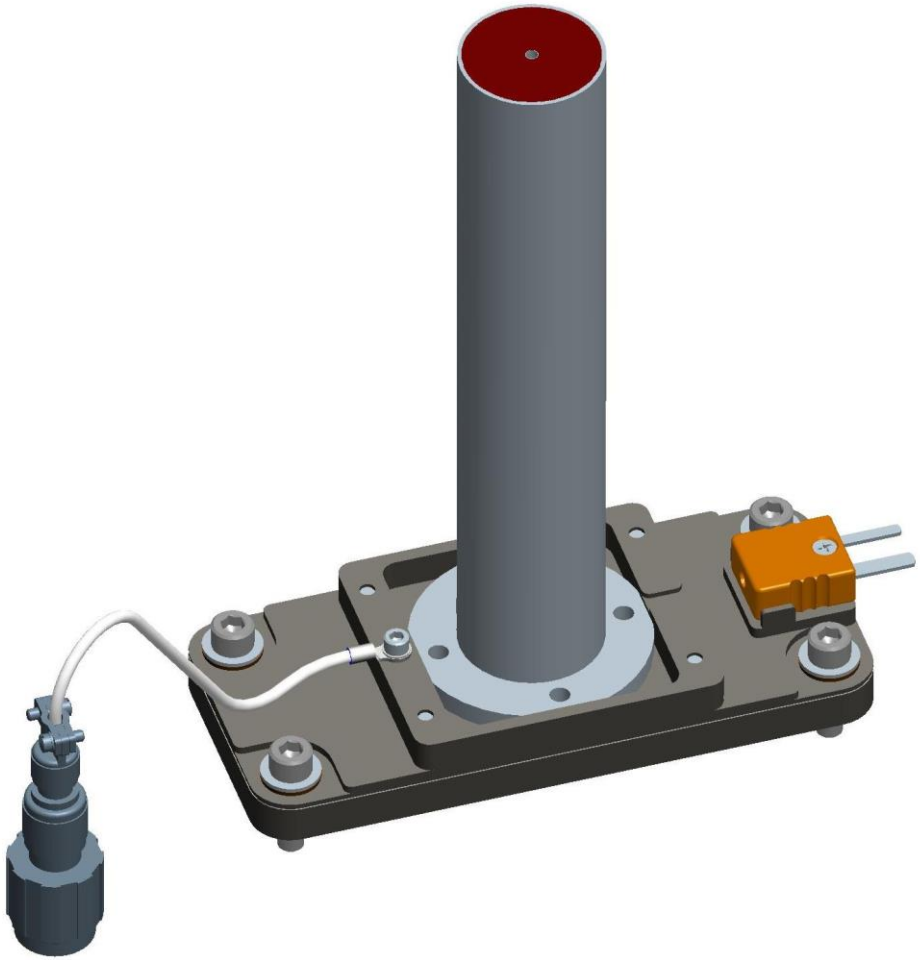
The upper part of the spherical burner will be supplied by Princeton University under the following conditions:

1) **Princeton University**: Fabricate a 1/4”, 3/8”, and 1/2” spherical burner head with supply tube per ACME ISRD 1.2.1, 1.2.2, 1.2.5, and 1.2.6.
2) **Princeton University**: Perform preliminary tests on the above spherical burner heads following ACME ISRD 1.2.3 as a guide and conferring with NASA ACME Project Scientist.
3) **Princeton University**: Deliver accepted spherical burner heads to NASA Glenn Research Center (GRC).
4) **NASA GRC** performs additional acceptance tests as necessary (microgravity).
5) **NASA GRC** deliver acceptable spherical burner heads to ZIN Technologies Inc. as government furnished equipment (GFE.) via C727.
6) **Zin Technologies Inc**.: Accept and serialize spherical burner heads from NASA GRC.
7) **Zin Technologies Inc.**: Perform additional fabrication on S1101MFA1218, S1101MFA1223, and S1101MFA1228 per ACME ISRD 1.2.4, 1.2.6, and 1.2.7. This fabrication will produce the following top level assemblies: S1101MFA1216: 1/4” spherical burner assembly, S1101MFA1220: 3/8” spherical burner assembly, S1101MFA1225: 1/2” spherical burner assembly
8) **Zin Technologies Inc.**: Deliver S1101MFA1216, S1101MFA1220, and S1101MFA1225 to NASA GRC via a DD1149.
9) **NASA GRC:** Test and verify S1101MFA1216, S1101MFA1220, and S1101MFA1225 per ACME ISRD 1.2.3.
10) **NASA GRC:** Deliver verified spherical burner assemblies along with verification certification documents to Zin Technologies Inc. as GFE via C727.

Co-Flow Burner Status

Co-Flow Design Completion: 0%
Co-Flow Drawing Count: 6
Co-Flow Drawing Completion: 0%

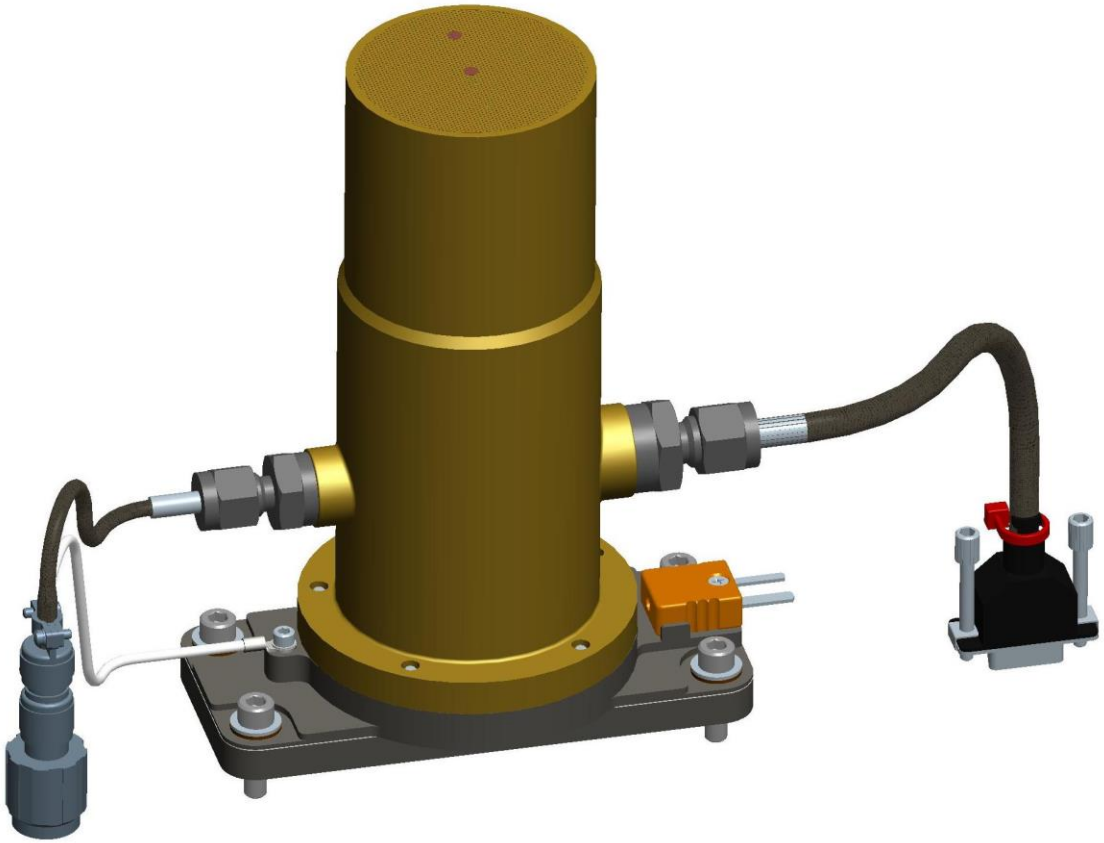
EM Build: 0%
Flight Build: 0%
Requirement Compliance: 100%



BRE Burner Status

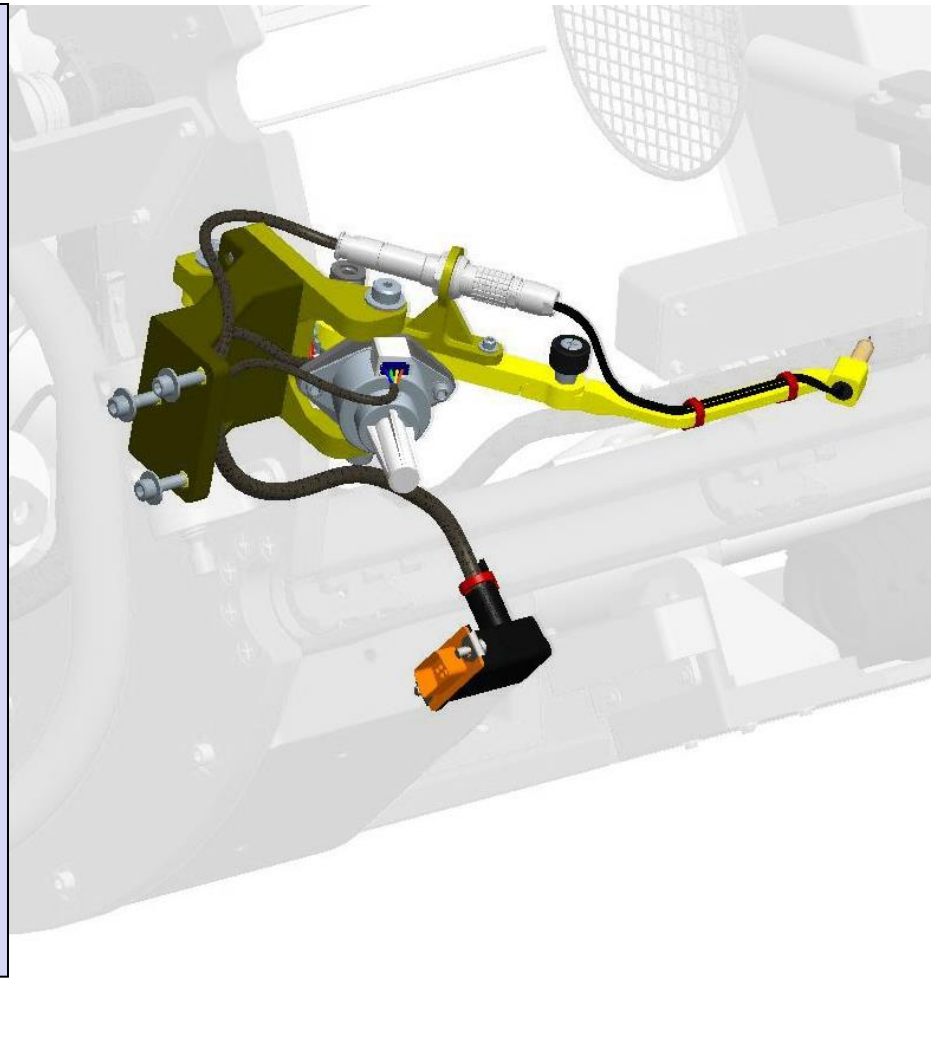
BRE Design Completion: 50%
BRE Drawing Count: 20
BRE Drawing Completion: 0%

EM Build: 0%
Flight Build: 0%
Requirement Compliance: 100%



Igniter System

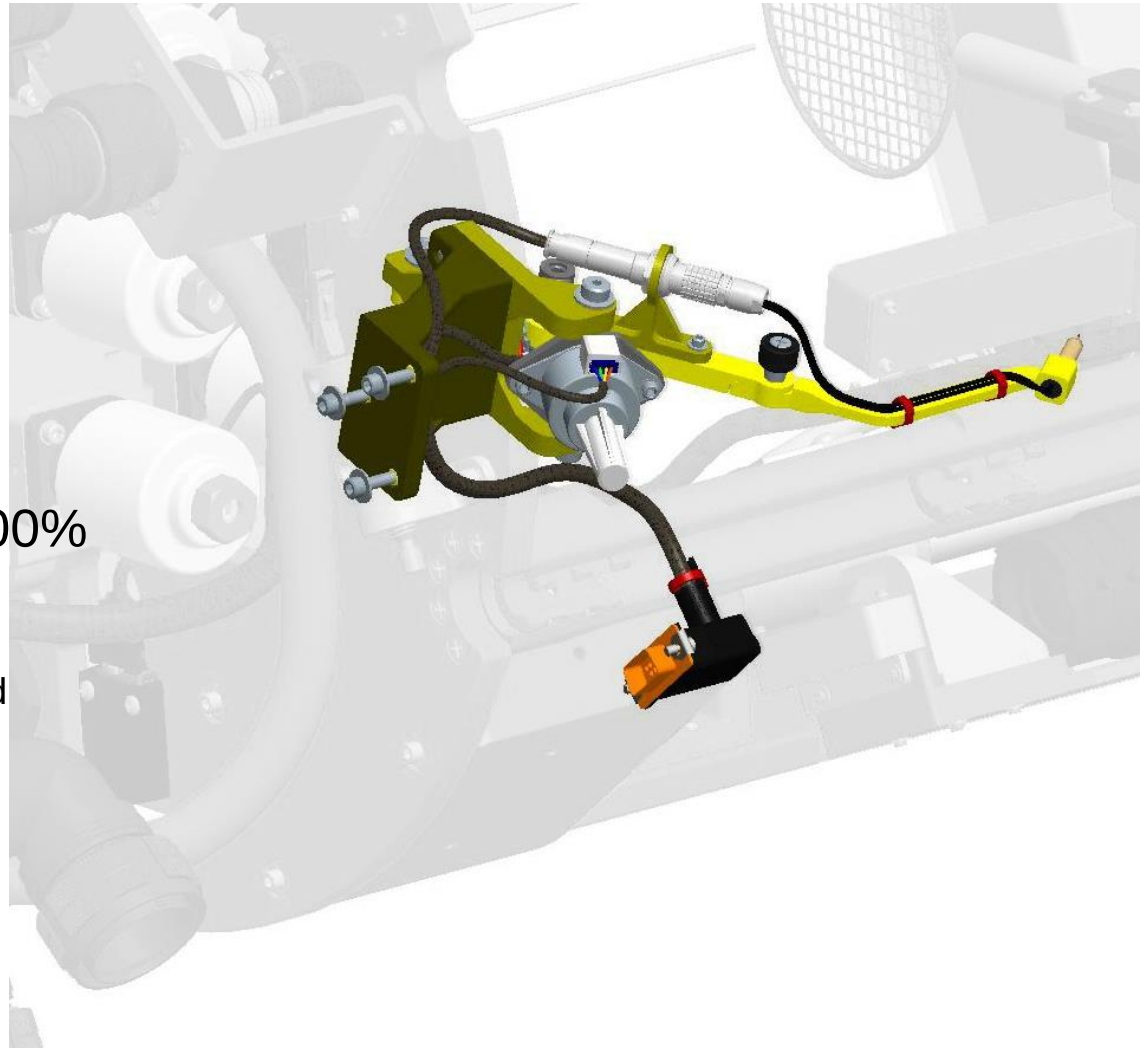
- ACME will use the same hot wire igniter design as MDCA. The concept has been tested and can ignite gaseous flow
- Haydon-Kerk G4 25000 Series Stepper Motor moves the arm
- Pivoting arm design clears the burner tip while minimizing rotation effects
- Igniter tips will be swapped by the crew depending on burner type and size to relocate the igniter tip into the ignition zone



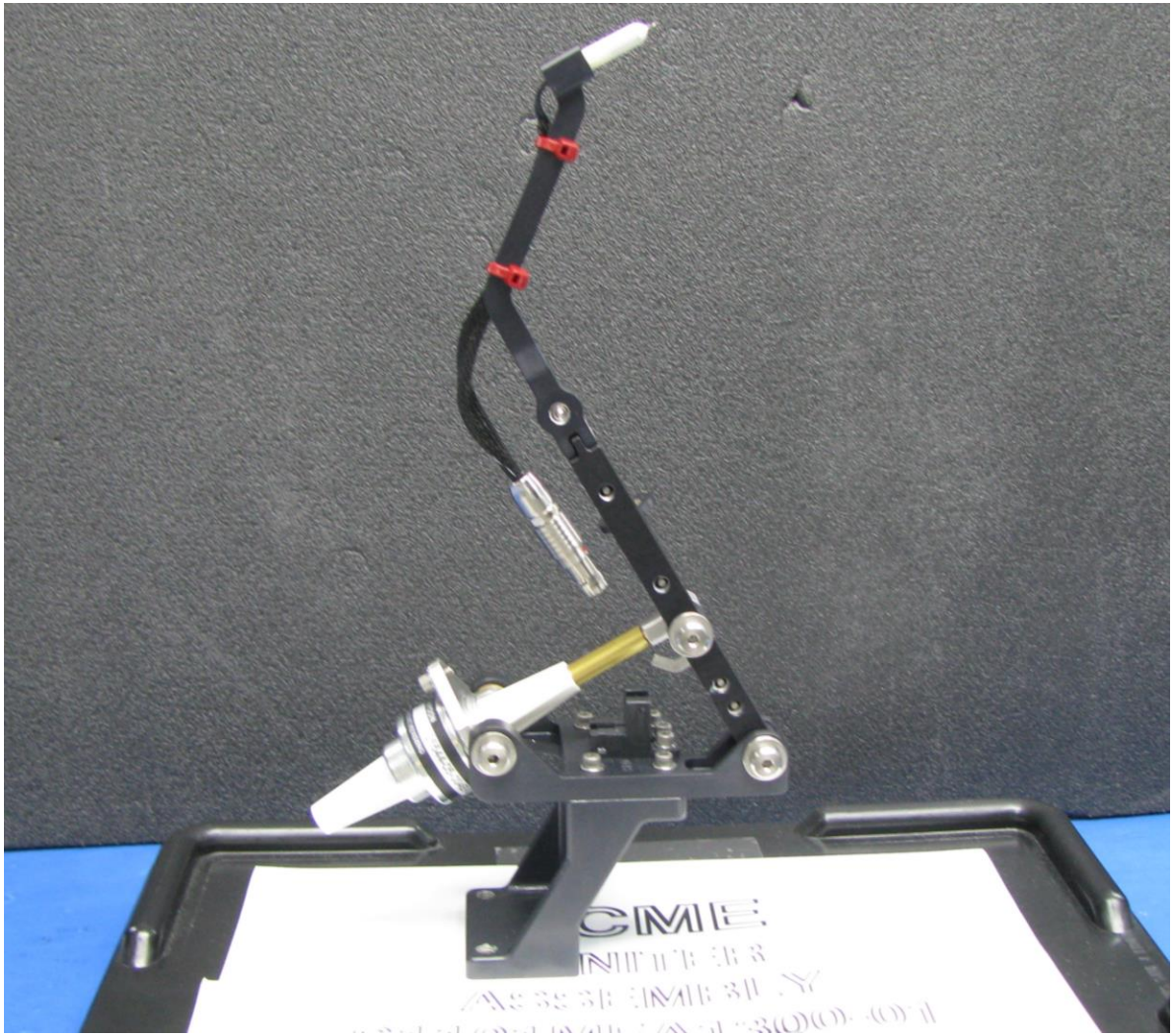
Igniter System

Subsystem Status:
Design Completion: 100%
Drawing Count: 22
Drawing Completion: 100%
EM Build: 100%
Flight Build: 95%
Requirement Compliance: 100%
Test Status:

- *Stepper Motor Test:*
  ACME-TST-010 – EM Passed

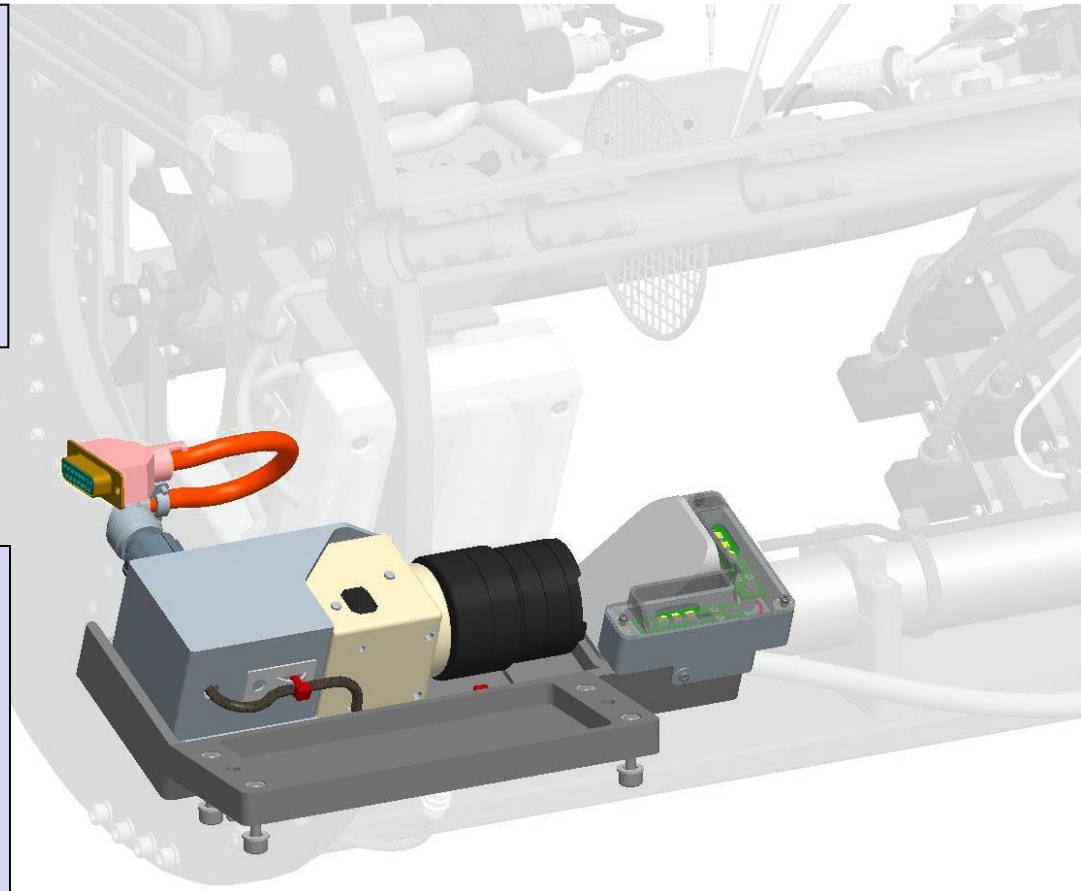


Igniter System Hardware



Analog Camera

The Analog Camera System satisfies requirements for near real-time visual information regarding experiment setup and progress.

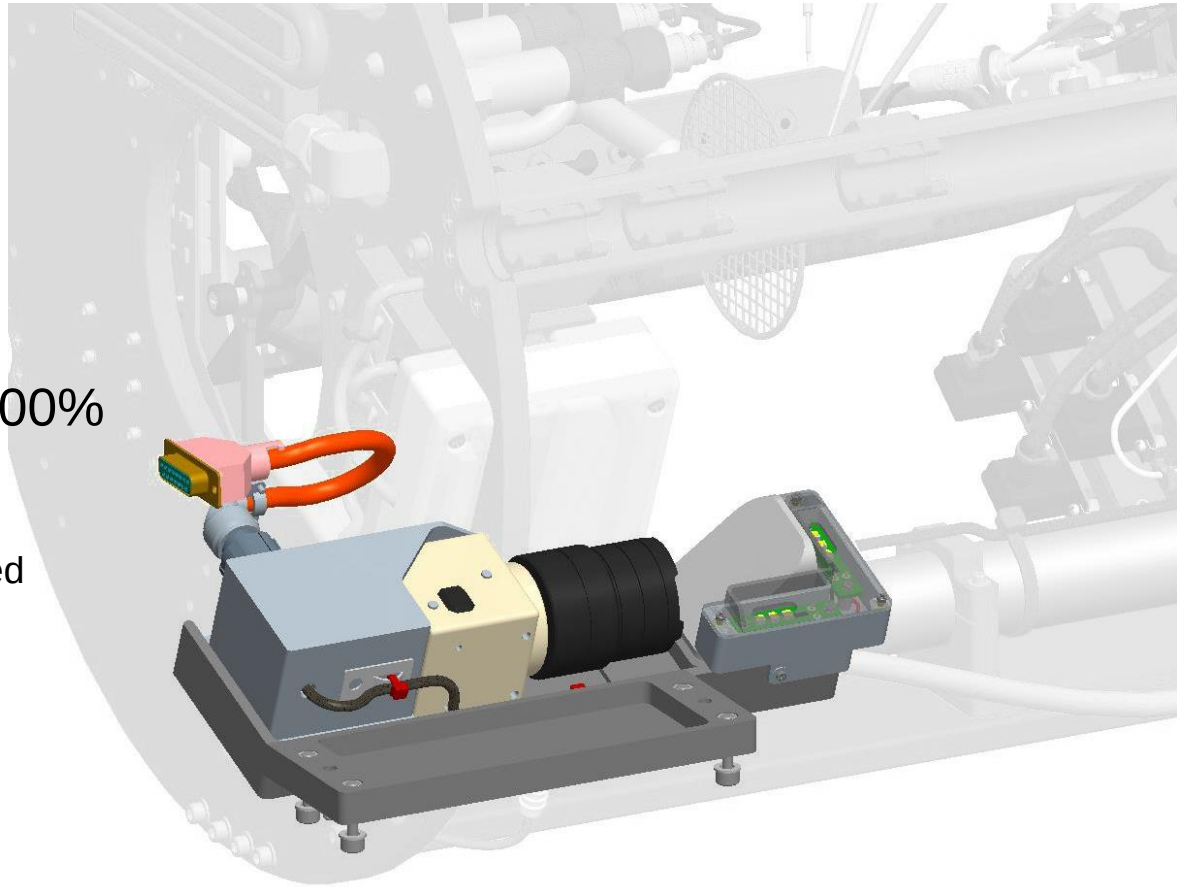


- Hitachi KP-D20B Analog Camera
- 45 degree turning mirror
- Two banks of LEDs for illumination with adjustable intensity to minimize saturation of diagnostic sensors

Analog Camera

Subsystem Status:
Design Completion: 100%
Drawing Count: 18
Drawing Completion: 100%
EM Build: 100%
Flight Build: 39%
Requirement Compliance: 100%
Test status:

- *LED Board Test:*
  ACME-TST-015 – EM Passed

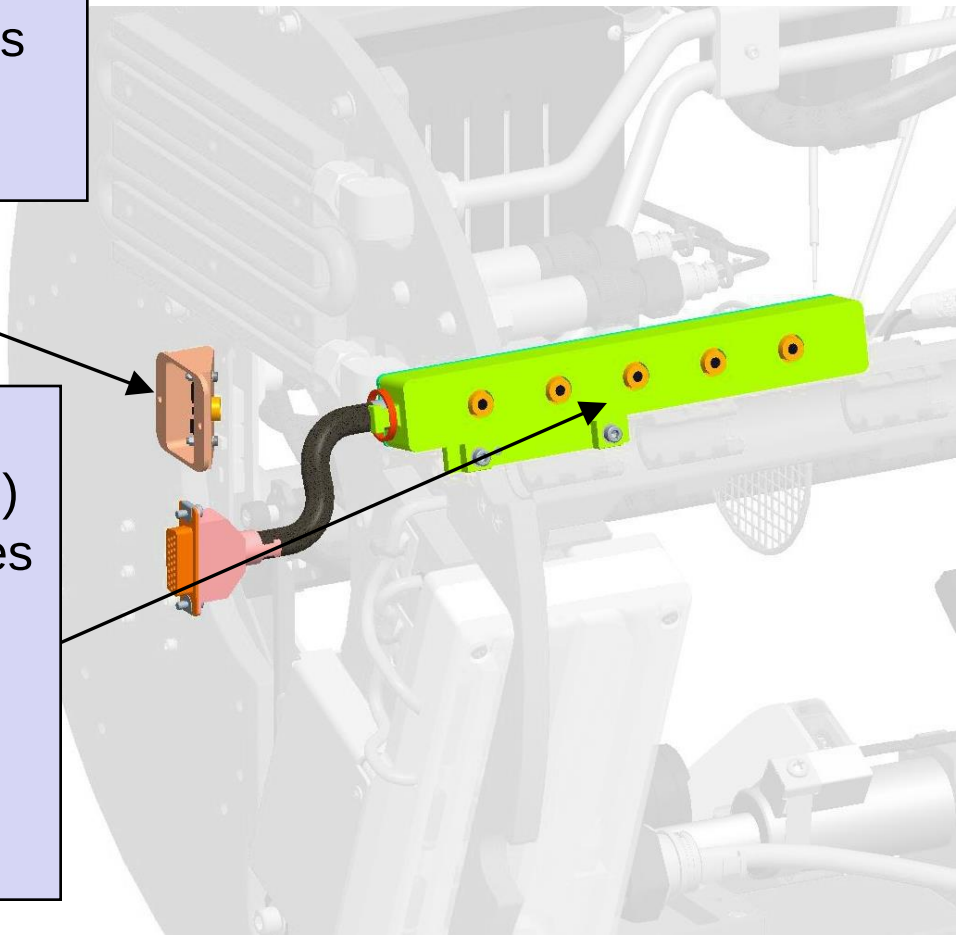


Analog Camera Hardware



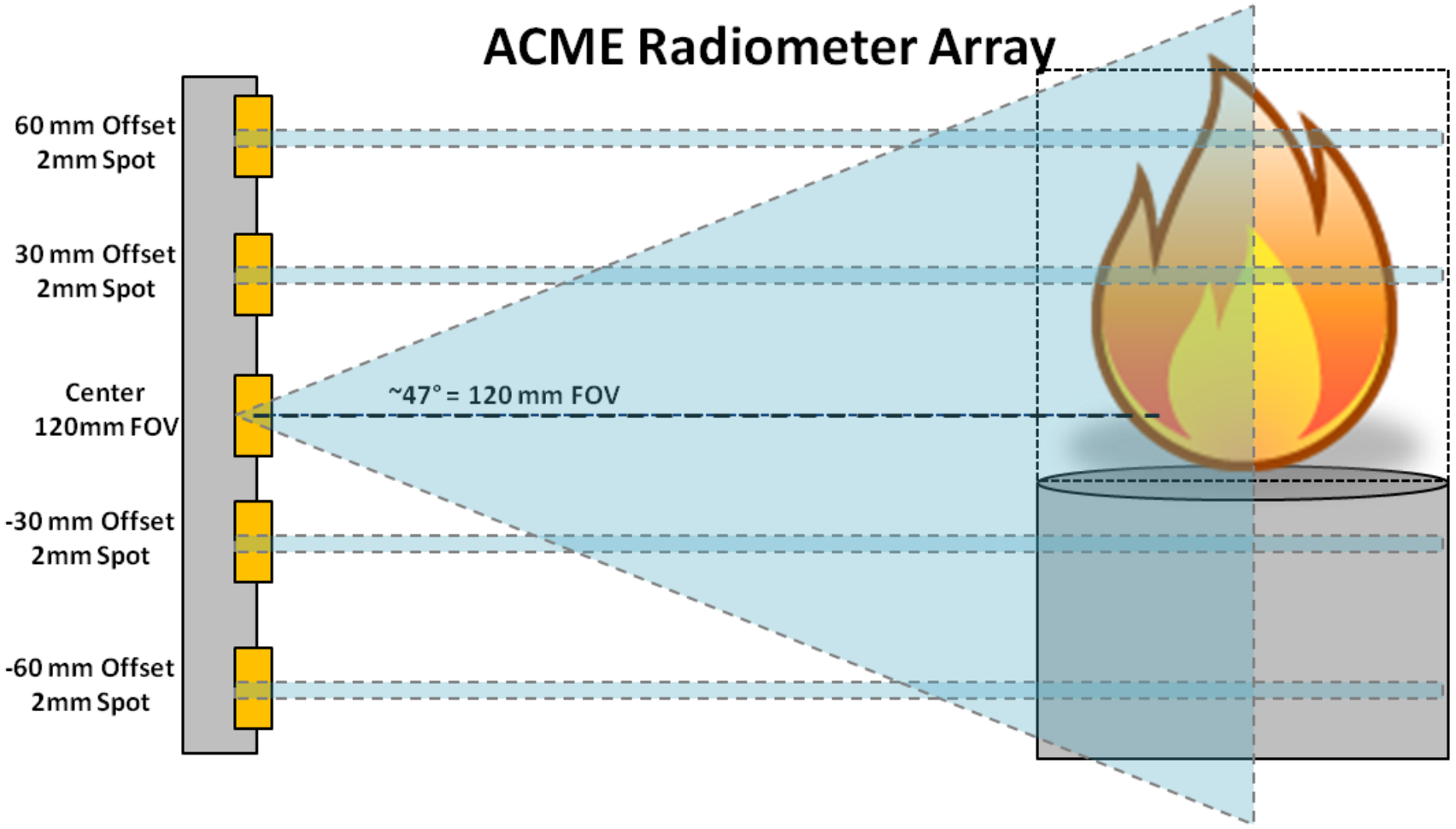
Diagnostics - Radiometers

Offset Single Radiometer
- Offset at 60° to the chamber axis
- Dexter S60M thermopile
- Faster Response time

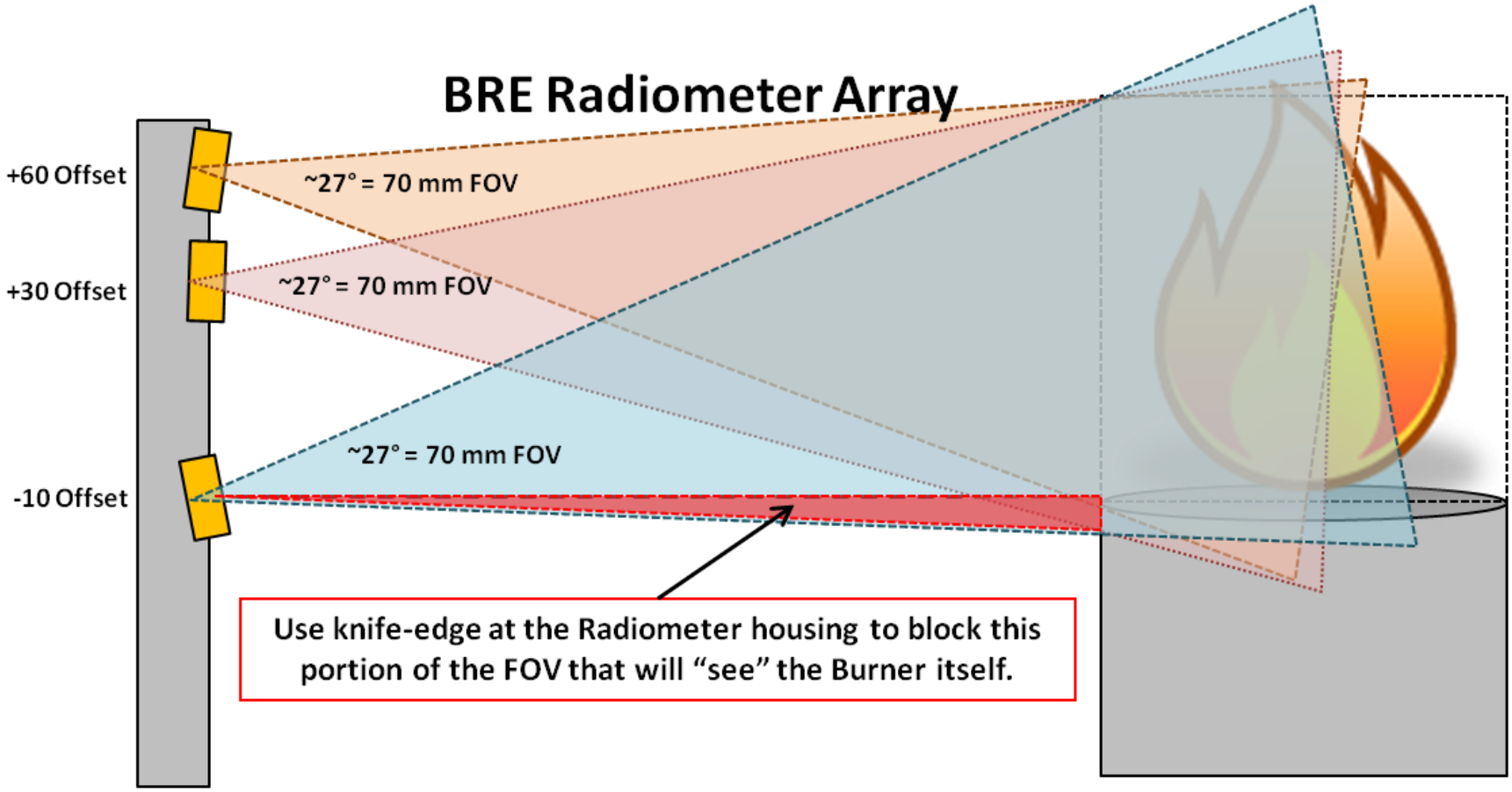


Radiometer Array Assembly
- ORU (to accommodate BRE design)
- Consists of 5: Dexter 2M thermopiles
- High S/N ratio model
- Center: 120° Wide Angle
- Outers: 2mm spot FOV, spaced 30mm/60mm either side of center
- BRE has its own Array design

Diagnostics - Radiometers



Diagnostics - Radiometers



Diagnostics - Radiometers

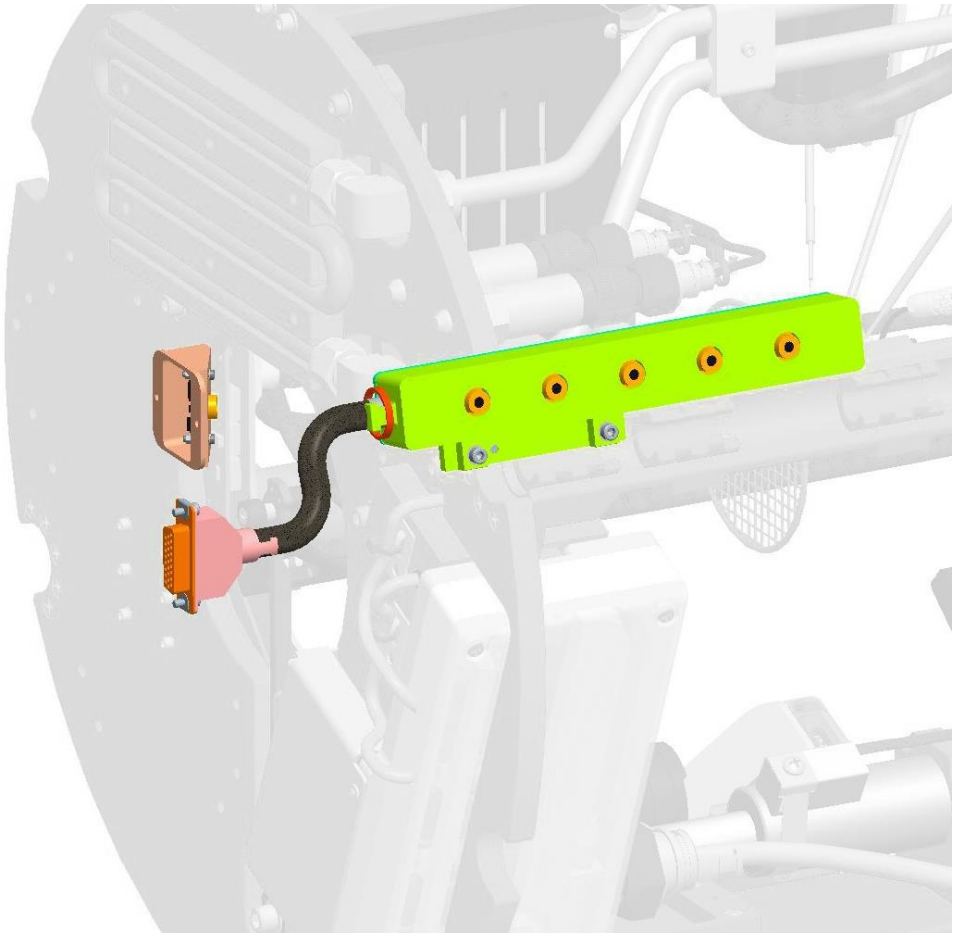
Subsystem Status:

Standard Design Completion: 100%
Drawing Count: 15
Drawing Completion: 100%
EM Build: 73%
Flight Build: 20%
Requirement Compliance: 100%

BRE Subsystem Status

BRE Design Completion: 25%
Drawing Count: 8
Drawing Completion: 0%
EM Build: 0%
Flight Build: 0%
Requirement Compliance: 100%
Test Status:

- *Radiometer Board Test:*
  ACME-TST-022 - Draft

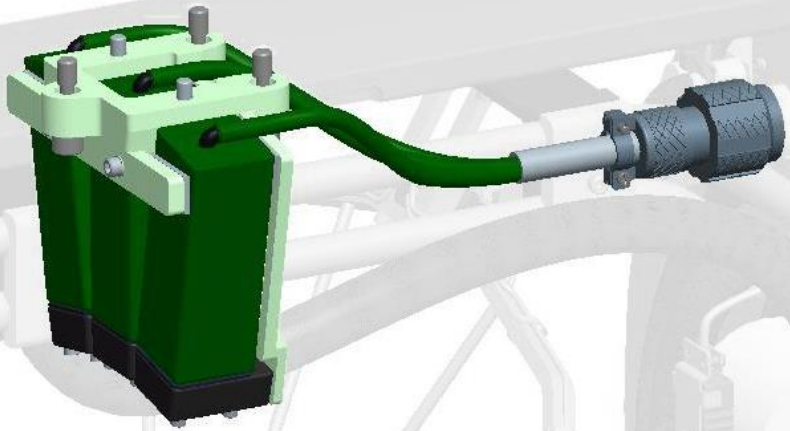


Diagnostics - Photomultipliers

Photomultiplier Array System

- Sensors used for Radiative & Chemiluminescent emissions measurement requirements
- Flame extinction detection

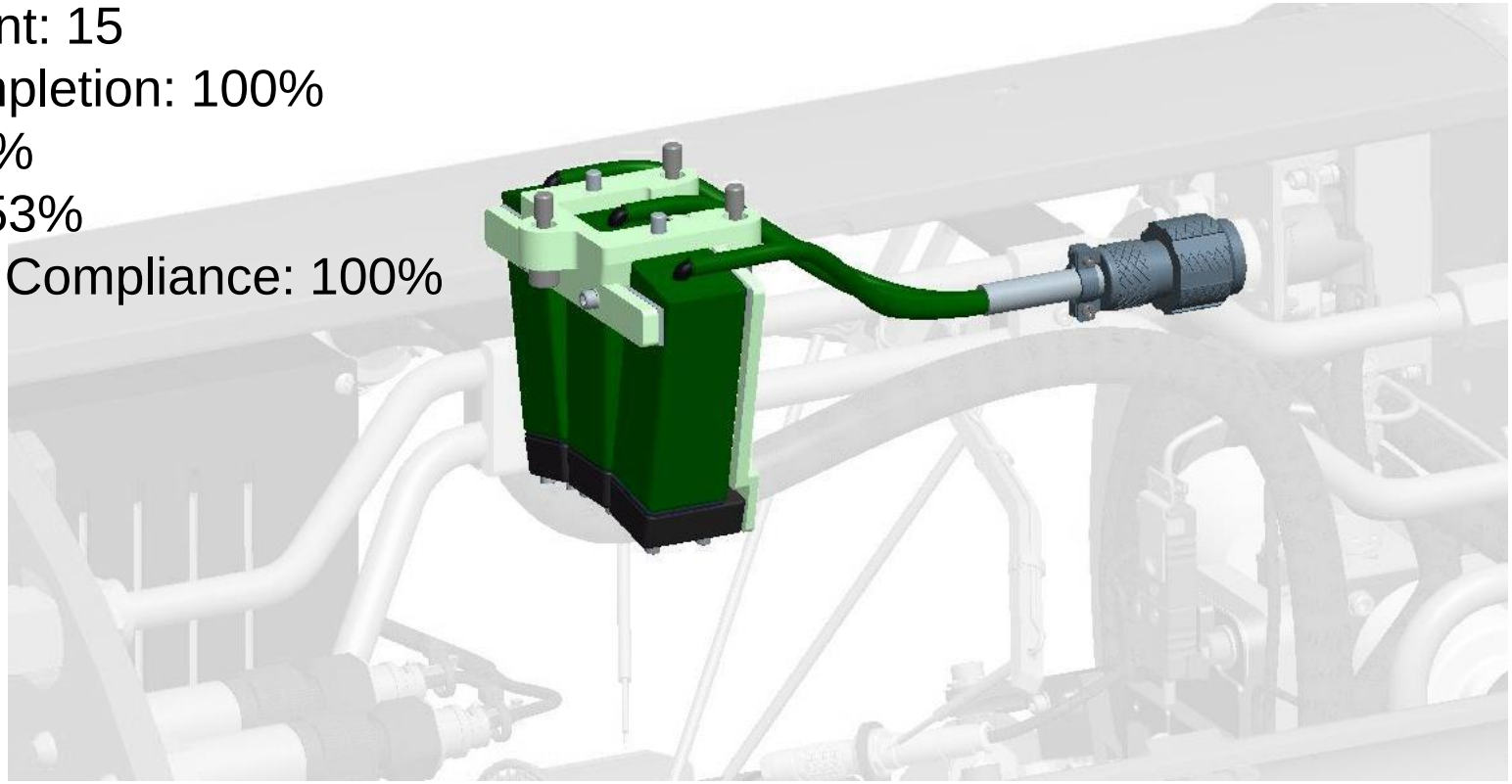




- PMT Array is an ORU
- 3 individual PMT modules
- Hamamatsu H10722-110, Super Bi-Alkali high sensitivity
- Filters: 310 nm (OH*), 431 nm (CH*), Neutral Density

Diagnostics - Photomultipliers

Subsystem Status:
Design Completion: 100%
Drawing Count: 15
Drawing Completion: 100%
EM Build: 73%
Flight Build: 53%
Requirement Compliance: 100%

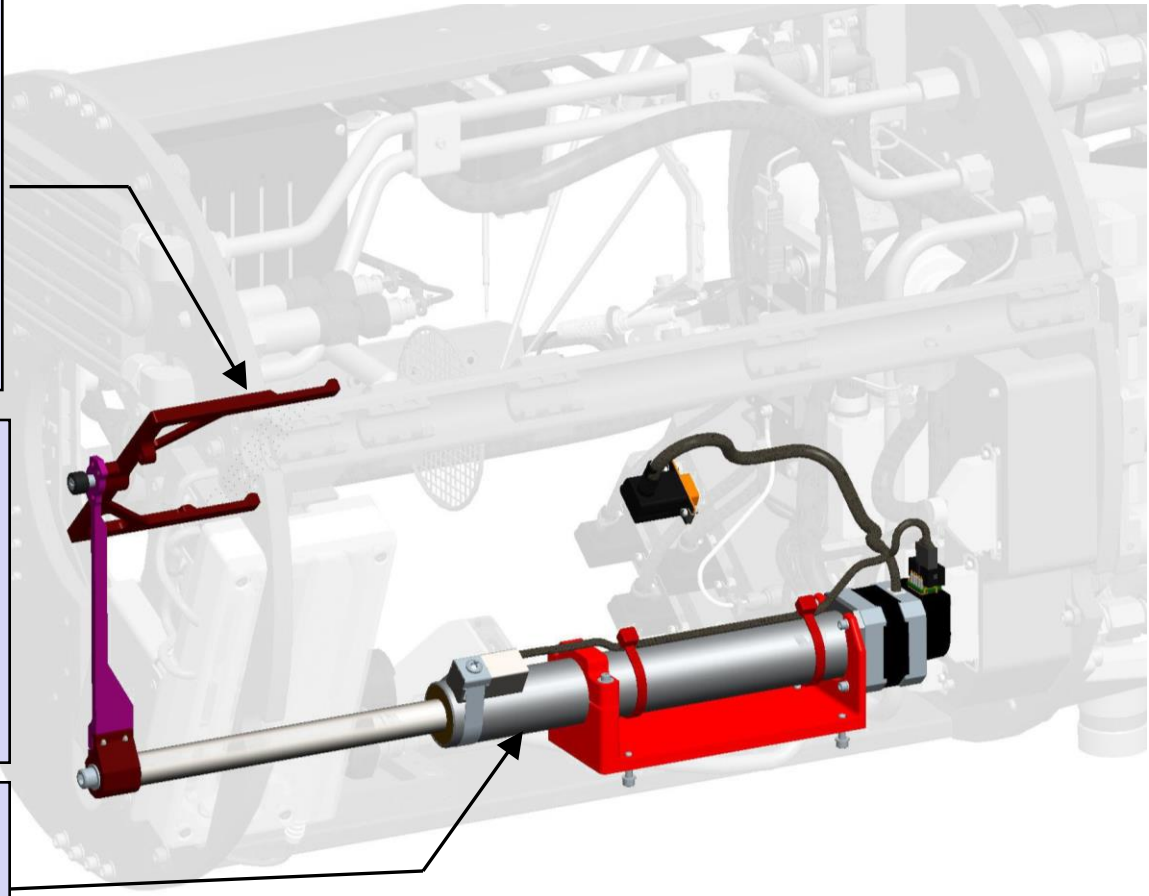


Diagnostics – Thin Fiber Pyrometry

The Fiber Array itself is an ORU in case of damage or wear, and can be rotated to position it to face either the Color Camera or the HiBMS-2

The Array translates out of the field of view when not needed and can be adjusted to any position within the flame

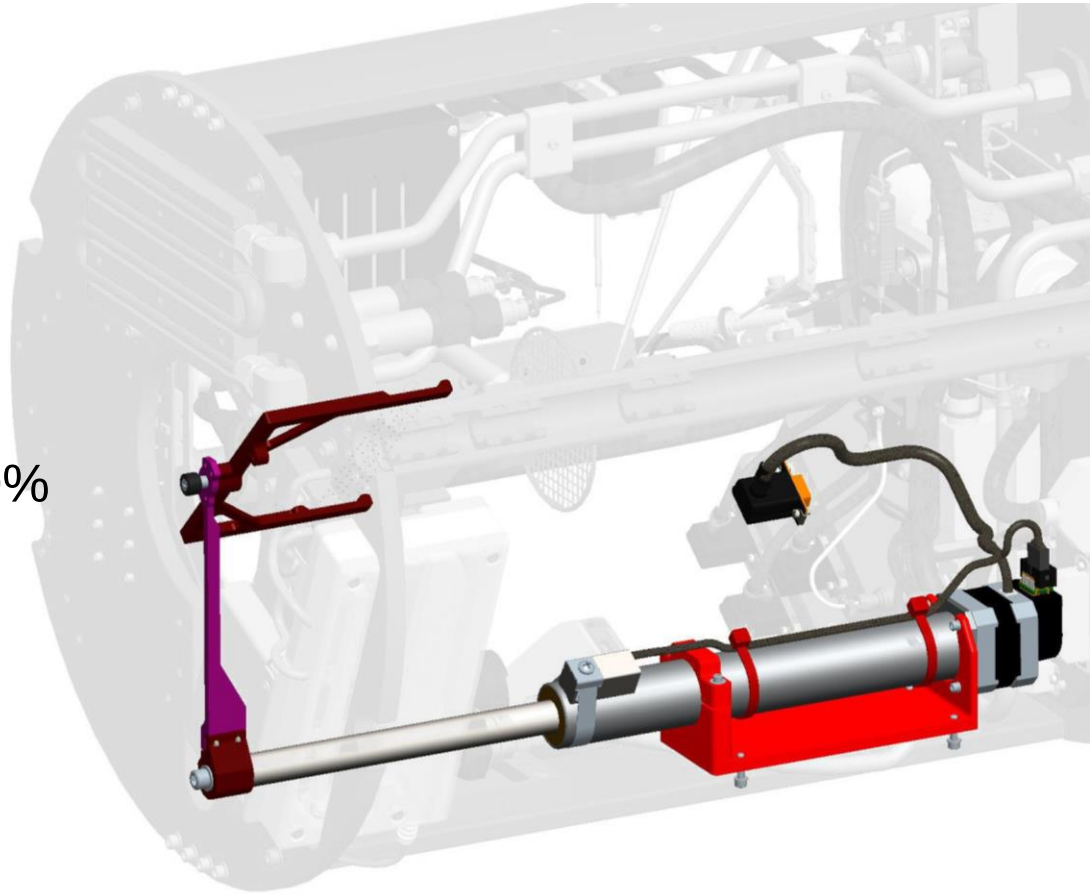
The TFP Motor is an Ultramotion Digit HT17 Series stepper motor



Diagnostics – Thin Fiber Pyrometry

Subsystem Status:
Design Completion: 100%
Drawing Count: 19
Drawing Completion: 89%
EM Build: 79%
Flight Build: 47%
Requirement Compliance: 100%
Test Status:

- *Stepper Motor Test:*
  ACME-TST-010 – EM Passed

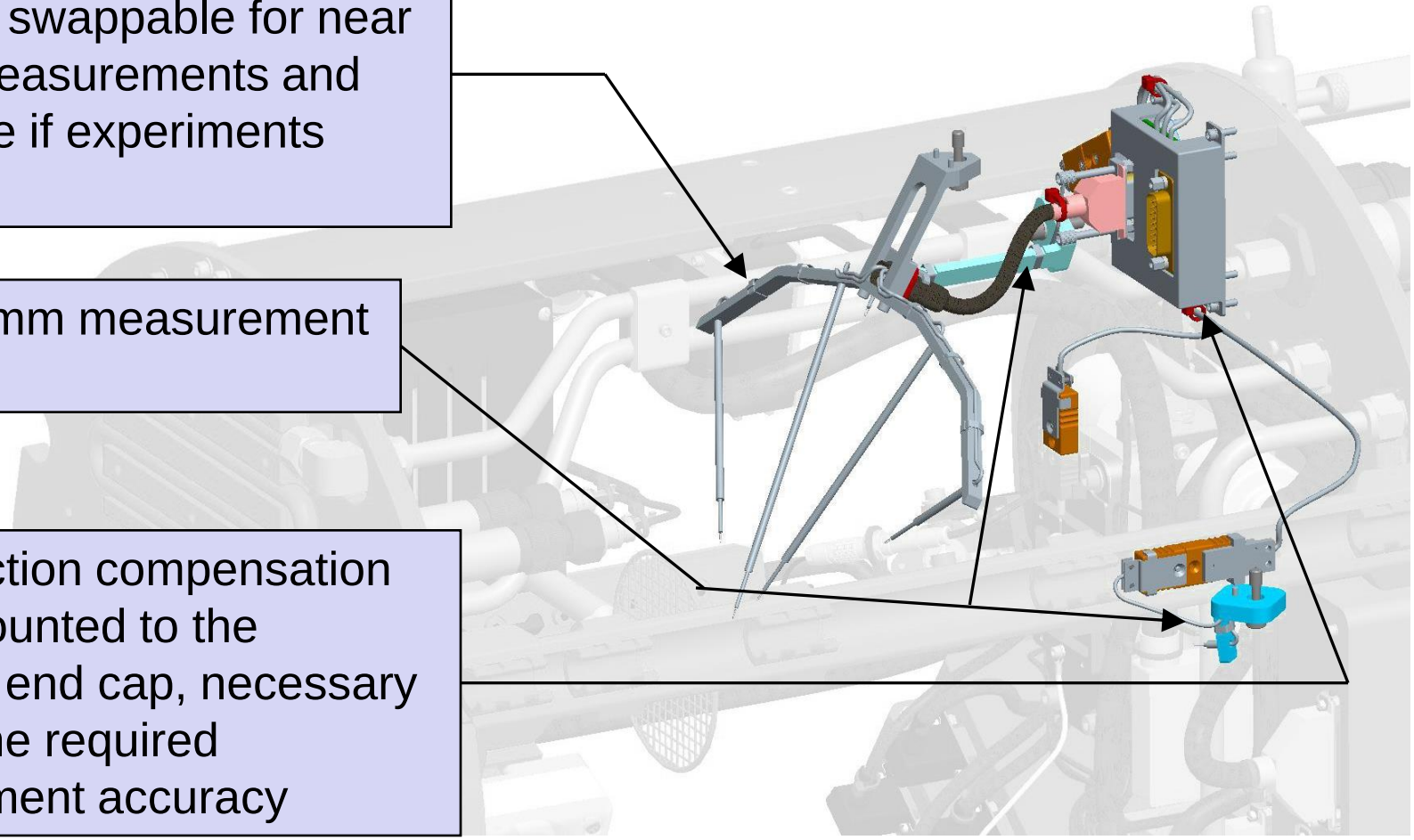


Far Field Thermocouples

A rake of four measurement positions, swappable for near and far measurements and removable if experiments require

Two 200mm measurement positions

Cold Junction compensation board, mounted to the upstream end cap, necessary to meet the required measurement accuracy



Far Field Thermocouples

Subsystem Status:

Design Completion: 100%

Drawing Count: 34

Drawing Completion: 100%

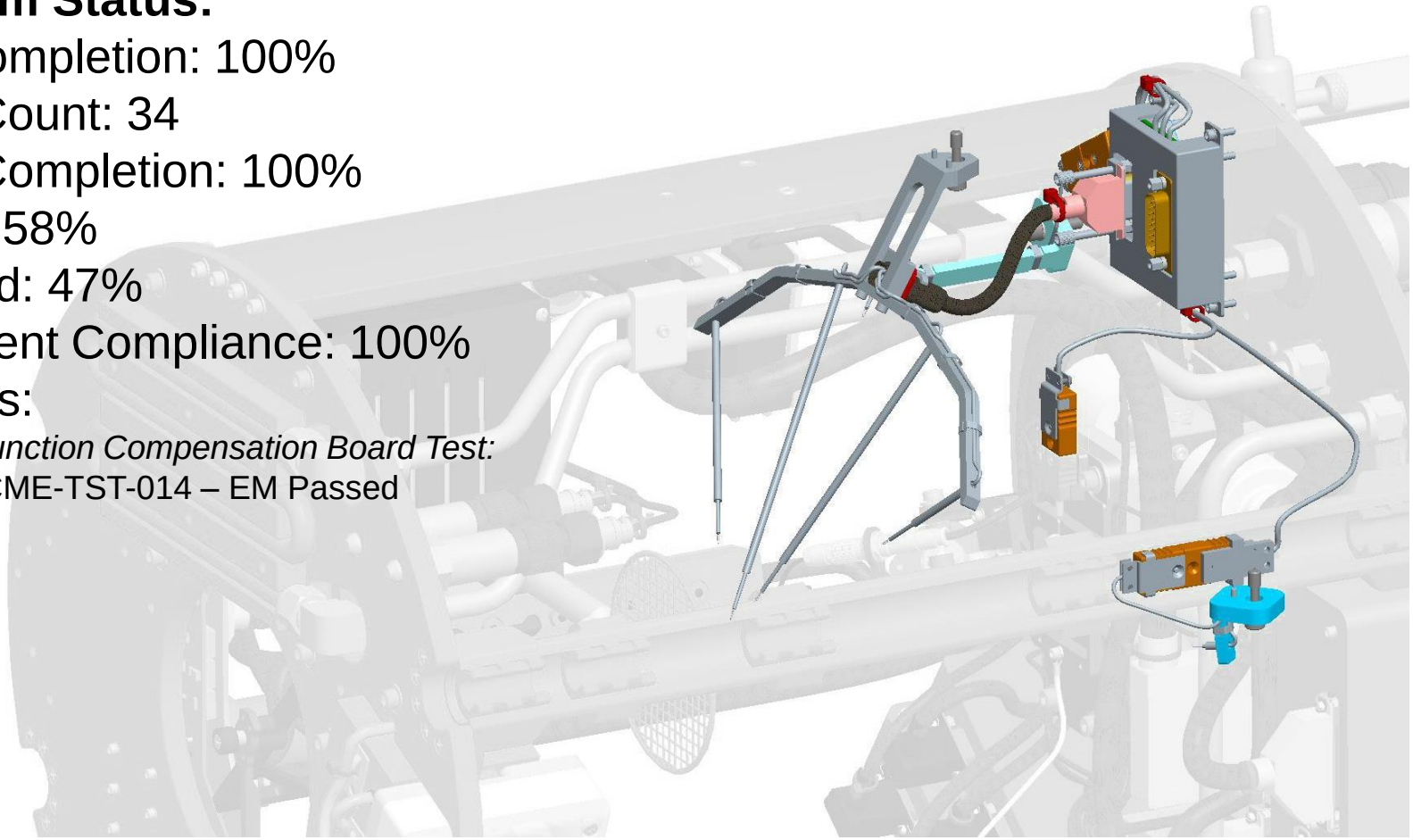
EM Build: 58%

Flight Build: 47%

Requirement Compliance: 100%

Test Status:

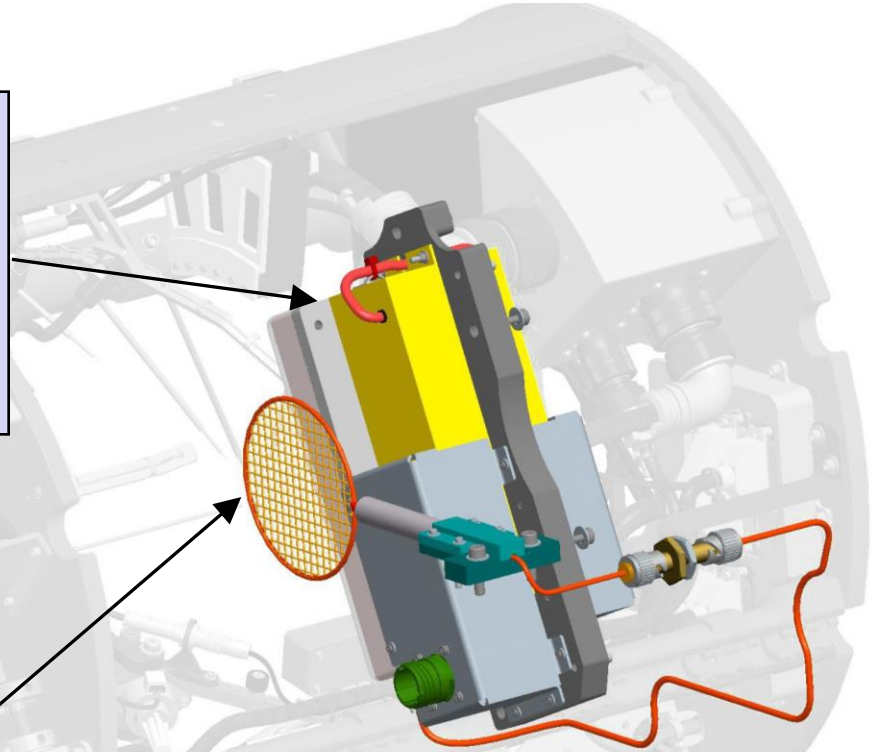
- *TC Cold Junction Compensation Board Test:*
  ACME-TST-014 – EM Passed



E-Field System

ORU High Voltage Module consists of 2 20A series Ultravolt power sources and passive electronics. Unless requested, this module is not removed.

ORU Copper Mesh positionable in two locations to meet requirements and removed when E-Field experiments are not performed



E-Field System

Subsystem Status:

Design Completion: 100%

Drawing Count: 35

Drawing Completion: 83%

EM Build: 83%

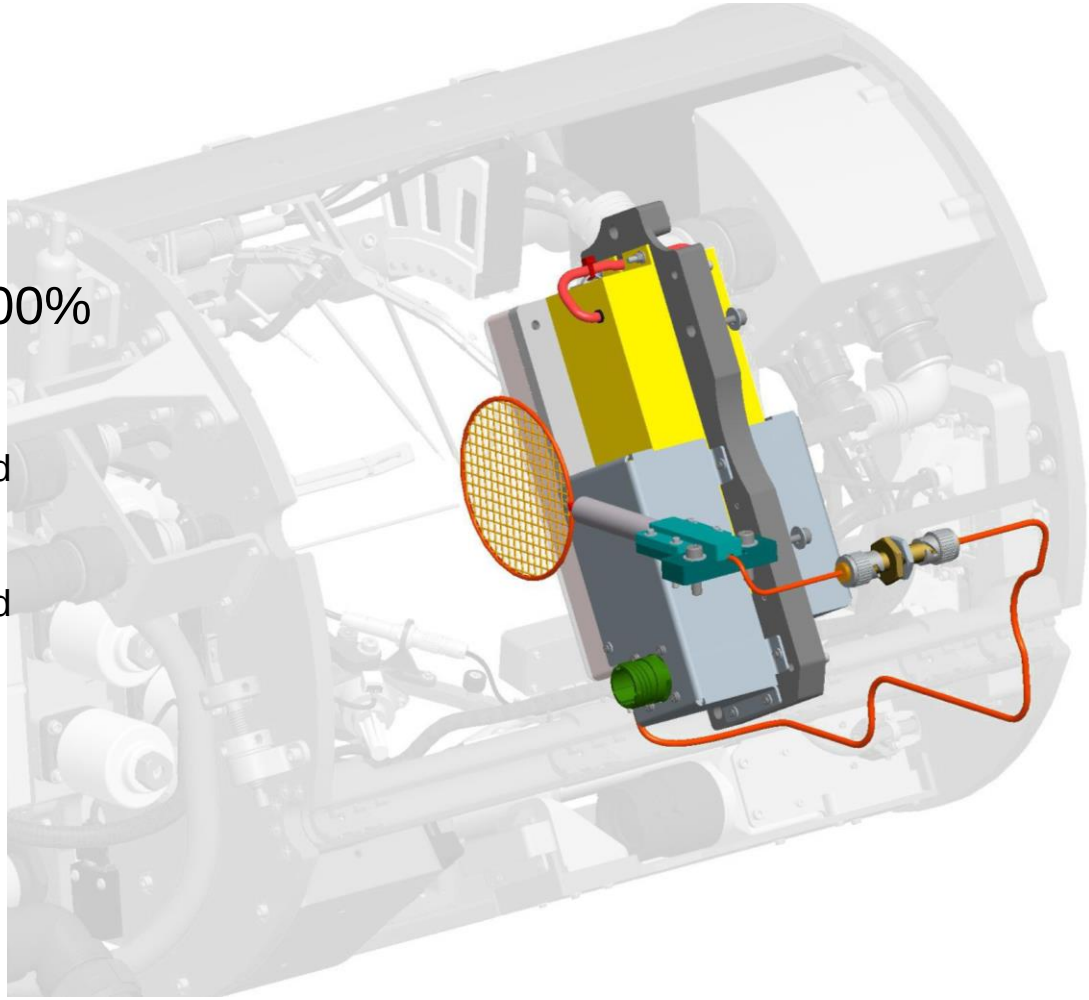
Flight Build: 48%

Requirement Compliance: 100%

Test Status:

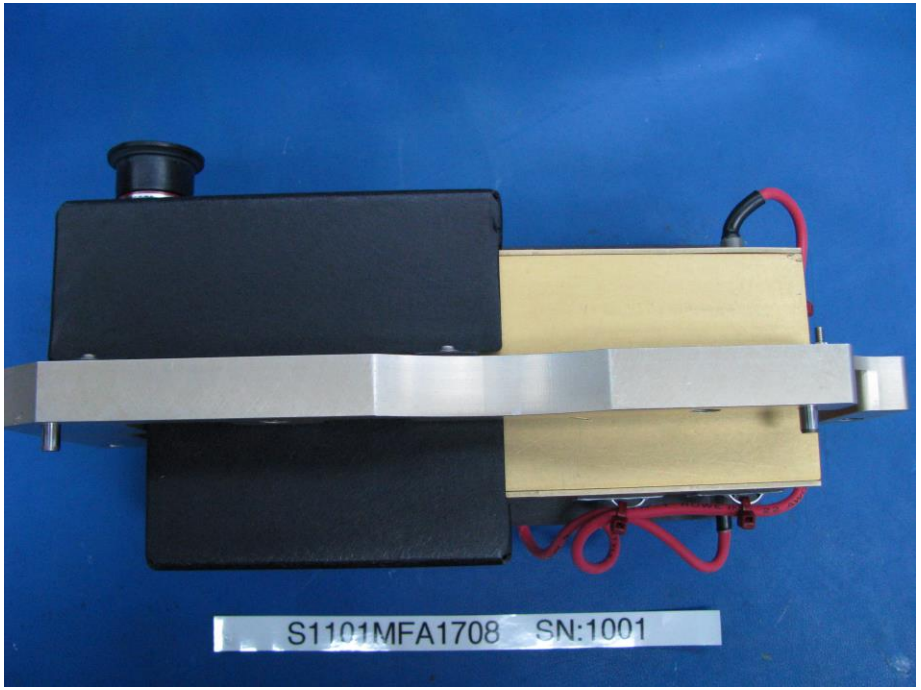
- *E-Field Mesh Test:*
  ACME-TST-004 – EM Passed

- *E-Field Power Supply Test:*
  ACME-TST-005 – EM Passed



E-Field System Hardware



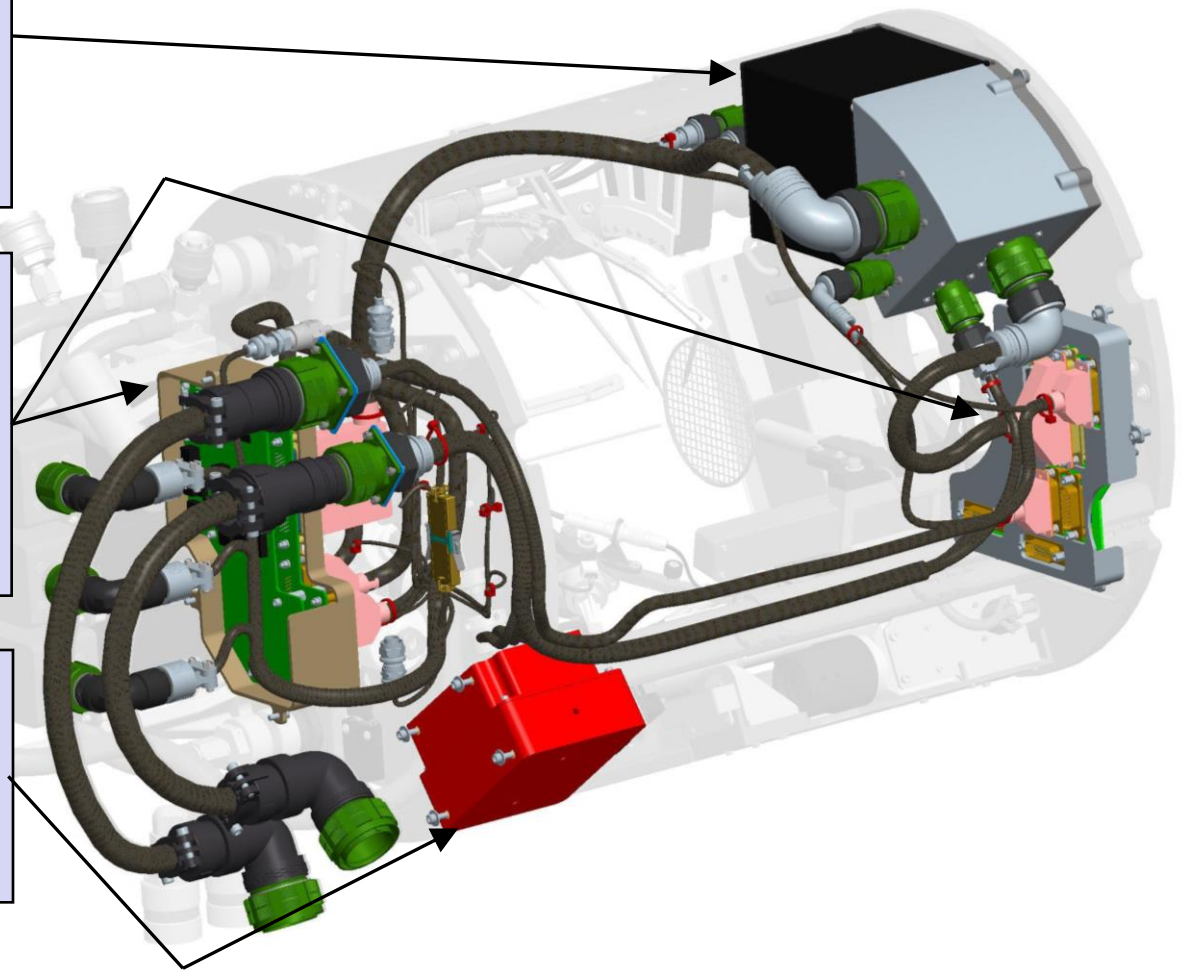


Control and Data Handling

UEI PowerDNA Cube Data Acquisition Unit handles all inputs and outputs inside the CIA except for motor control

Relay Boards handle component switching, power distribution and data routing to the Data Acquisition Unit

Driver boards controlled by the Motion Control Board inside the AVP control Igniter and TFP Motors



Cube Design Details



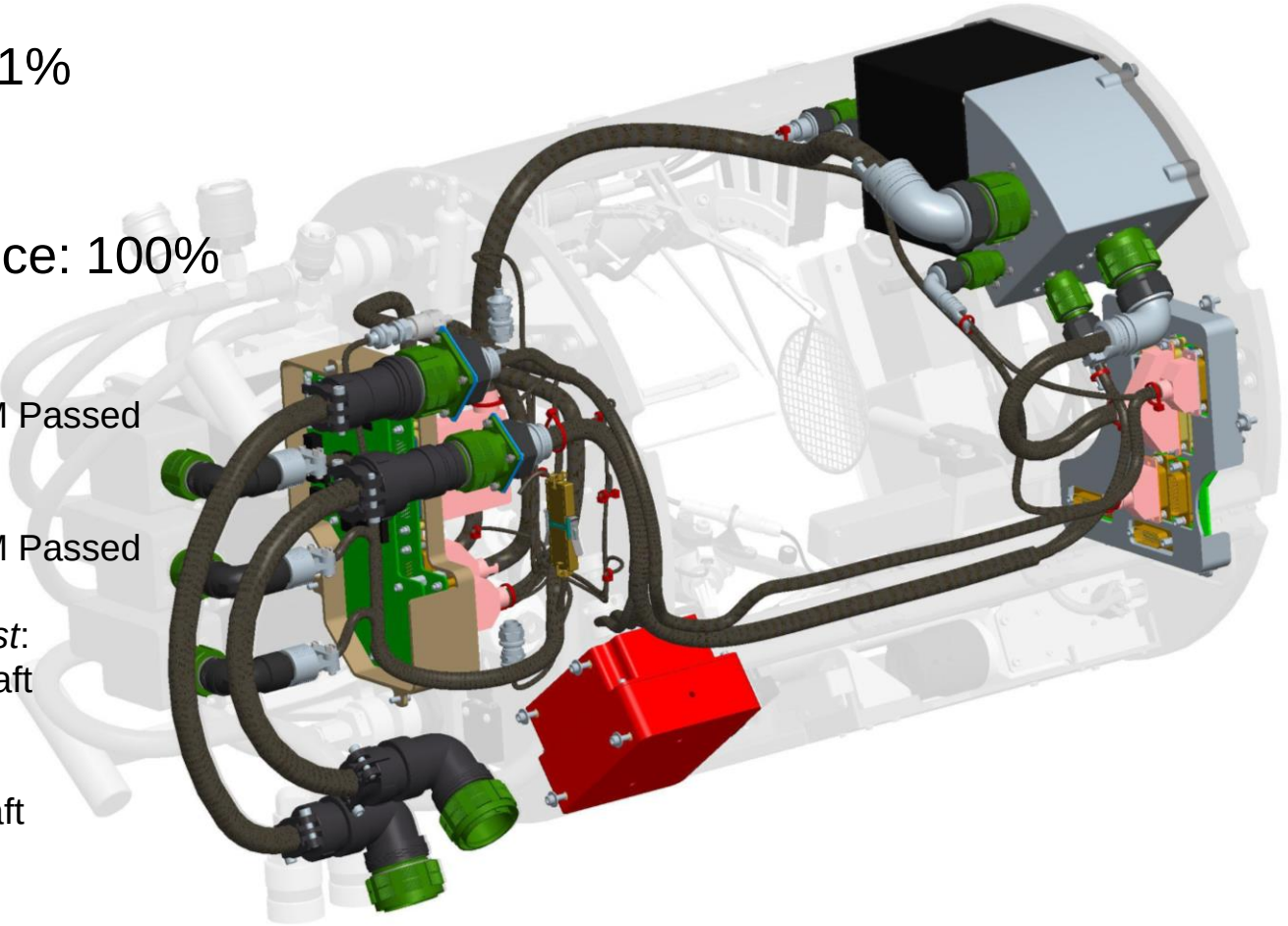
Design makes it into an ORU. Contained in a single housing.
Extending connections out to circular connectors .

Control and Data Handling

Subsystem Status:

Design Completion: 100%
Drawing Count: 54
Drawing Completion: 91%
EM Build: 52%
Flight Build: 18%
Requirement Compliance: 100%
Test Status:

- *Cube Test:*
  ACME-TST-018 – EM Passed
- *Battery Vacuum Test:*
  ACME-TST-013 – EM Passed
- *Downstream Relay Board Test*:
  ACME-TST-026 – Draft
- *Upstream Relay Board Test:*
  ACME-TST-028 - Draft



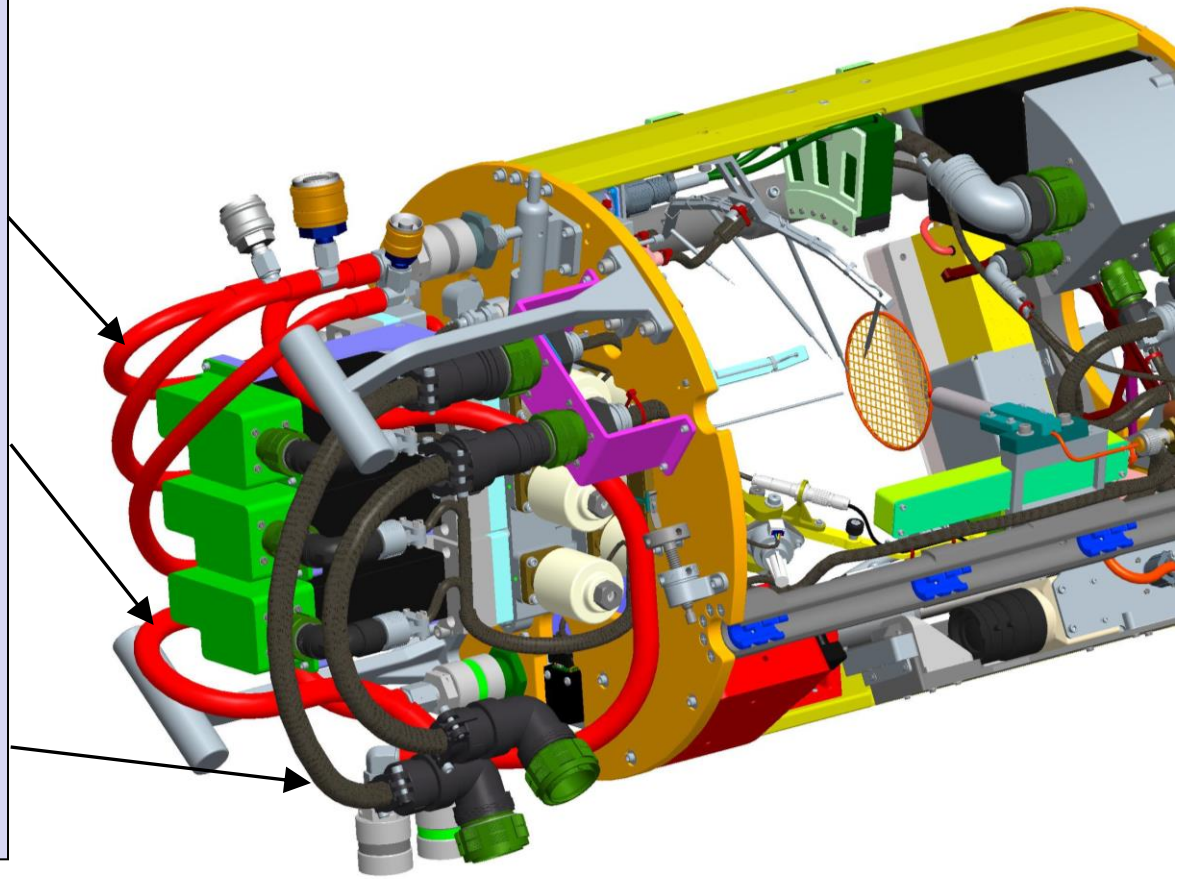
Cube Hardware



Chamber Connections

Consists of the following removable hoses and cables:

- Three gas hoses – Parker Stratoflex 101
- Two water cooling hoses – Parker Stratoflex 101
- Two Electrical Connections to provide power, communication and control for the insert

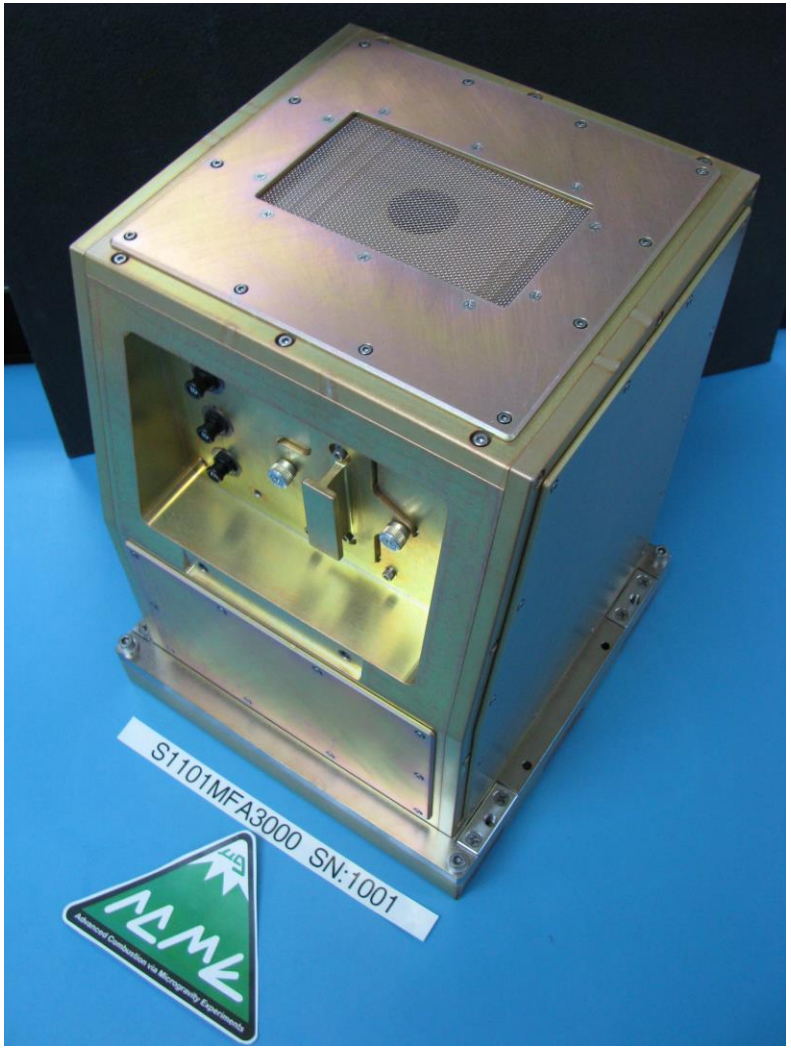


Avionics Package Design

Tim Gobeli – Electrical Engineering Lead

Avionics Package Hardware

Avionics Package contains the hardware that operates the experiment and provides control for the ACME Color Camera



Avionics Package Hardware

ORU Hard Drive contains the Operating System of the AVP and stores image data for the ACME Data Camera

Hard Drive is a 512Gb Solid State drive

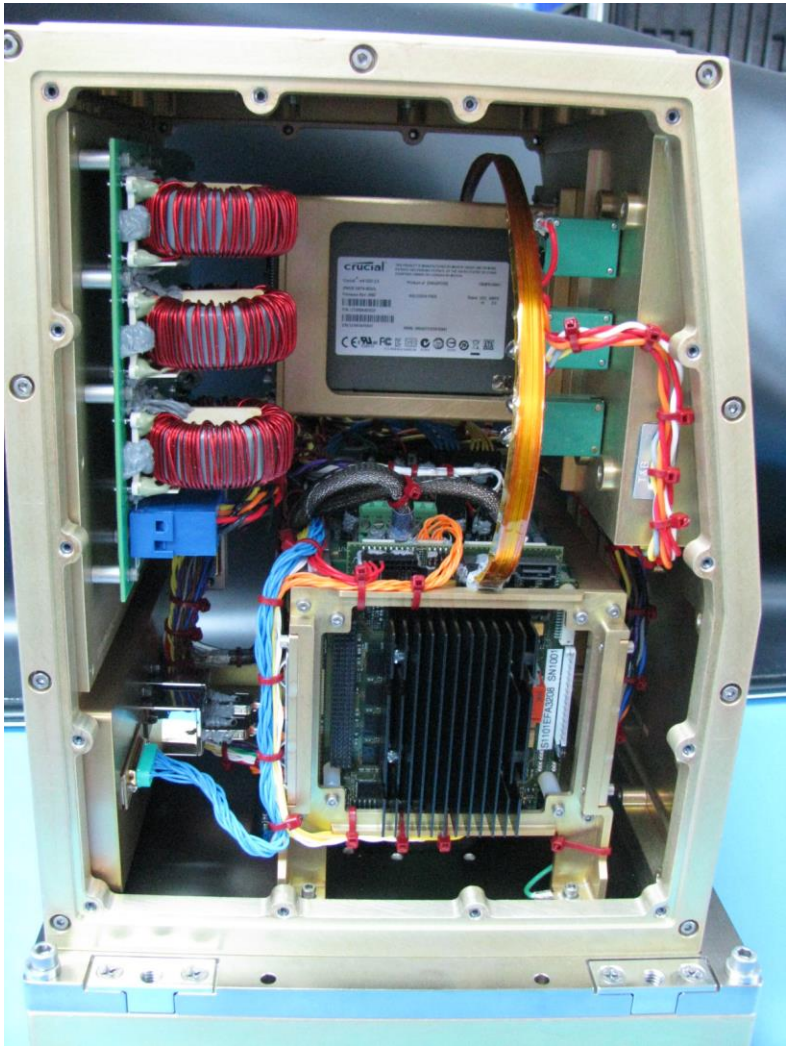




Avionics Package Hardware

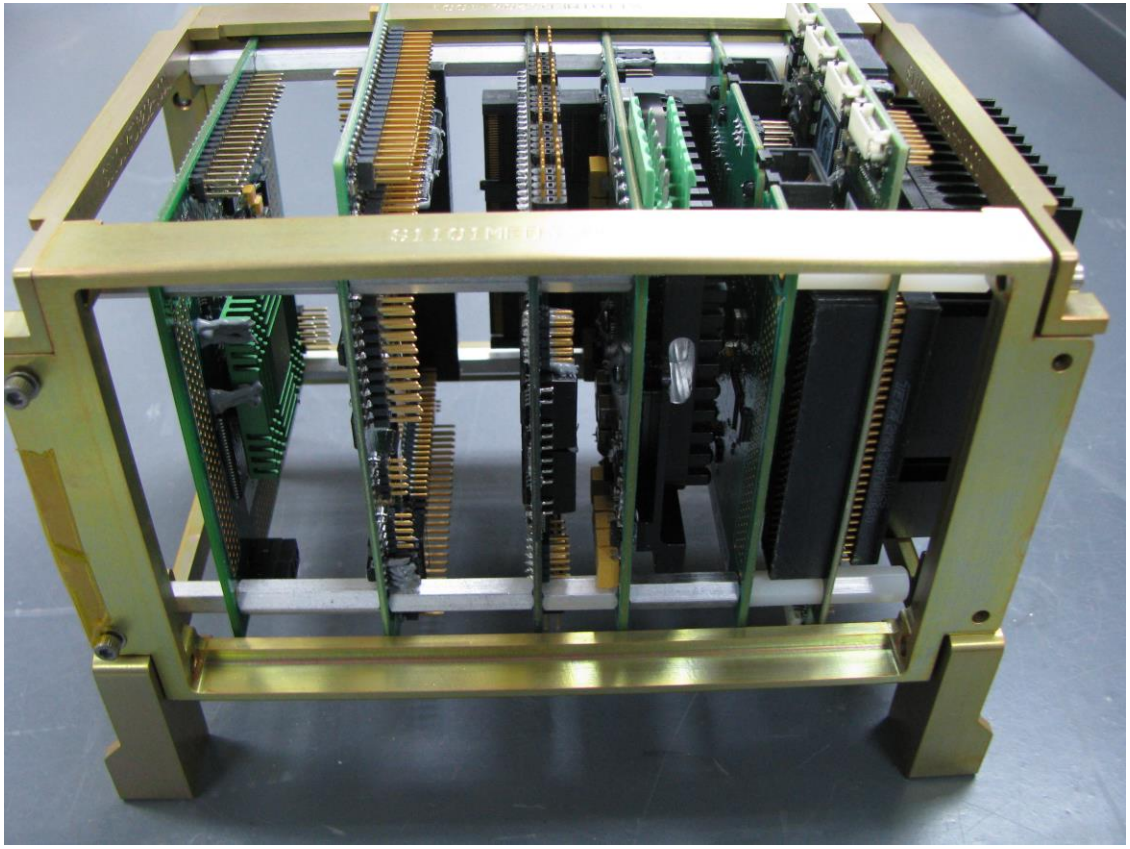
Internal Hardware:

- Power Distribution / Filtering
- Main Computer System



Avionics Package Hardware

Main Computer System Hardware



AVP Command and Control

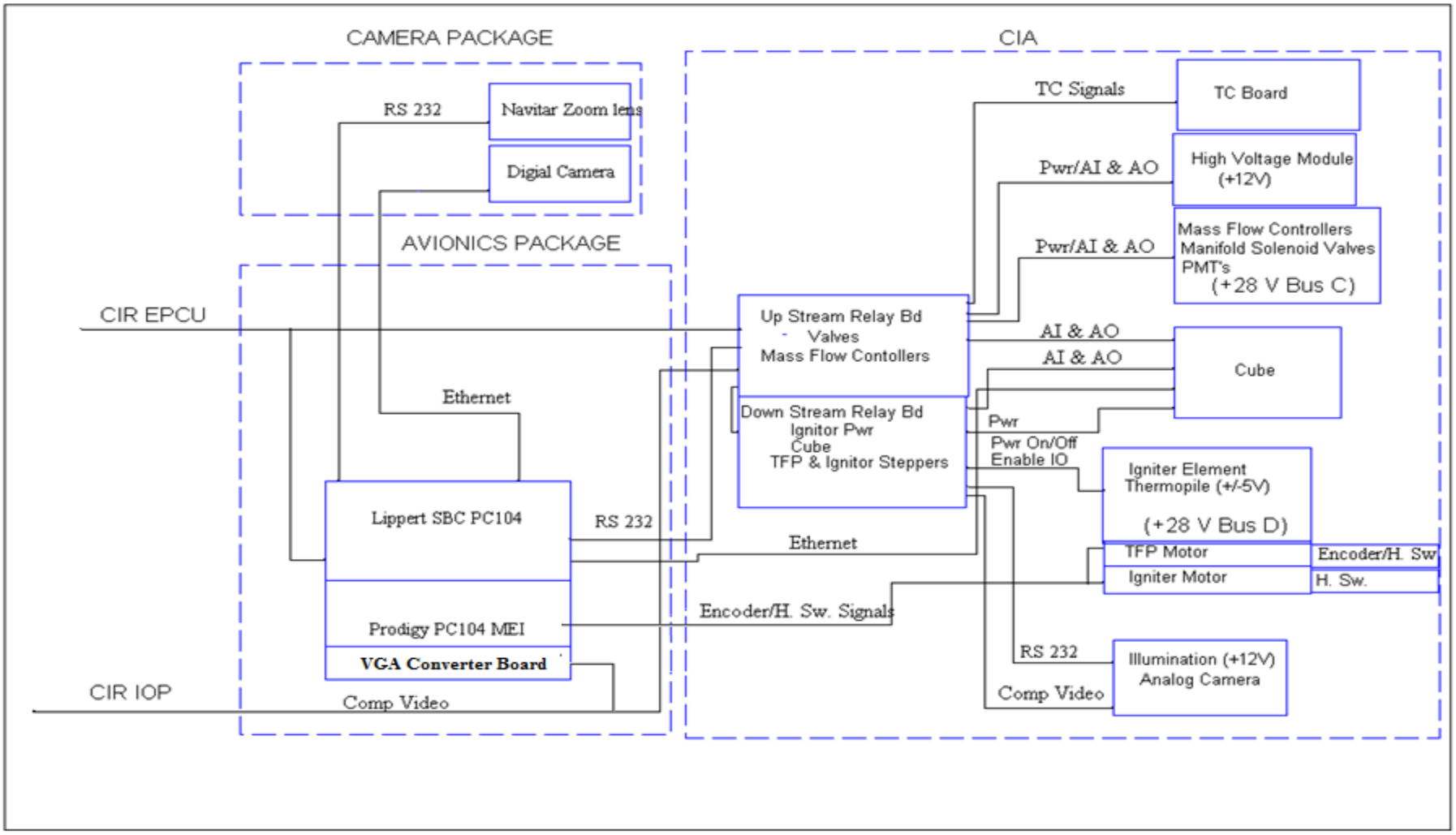
ACME Single Board Computer will coordinate CIR diagnostic hardware and control ACME hardware:

- •Intel Core 2 Duo Processor 1066 MHz
- •1GB DDR3 RAM onboard
- • Ethernet Ports
  - •1 port receives commands and transmits telemetry from and to the CIR Input Output Processor (IOP)
  - •1 port sends control signals and receives data acquisition packets from data acquisition “Cube” located in ACME Chamber Insert Assembly.
  - • 1 port receives video from digital camera package.
- •RS 232 Ports
  - •Transmits commands to ACME Chamber insert Color camera.
  - •Controls Motion Control Boards/Stepper Motors.
- •RS 422 Port
  - • Communicates to Mass Flow Controllers – monitoring operation/functionality.

AVP Control and Data Management

- Two 4 Axis Motor Control Boards coordinate actuation of ACME motors to perform input/output functions:
  - Insert and retract position of igniter and Thin Film Pyrometry (TFM)
  - Digital camera zoom, iris, focus control and filter insertion
- Cube Analog input (differential inputs) and outputs (single ended) collect experiment telemetry and provide hardware control.
- Analog Outputs
  - D/A Cube +5 – 0 outputs to relay boards provide the discrete I/O channels to chamber – on/off, enable/disable to MFCs, valves, igniters, both stepper drives, illumination, High Voltage and a variable output to control MFC, High Voltage, Photomultiplier gains.
- Analog Inputs
  - Collects Radiometer, Photomultiplier, Gas flow rates, Multiple Temperature points data, Ion Current and High Voltage Grid measurements.

Data and Control Diagram

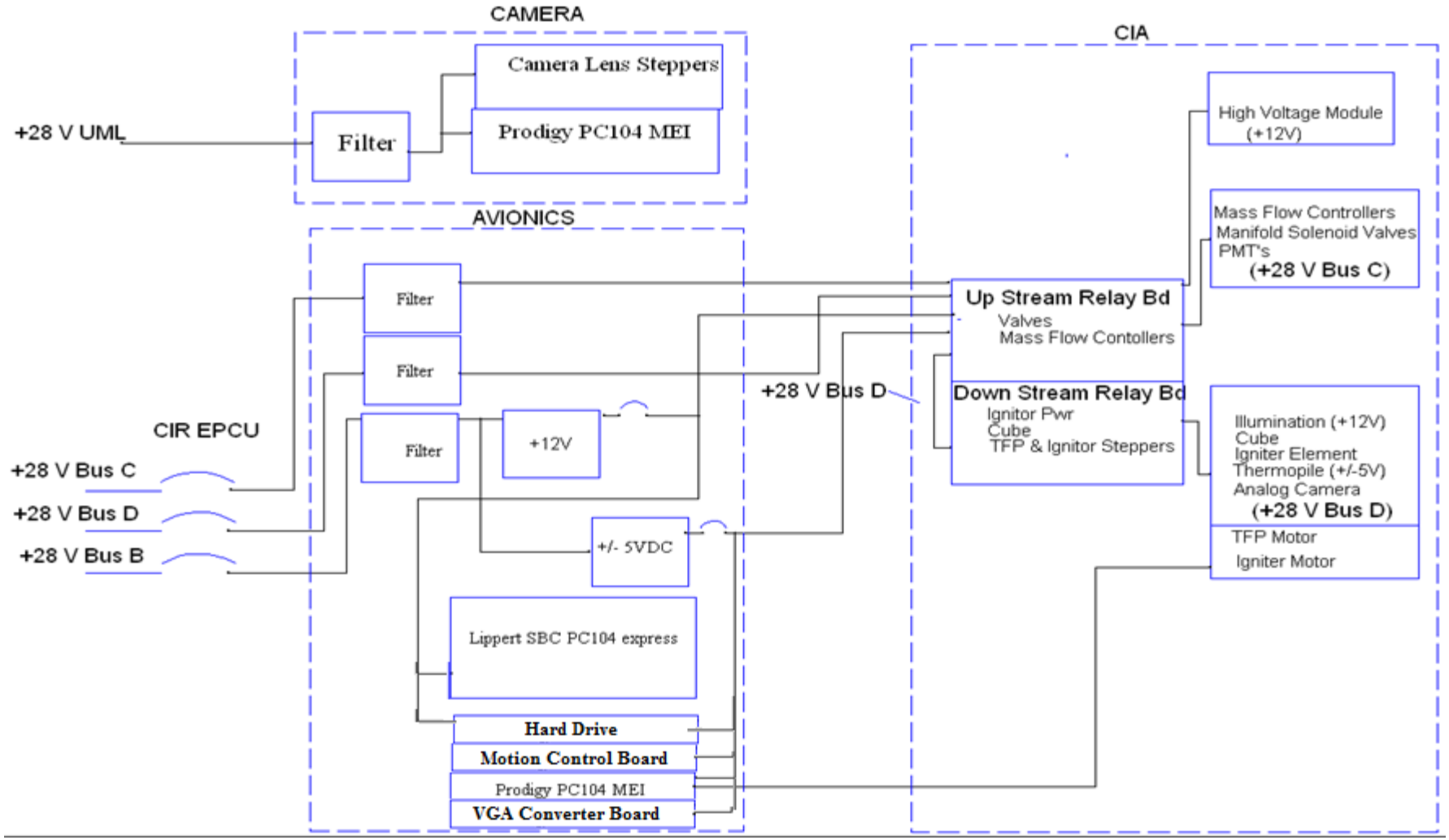


AVP Power Distribution

Primary power provided by CIR from three 8 amp 28 VDC circuits from PIL connector.

- **Power Board**
  - DC/DC converters provide secondary power for:
    - ±5 volt
      Relay Boards
      Photo Multiplier, Hard Drive, T/C Board,
      Radiometer, E-Field Interface Boards, Pressure Transducers,
      TFP Encoder, Igniter Home Switch, HV Interface Boards
    - +12 volt levels
      Analog Camera
      Ultravolt High Voltage supply, Illumination
  - Passes through +28 V EPCU power for Chamber Insert Motors, Valves, Cube – provides filters for this power (isolated from ACME secondary power system)
  - Also provides power and circuit wire protection to CIA through CIR interconnection cabling.

Power System Block Diagram



AVP Load Calculations

Bus D

Cube +28 V cube		
ppc5		4.0 watts
AI-207		2.2 watts max
AI-225		3.9 watts
AO-333		3.0 watts
Subtotal =		13.1 watts @ 28V = .467A
TFP Stepper	1.4A	
Igniter Stepper	.4 A	
Igniter	2.0 A	
Bus D Total Current	**4.267 A**	

Bus C

+28 V Solenoid valves (10)	.35A ea =	3.5 A
MFC (3)	.28A ea =	.84 A
Bus C Total Current		**4.34 A**

AVP Load Calculations

Bus B

+12 V

HV supplies (2) - .6a ea @ 12V = 1.2 A	
Illumination	.03 amps
Analog Camera	.220 amps
hard drive	.3 amps
CPU - 40 watts	3.33 amps
Serial Board	.51
GigE Board	.23

12 volt Current total 5.81 amps (Total available 7.5 amps)

Reflected to 28 Volt side (.79 efficiency) = 3.06 Amp

AVP Load Calculations

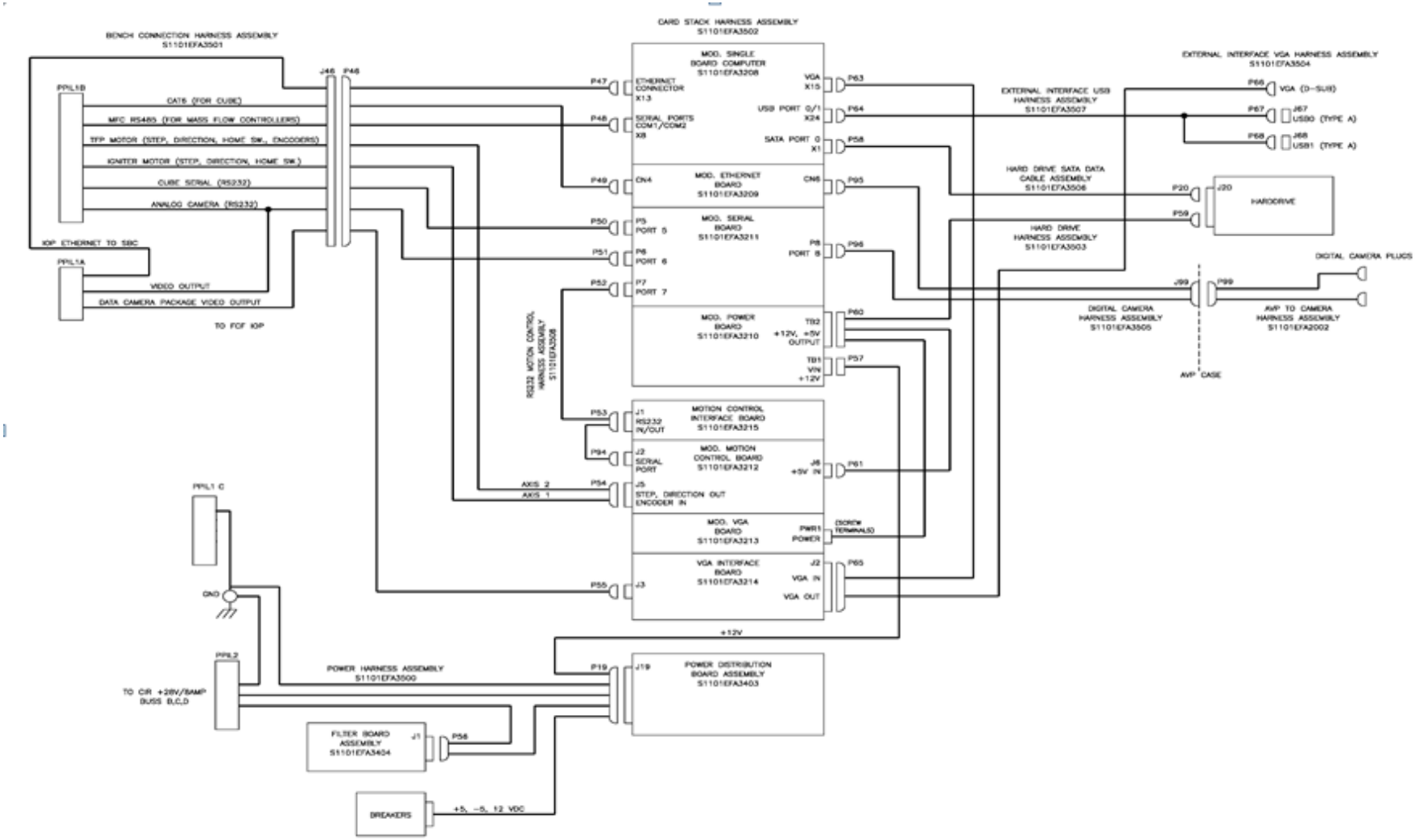
Bus B (cont.)

+/-5v

VGA Converter Board = .8A
Motion Controller Board = .8A
PMT.062 a @ +/-5V (3) = .186 A
Radiometer +5v .0012 (3) = .0036
Hard drive .95 amps
Pressure Transducer = .01A
TFP Encoder = .005A
Igniter Home Switch = .025A
+/- 5 Volt Current 1.196 A (~Total available 7.5 amps)
Reflected to 28 Volt side (.79 efficiency) = 1.51 amps

Bus B Total Current = 4.57 Amp

AVP Interconnect/Bonding



Avionics Package Hardware

Status:

Design Completion: 100%

Drawing Count: 92

Drawing Completion: 100%

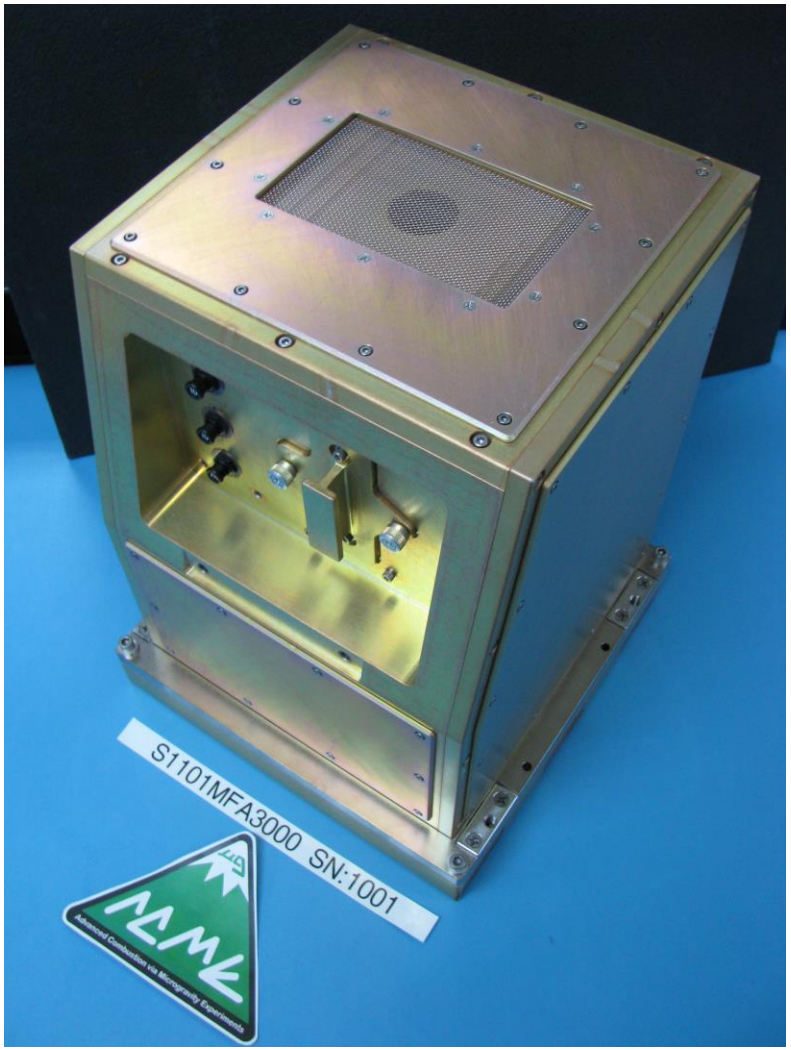
EM Build: 100%

Flight Build: 37%

Requirement Compliance: 100%

Test Status:

- *HD Tray/SATA Cable Test:*
  ACME-TST-006 – EM Passed
- *Power Distribution Test:*
  ACME-TST-011 – EM Passed
- *Motion Control Board Interface Test:*
  ACME-TST-023 – EM Passed
- *VGA Interface Board Test:*
  ACME-TST-024 – EM Passed
- *AVP Assembly Test:*
  ACME-TST-009 – EM Passed
- *Filter Board Test:*
  ACME-TST-016- EM Passed



Data Camera Package Design

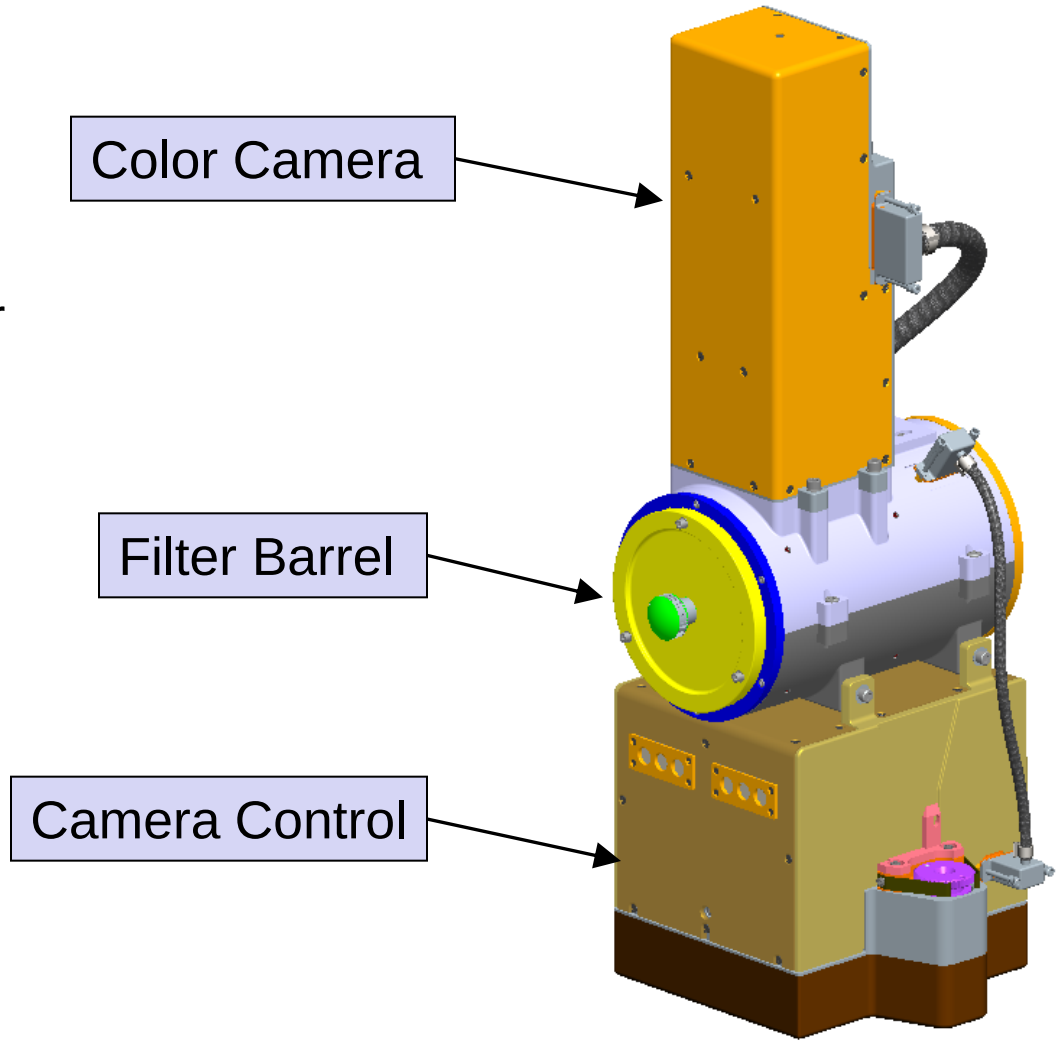
Steve Lawn – Diagnostic Engineering Lead

ACME Data Camera Package Hardware

ORU ACME Provided Diagnostic Package for Color Flame Imaging

Located outside chamber at CIR UML-6, providing forced air cooling and power

Camera Control includes power distribution and motion control

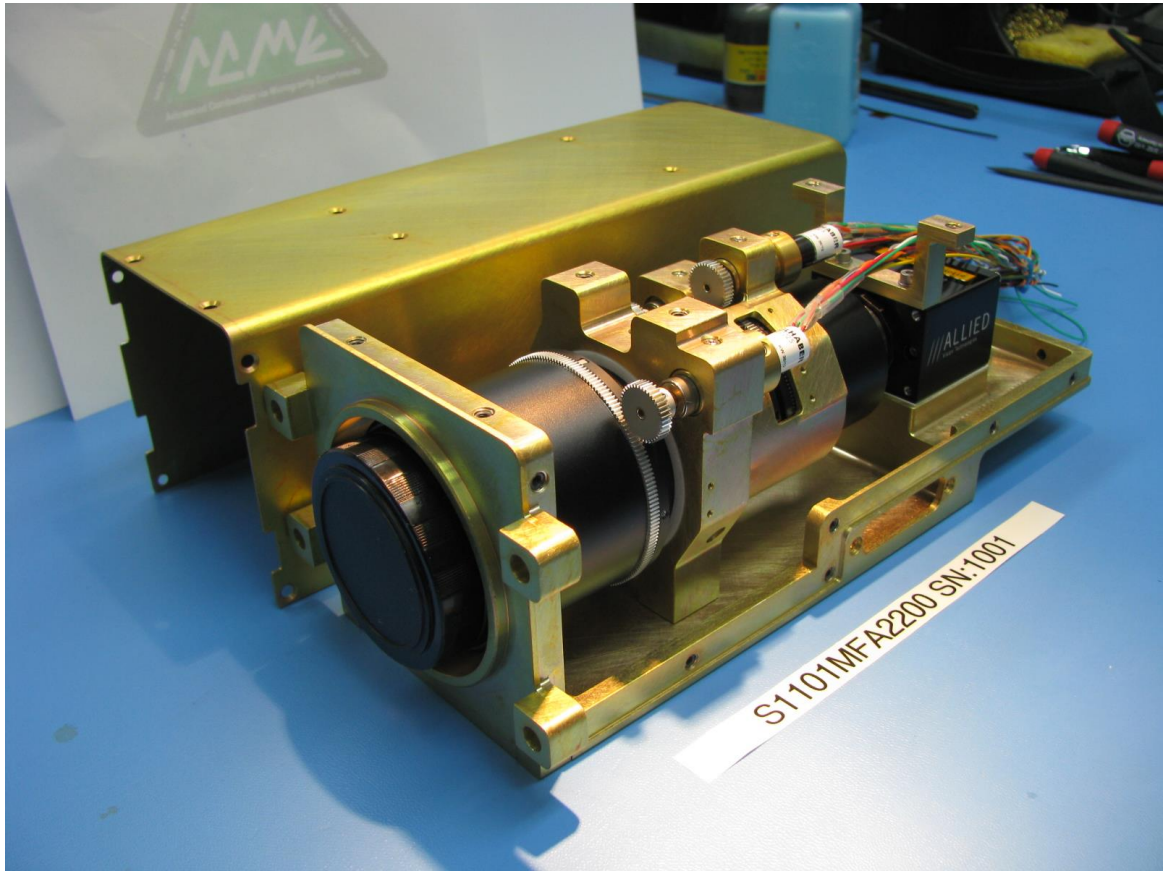


ACME Data Camera Package Hardware

Color Camera subassembly is ORU in the event of pixel loss.

- Allied Vision GC-1380CH Camera

- Zoom Lens – Navitar Zoom 7000 with motorized Zoom, Iris and Focus



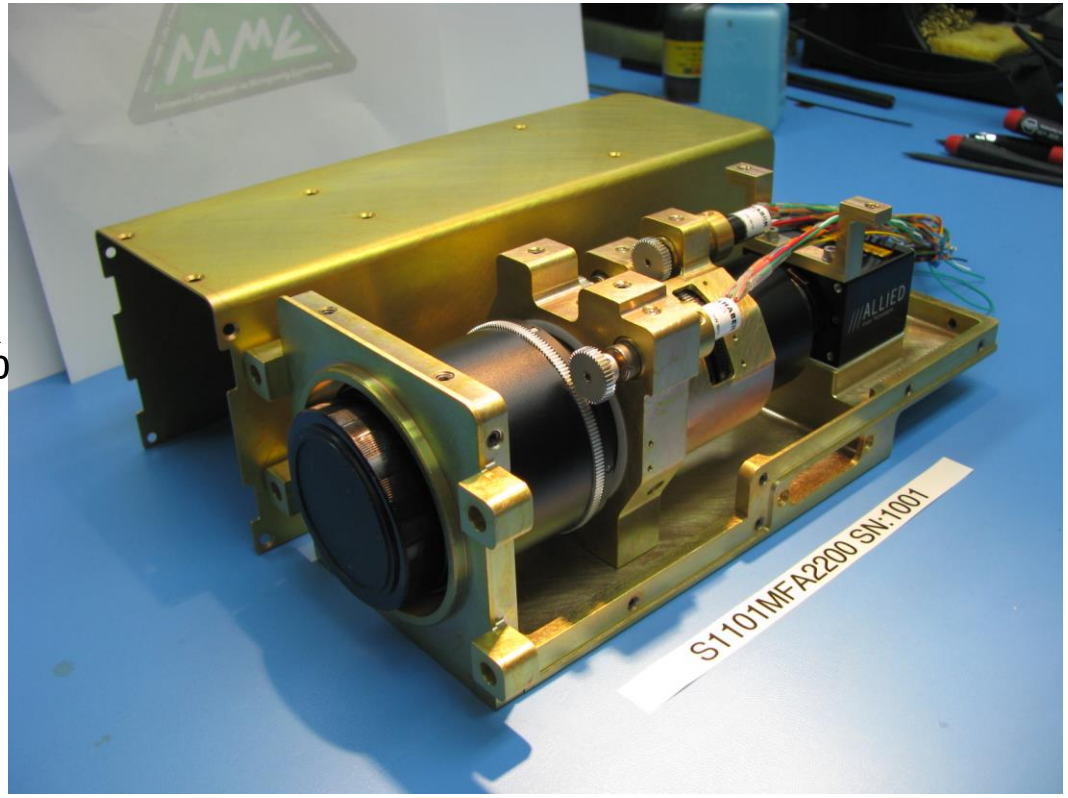
ACME Data Camera Package Hardware

Status:

Design Completion: 100%
Drawing Count: 28
Drawing Completion: 100%
EM Build: 93%
Flight Build: 39%
Requirement Compliance: 100%
Test Status:

- *Hall Effect Sensor Board Test:*
  ACME-TST-008 – EM Passed

- *Camera Enclosure Test:*
  ACME-TST-019 – Draft



ACME Data Camera Package Hardware

Glenn Research Center

Filter Barrel Assembly includes:

- 45° Turning Mirror
- Front Window: Fused Silica
- Motor to rotate Filters (Note: Does not have Continuous rotation capability)
- 4 Configurable filter locations
  - Clear – Fused Silica
  - Schott BG-7 (Blue-Green)
  - 430 nm
  - 450 nm



ACME Data Camera Package Hardware

Filter Barrel Status:
Design Completion: 100%
Drawing Count: 38
Drawing Completion: 95%
EM Build: 89%
Flight Build: 47%
Requirement Compliance: 100%
Test Status:

- *Hall Effect Sensor Board Test:*
  ACME-TST-008 – EM Passed



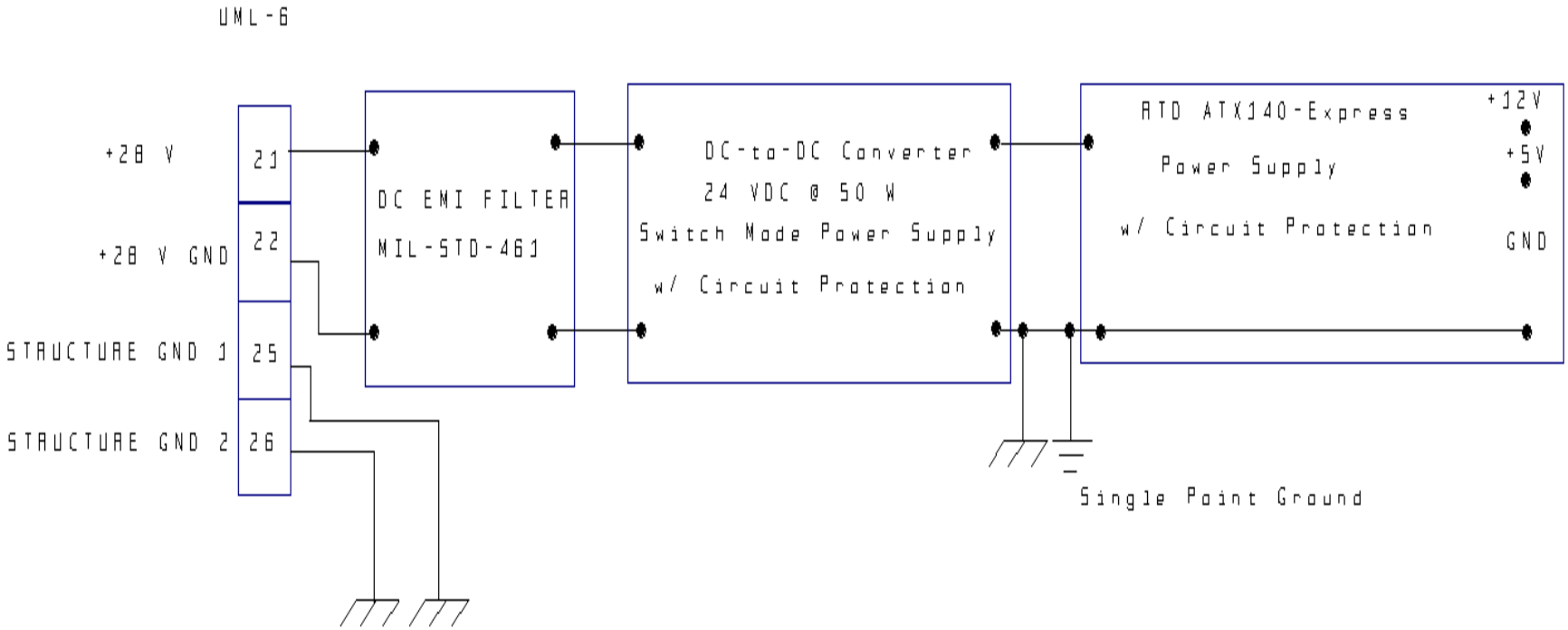
Data Camera Package Bonding/Grounding

- Payload components shall use the chassis ground by tying each pin to the payload hardware chassis at separate points to ensure crew safety. (ICD Paragraph 6.1.1.2.6)
- Pinout Configurations for UMLs 1,2,4,6 and 8 Bay B

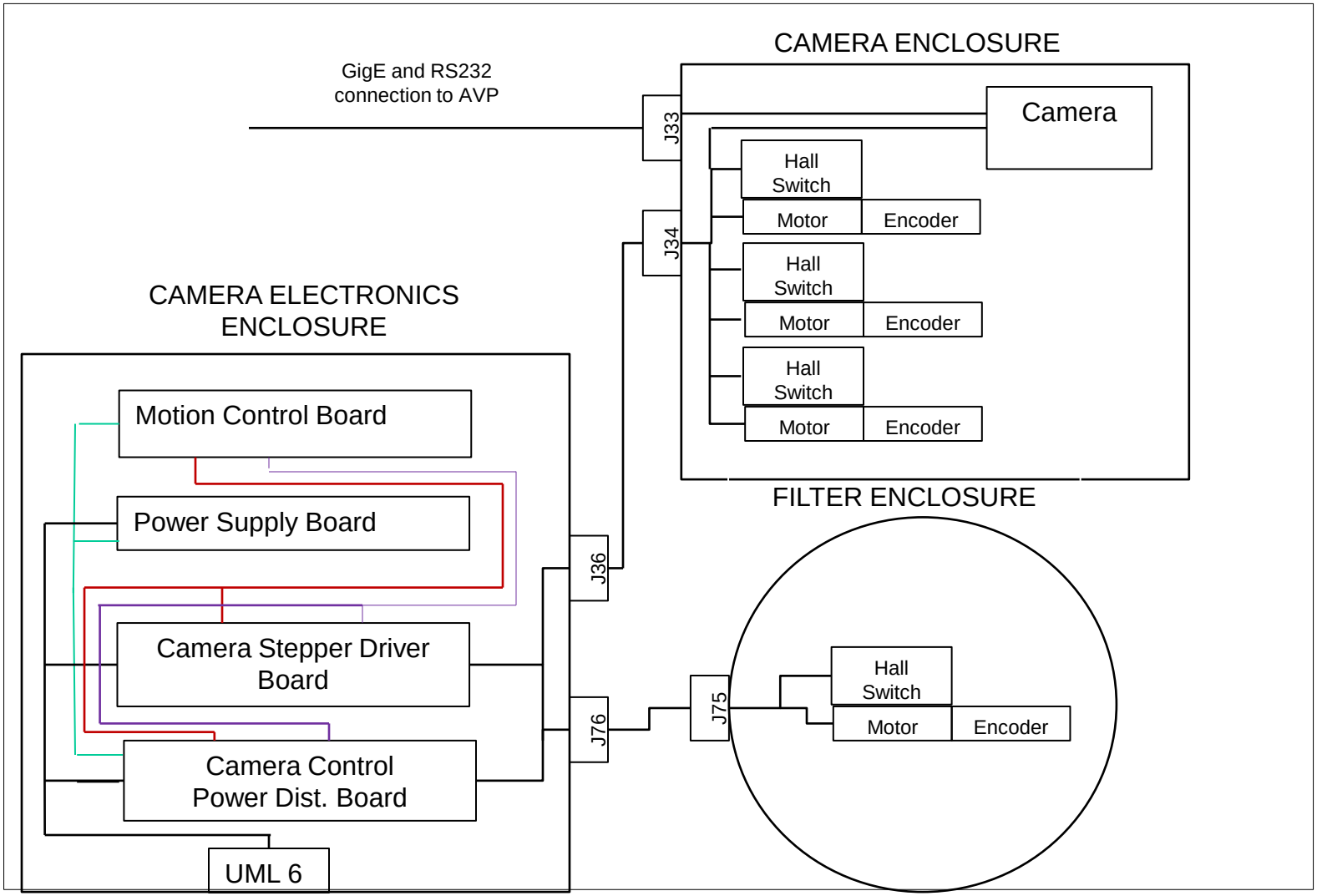
Pin #	Signal
21	+28V 8A
22	+28V 8A Return
25	Structure Ground 1
26	Structure Ground 2

- Pins 25 and 26 attach directly to mounting structure along with single point ground

Data Camera Package Bonding/Grounding



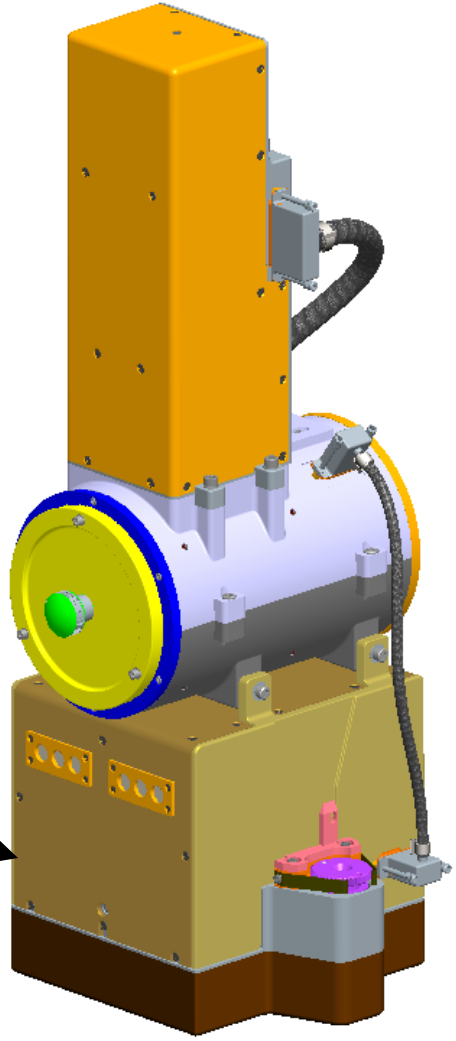
Data Camera Package Interconnect Diagram



ACME Data Camera Package Hardware

Camera Package Assembly Status:
Design Completion: 100%
Drawing Count: 43
Drawing Completion: 79%
EM Build: 0%
Flight Build: 0%
Requirement Compliance: 100%
Test Status:

- *Camera Stepper Driver Board Test:*
  ACME-TST-017 – Draft
- *Camera Power Distribution Board Test:*
  ACME-TST-021 – Draft
- *Camera Control Test:*
  ACME-TST-027 - Draft



Software

Mike Medved – Software Engineering Lead

ACME Software

- ACME has two CSCIs:
  - ACME Main Processor (AMP) CSCI
    - Flight code in AVP
  - Acme Ground Software (AGS) CSCI
    - TSC Workstation

- Both are Class C software
  - 7150.2A – “Flight or ground software that is necessary for the science return from a single (non-primary) instrument

- Software recently reclassified as safety-critical

ACME Software Safety

- ACME Software Litmus Test
  - ACME MP – Class C, Safety Critical
    - Software Control Risk Level – 2 (Medium)
    - Likelihood of Occurrence – Improbable
    - Software Prioritization Risk Level - 5

Software Control Risk Levels	**Likelihood of Occurrence**				
	Improbable	Unlikely	Possible	Probable	Likely
SW 1 (High)	4	3	2	1	1
SW 2 (Moderate)	5	4	3	2	1
SW 3 (Medium)	6	5	4	3	2
SW 4 (Marginal)	7	6	5	4	3
SW 5 (Low)	7	7	6	5	4

ACME Software Safety

- From 8719.13C:
  - “In general, it is expected that for Low Software Prioritization Risks (7-5s and Green in chart above) the main requirements for both acquirer and provider would be to perform the Software Safety Criticality Assessment, keep a record of it, and revisit the SSCA upon any changes to the software functionality or design to assure the software maintains its low risk status.”

Software Safety

This is applicable to Class C software with a safety risk index of 5, 6 or 7.

- ZIN writing waivers to the following requirements:
  - for detailed design (SWE-058)
  - Coding standards (SEW-061)
  - traceability (SSS-011)
  - software maintenance plan (SWE-105)
  - software user manual (SWE-115)
  - Verifying S/W hazards controlled (SSS-055)
  - Verification of S/W safety requirements (SSS-059)
- ZIN writing deviations to:
  - change traceability from detailed design to architectural design(SWE-049, SWE-064)
- Create software safety plan
- Addition of acquirer SMA to document workflow for software related documents
- Updates to current documentation
- Additional work with PSRP software panel

Software Documentation

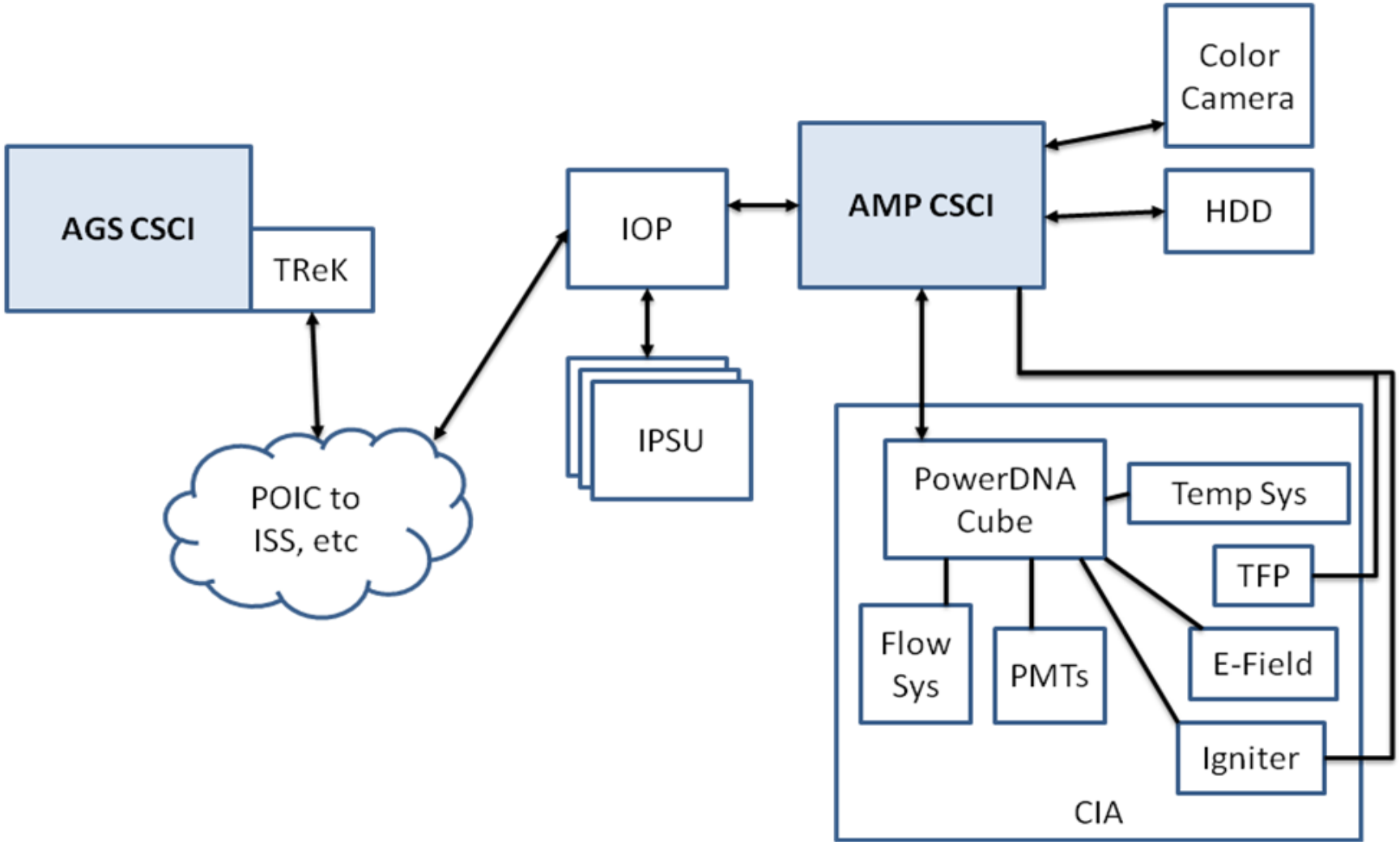
- 7150.2A Software Documentation
  - Software Management Plan (SMP) (ACME-PLN-004)
  - Software Requirement Specification (SRS)
    - All requirements in Section 4 of ERD
  - Software Design Document (SDD) (ACME-DOC-005)
  - Software Assurance Plan (SQAP) (ACME-PLN-005)
  - Interface Design Description (IDD) (ACME-DOC-010)
  - Software Test Plan (STP) (ACME-PLN-009)
  - Software Test Procedures

- Some documents are covered at the contract level
  - SpaceDOC CM Plan (P30016)
  - SpaceDOC Risk Management Plan (P30015)

Software Development

- Development occurs according to a basic waterfall
  - In step with the hardware development & NASA review cycle
  - Software development tracked at the project level by ZIN PM
- Requirements documented in “shall” statements in ERD
- Design captured in Doxygen
- Requirements will be verified and validated

System Context



AMP Capabilities

- Communication with CIR IOP
  - Send/Recv commands and telemetry
  - Send imagery and data files
  - Activate IPSUs
- Color Camera
  - Send commands and change parameters
  - Capture and store images
- Ops Camera
  - Send configuration commands (gain)

AMP Capabilities

- Communicate with PowerDNA Cube
  - Handle Analog Input and Output to Devices
- Communicate with PMD motion control boards
  - Configure different axis for movement
  - Command motion
- Handle experiment execution
  - Sequences of commands
  - Implemented via a scripting engine

AGS Capabilities

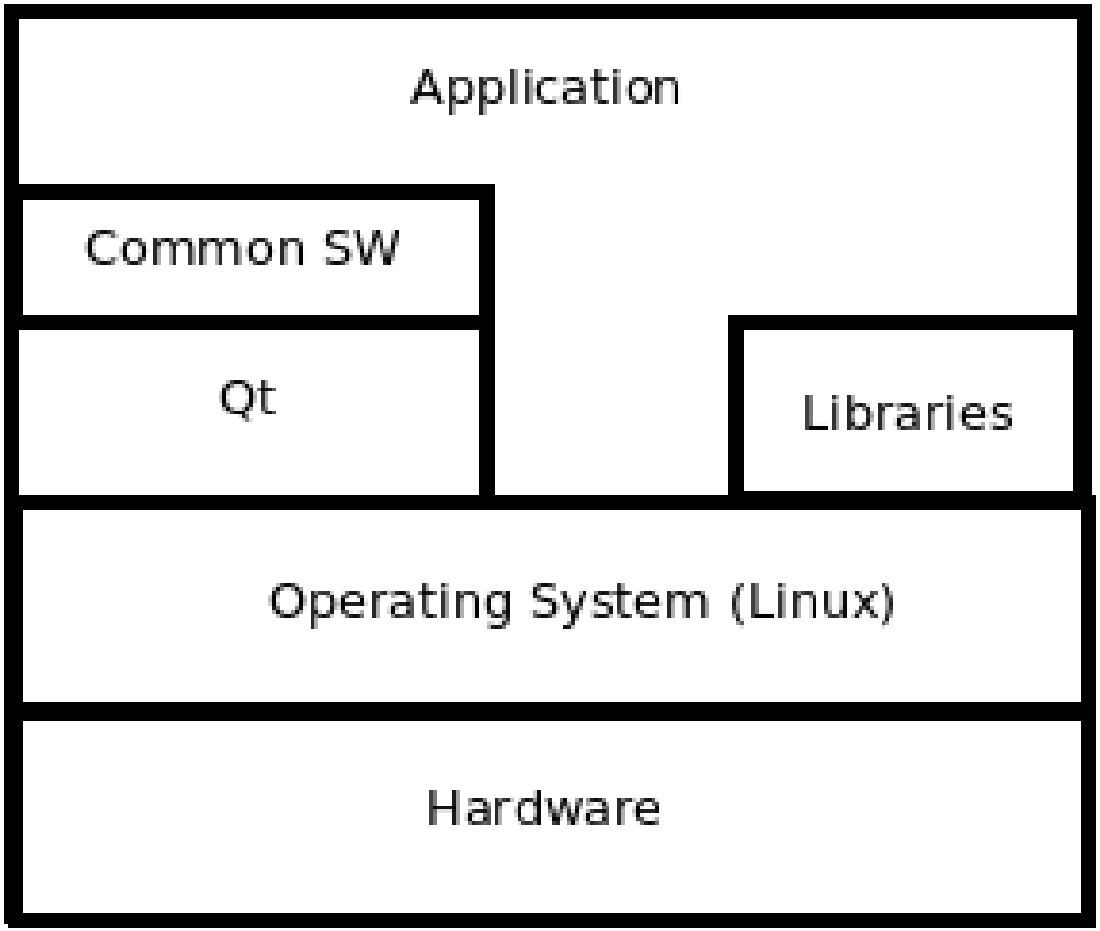
- Runs on a TSC workstation (Windows)
- Uses the TReK network to communicate
- Development is delayed in comparison to AMP CSCI

- Capabilities:
  - Send single commands
  - Send sequences of commands
  - Request and display telemetry
  - Display health and status

Common Platform

- Projects all have similar needs:
  - Operating system, libraries
  - Command and Telemetry Handling, Communications
  - Logging, Serial Port Communication
  - Hardware control, experiment sequencing
  - Development environment
- The Common Software Platform provides a standard, easy to use, free platform for NASA projects
- Relies heavily on free and open source software (FOSS)
- Provides a directly reusable common software architecture

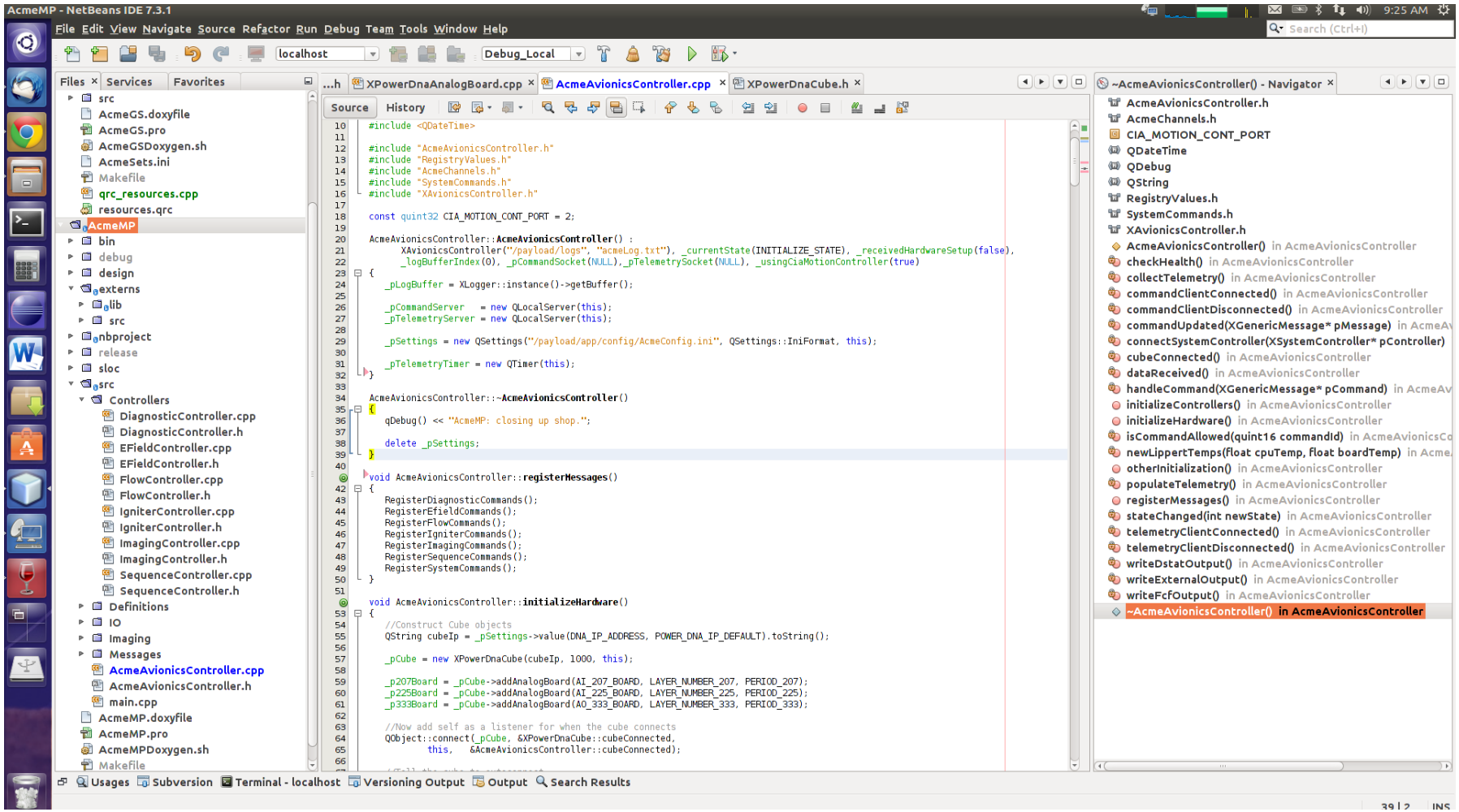
Common Platform



Development Environment

- Source control – Subversion
- Operating System – Linux (Ubuntu)
- GNU build tools (compiler, linker, make, etc)
- Netbeans IDE
- Qt Designer for GUIs
- All free, stable, and open source
  - Not subject to budget fluctuations
  - Upgradeable
  - Maintained

Development Environment



Linux OS

- Linux – Operating System
  - Highly successful, stable OS
  - Highly configurable
    - Many distributions
  - Lots of free libraries and tools
  - Easy to upgrade
- Lubuntu – A lighter Ubuntu distribution with LXDE desktop

Qt Framework

- Qt is a cross-platform application and UI framework
- Cross-platform means programs built on Qt can be compiled on other OSes with little to no work
  - Linux, Windows, QNX, VXWorks, Android, Mac, iOS
  - “Write once, build everywhere”
- Provides many capabilities:
  - Threading and Concurrency
  - Networking and Sockets
  - Timers
  - String manipulation, File Systems, etc
- Event driven architecture
  - Signals and slots
- Hundreds of 3rd party Qt libraries

Common Classes

- Common C++ Software – developed to share between projects
- All directly useable without modification
- Very low coupling due to Qt signals/slots
- Hardware interface classes
  - AVT GigE Camera
  - PowerDNA Cube
  - PMD Motion Control
- Software Architecture
  - Avionics controller, System Controller, hardware interface
  - Command and Telemetry message serialization/deserialization
  - Logging
  - GUI elements

Common Classes

- Used on multiple projects
  - ACME, PBRE, OASIS, IPSU-G, ACE
  - More testers and maintainers means fewer bugs
  - Code review benefits all
- Around 5200 SLOC

Software Status

- AMP CSCI
  - MDCA was 10 kSLOC
  - Estimated ACME size 4-5 kSLOC
  - Current 2.5 kSLOC

- AGS CSCI
  - MDCA was 20 kSLOC
  - Estimated ACME size 3 kSLOC
  - Currently 1 kSLOC

Software Testing

- Unit Level
  - Informal, use Qt unit test framework as needed
- Engineering Model
  - Black box, send commands and ensure proper operation
- Integrated
  - With IOP and IPSU simulators
- GIU
  - Verify software requirements as part of system verifications as much as possible

Structural Analysis

Chris Werner – Structural Analyst

Structural Requirements

Req. #	Requirement
SSP-57000 3.1.1.3 B	Load Requirements Designed to withstand the on-orbit loads of 0.2 Gs acting in any direction.
SSP 50835 3.1.1.2.1.5.B	Pressure Loading Maintain positive margins of safety when exposed to a depressurization rate of 13.3 kPa/second (116 pounds per square inch (psi)/minute) as shown in Table E.2.7-1.
SSP-57000 3.9.1.1	Pressure Be safe when exposed to the pressure transitions between 0 to 104.8 kPa (0 to 15.2 psia).
SSP 50835 3.9.1.1.A	Pressure Operate properly after exposure to launch and ground handling pressures ranging from 60.0 to 129.3 kPa (8.7 to 18.8 psia).
GLPR 7120.5.30 Section 3.4.4	Margins of Safety All structural elements critical for safety and mission assurance shall be shown by analysis to have positive margins of safety or, in the case of containment devices, be structurally adequate against penetration.

Req. #	Requirement
GLPR 7120.5.30 Section 3.4.6	Pressurized Systems Pressurized systems to be flown on unmanned vehicles shall meet the requirements of ANSI/AIAA S-080, Space Systems-Metallic Pressure Vessels, Pressurized Structures, and Pressure Components, and ANSI/AIAA S-081, Space Systems-Composite Overwrapped Pressure Vessels (COPV).
Hazard Report STD-ACME-17.c.1	Vented Container Analysis to demonstrate the vented containers provide an internal effective vent area ratio that results in a differential pressure loading of no greater than 0.01 psid.
Hazard Report ACME-2 1.1.1	Pressure system hardware shall be designed to the following factors of safety (FS): Lines/Fittings FS = 4 x MDP. Other system components FS=2.5xMDP. Hardware shall be designed to withstand thermal and vibratory loads associated with both launch/landing and during on-orbit operation.

Note: requirements are paraphrased

Structural Analysis – Load Requirements

- On-Orbit Operation Pressures
  - Fuel system pressure
  - Water system pressure
- Crew-induced loading
  - CIA Handles
  - Color Camera Package
  - Avionics Package
- Launch loads
  - Quasi-static inertial loads

All pressurized systems will be proof tested to 1.5 x MDP and leak tested to 1.0 x MDP

Details of all structural analysis can be found in ACME-ANA-002

Structural Analysis – Safety Factors

- Safety Factors for Safety Critical Components are Defined in Section 5.1.2 and 5.1.3 of SSP-52005
  - Non pressurized metallic structures and fasteners will use 1.25 for yield and 2 for ultimate for margins of safety calculations
  - Pressurized components will have a safety factor greater than 2.5 to the ultimate
  - Pressurized Lines and Fittings will have a safety factor greater than 4.0

All Safety Critical Fasteners:

- Calculated using NSTS-08307 Pre-loaded Bolt Criteria and NASA TM 106943
- Torque based on MSFC-STD-486B
- Fitting Factor of 1.15 is applied to all Margins of Safety Calculations
- Joint separation of 1.2 is used for structural joints (1.4 for pressurized applications)

Structural Analysis – Analysis Methods

- Structures are analyzed for stress using a combination of hand calculations and Finite Element Analysis (FEA)
- FEA pre and post processing is performed using Femap Version 10.1 / 10.3
- Analysis models are run using NX NASTRAN 6.0
- The maximum von Mises stress is used for safety factor and margin calculations unless maximum principle stress is larger
- Metallic Material Properties are obtained from MMPDS-06
- Margins of Safety are calculated below:

$$MS_y = \frac{\sigma_y}{\sigma * FS_y} - 1$$

σ = component stress
MS_y = margin of safety to yield
σ_y = material yield strength
FS_y = yield safety factor

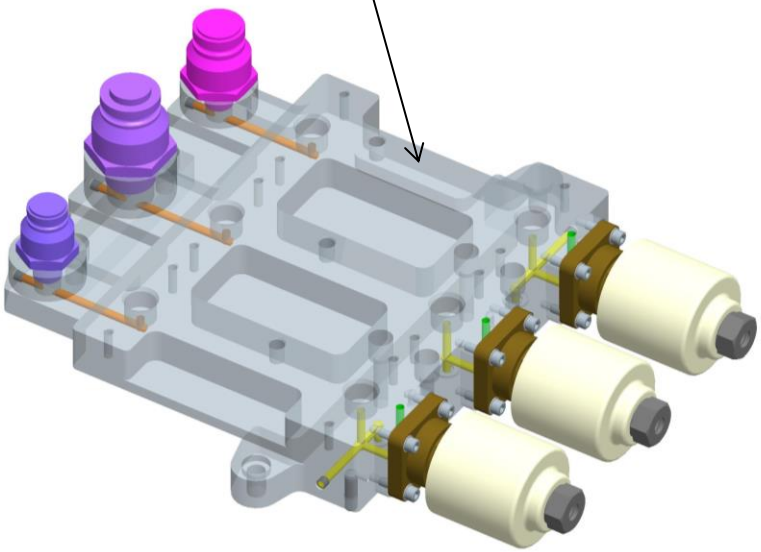
$$MS_u = \frac{\sigma_u}{\sigma * FS_u} - 1$$

σ = component stress
MS_u = margin of safety to ultimate
σ_u = material ultimate strength
FS_u = ultimate safety factor

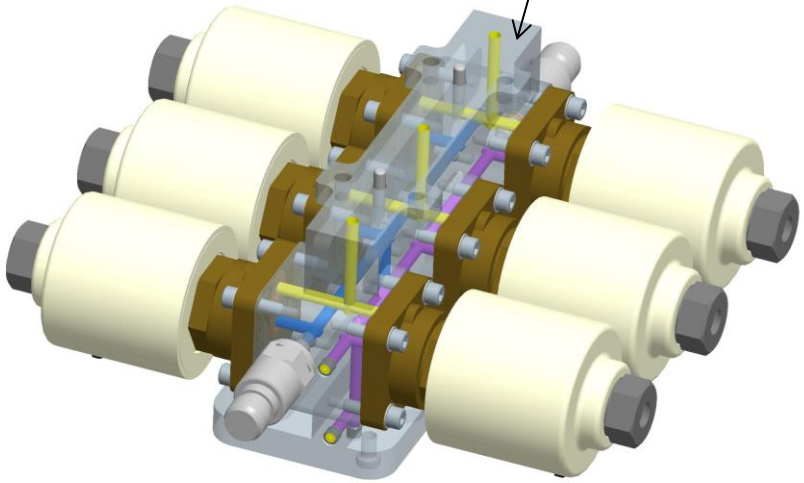
Structural Analysis - Pressure

ACME has two custom pressurized components exposed to an MDP of 135psia

Input Manifold



Switching Manifold

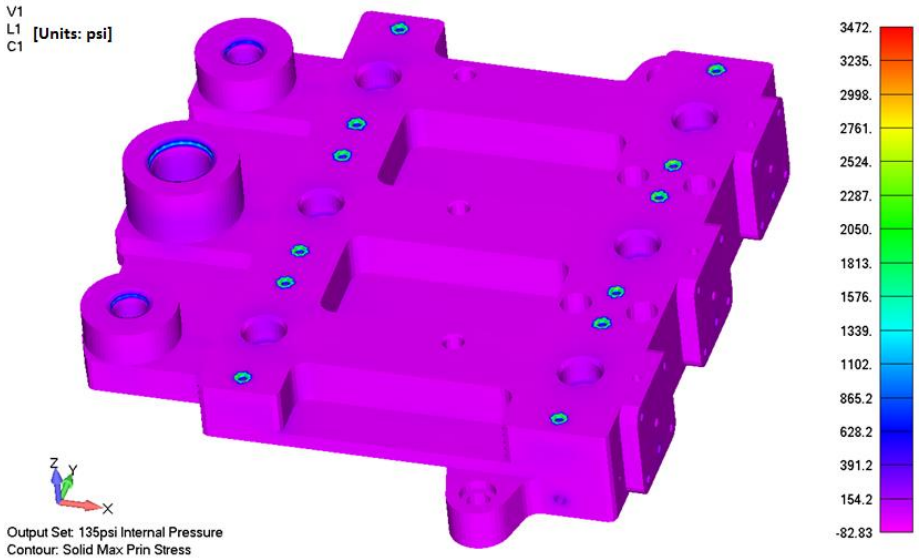


Both are made of 300 Series Stainless Steel
Ftu = 75,000 psi

Structural Analysis - Pressure

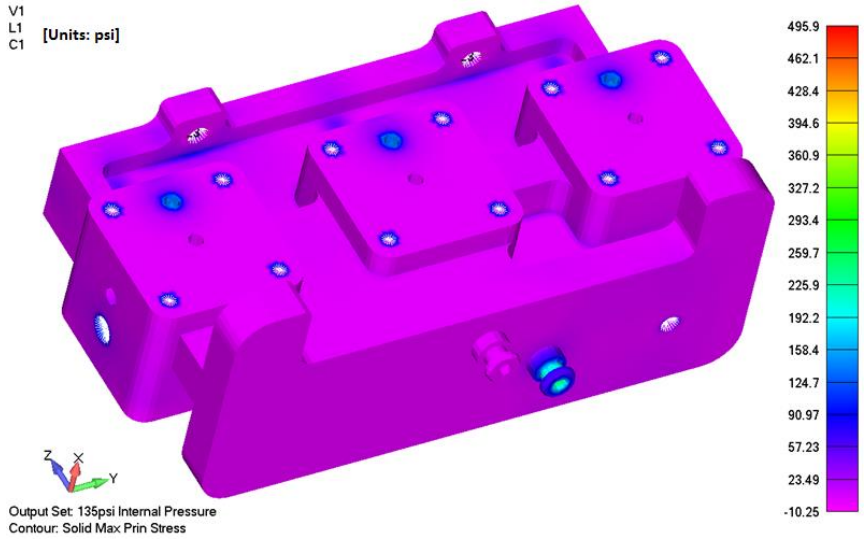
Both components are analyzed for stress using a finite element model

Input Manifold



Max principle stress = 3,472 psi
Safety Factor = 21

Switching Manifold

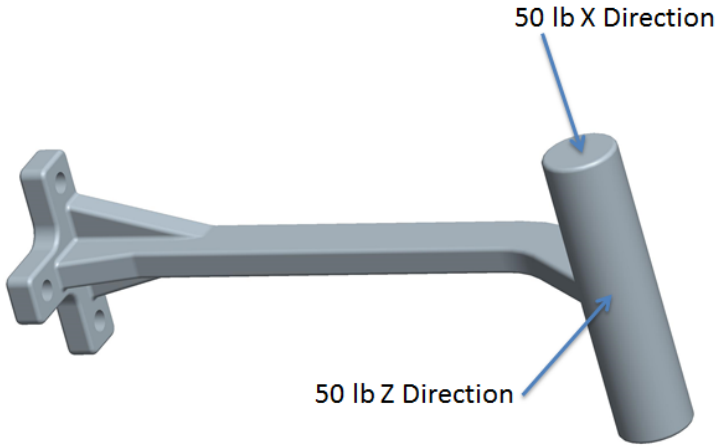


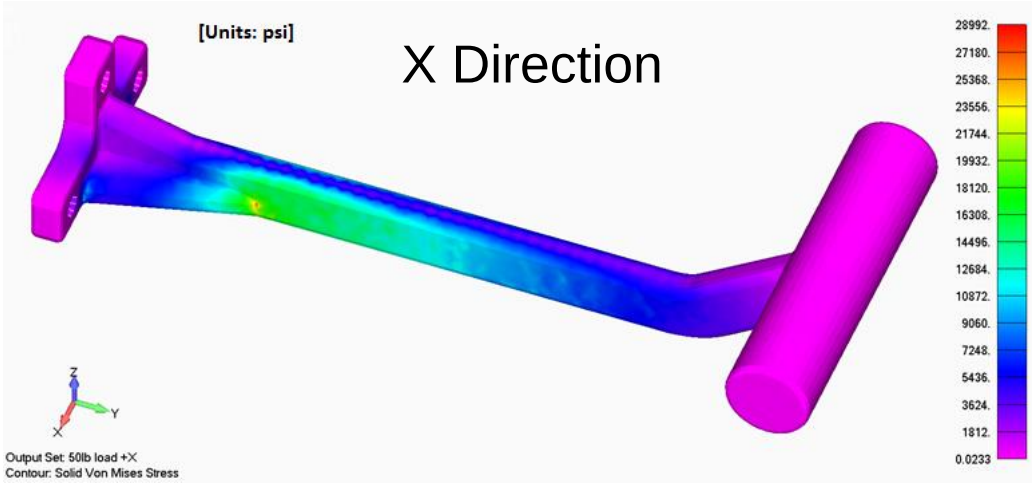
Max principle stress = 496 psi
Safety Factor = 151

Structural Analysis – Crew Loading

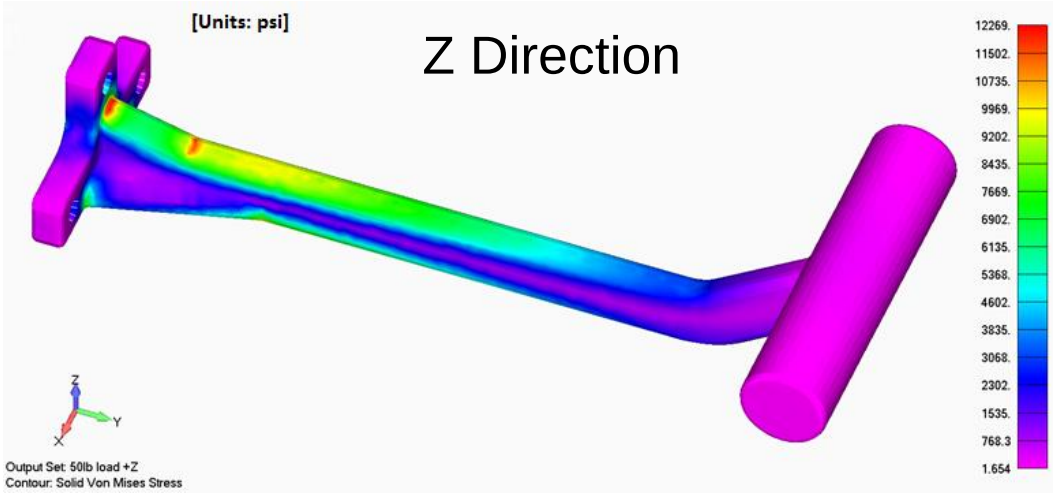
Glenn Research Center

ACME CIA Insert Handles can be loaded up to 50 lbf





Made of 7075 T7351 Aluminum

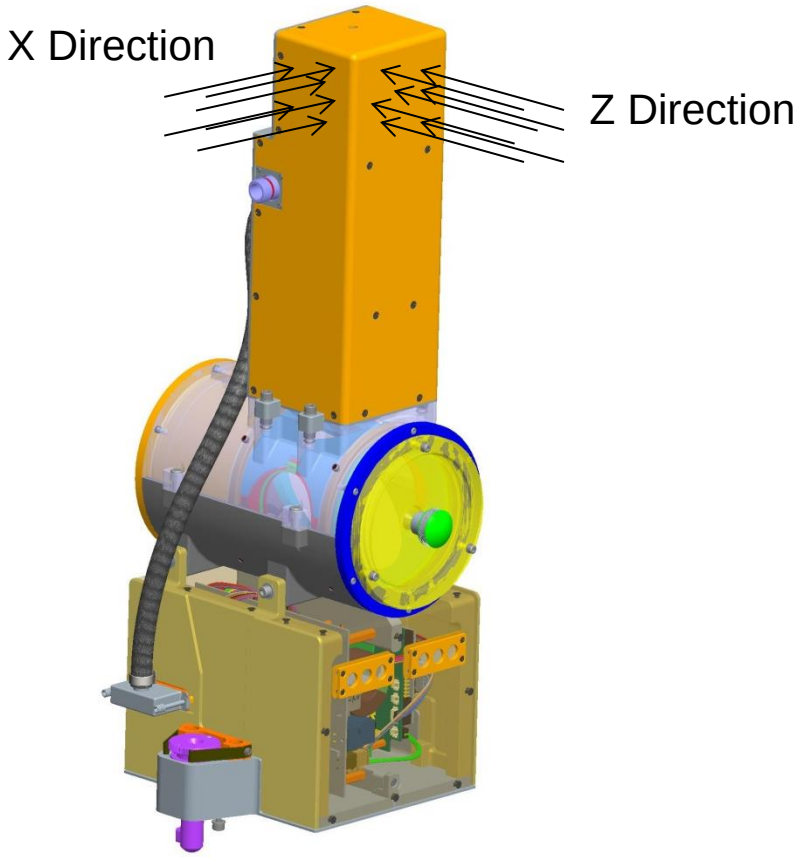


Max von Mises stress = 28,992 psi
Msu = 0.13

Structural Analysis – Crew Loading

Glenn Research Center

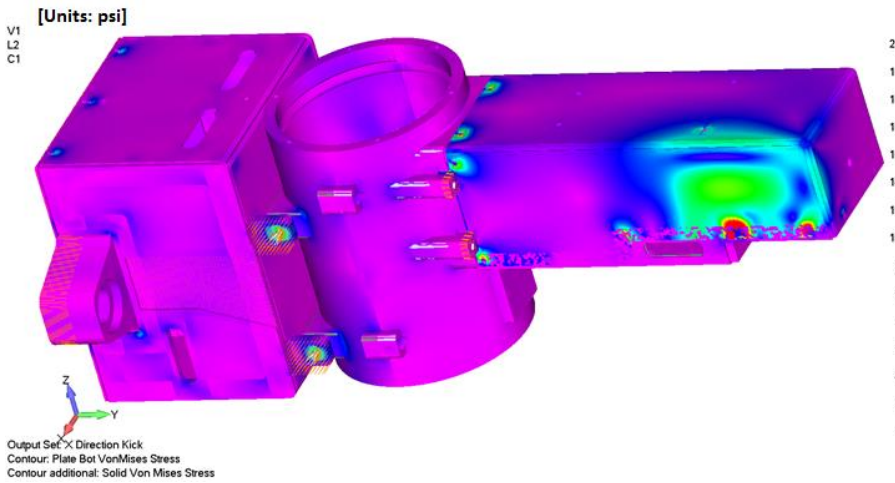
ACME Color Camera Package can be loaded up to 125 lbf



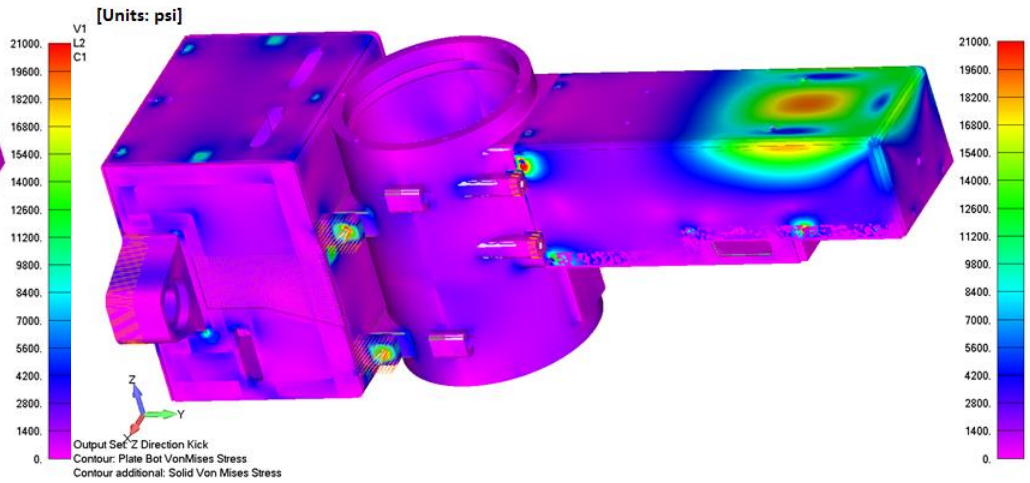
- Structural components created and assembled in FEA for X and Z direction kick loads
- Equivalent pressure load of 125 lbf applied at the top to generate the largest worst case moment
- Model constrained at the bottom to represent it's connection to the CIR optics bench
- Crew Loading will envelope loads due to launch

Structural Analysis – Crew Loading

Glenn Research Center



X Direction Kick Load

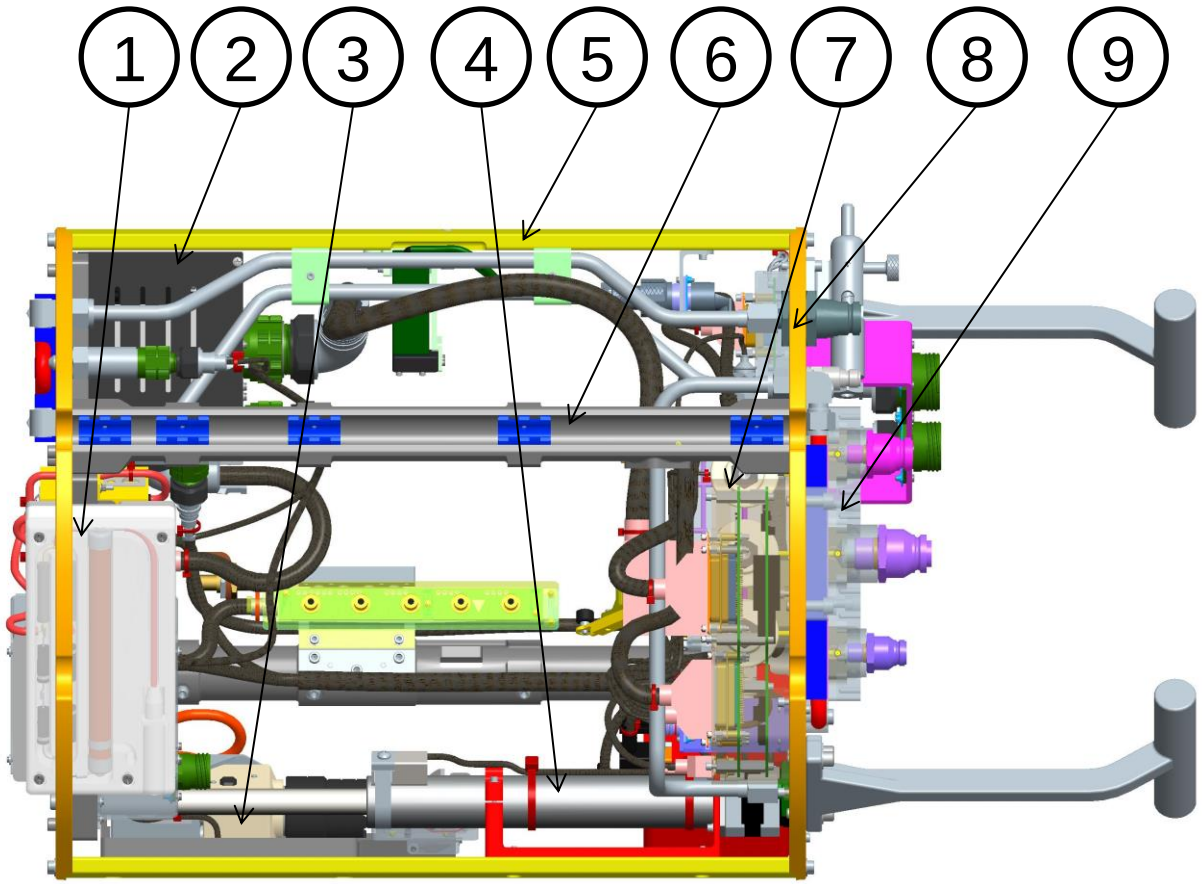


Z Direction Kick Load

Name	Part Number	Max Stress	Margin of Safety	Load Case
Camera Enclosure Cover	S1101MFD2212	19823	0.06	Z Direction
Camera Mounting Plate	S1101MFD2202	8876	1.37	X Direction
Camera Enclosure End Panel	S1101MFD2204	16050	0.31	Z Direction
Filter Enclosure Upper Housing	S1101MFD2304	4140	4.07	Z Direction
Filter Enclosure Lower Housing	S1101MFD2302	5725	2.67	Z Direction
Camera Control Side	S1101MFD2108	10161	1.07	Z Direction
Camera Control Cover	S1101MFD2118	14043	1.60	Z Direction
Camera Control Board Mount	S1101MFD2104	2591	7.10	Z Direction
Camera Control Base	S1101MFD2102	17875	0.17	X Direction

Structural Analysis – Launch

The components below are explicitly modeled in a finite element model of the CIA Structure and analyzed for launch loading.

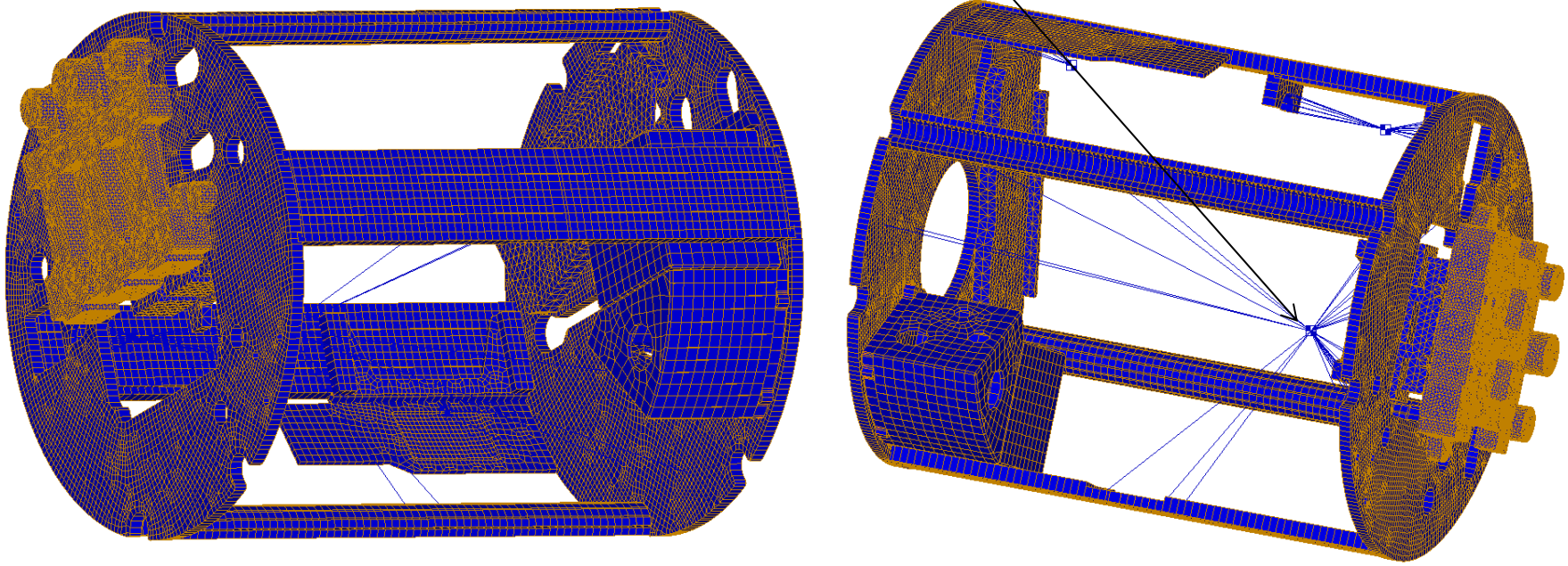


1) E Field Power Supply
2) Data Acquisition Cube
3) Camera Assembly
4) TFP Motor Assembly
5) Component Mount Bracket (2x)
6) Frame Rail (2x)
7) Switching Manifold
8) Bulkhead (2x)
9) Input Manifold

Structural Analysis – Launch

CIA Structure modeled using Solid and Plate elements with Rigid elements connecting the various components. Mass elements were used to adjust weights and CG of components. If the CG was relatively close to the component, then the material of that component was adjusted for the weight.

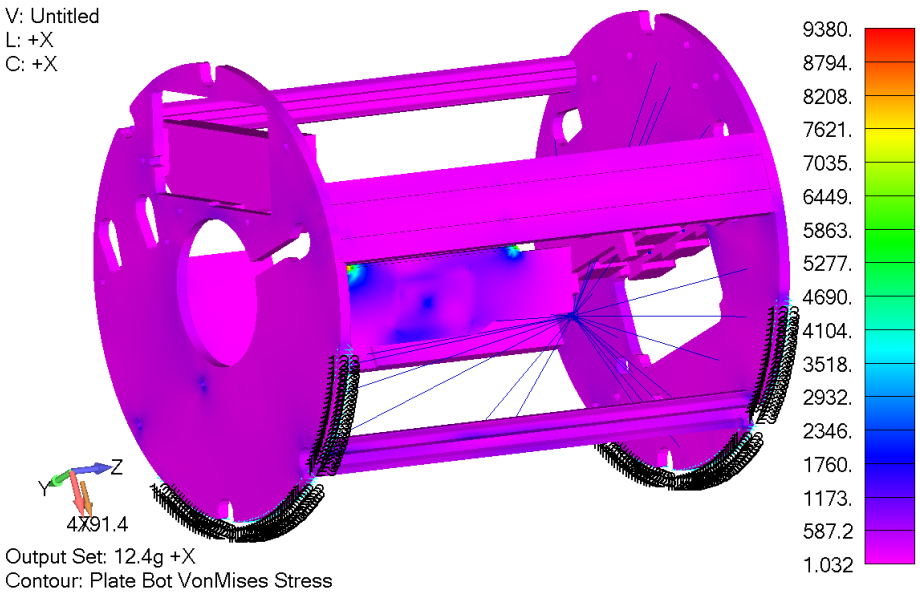
Ballast Weight for Correct Mass and CG from Pro-E Model



Structural Analysis – Launch

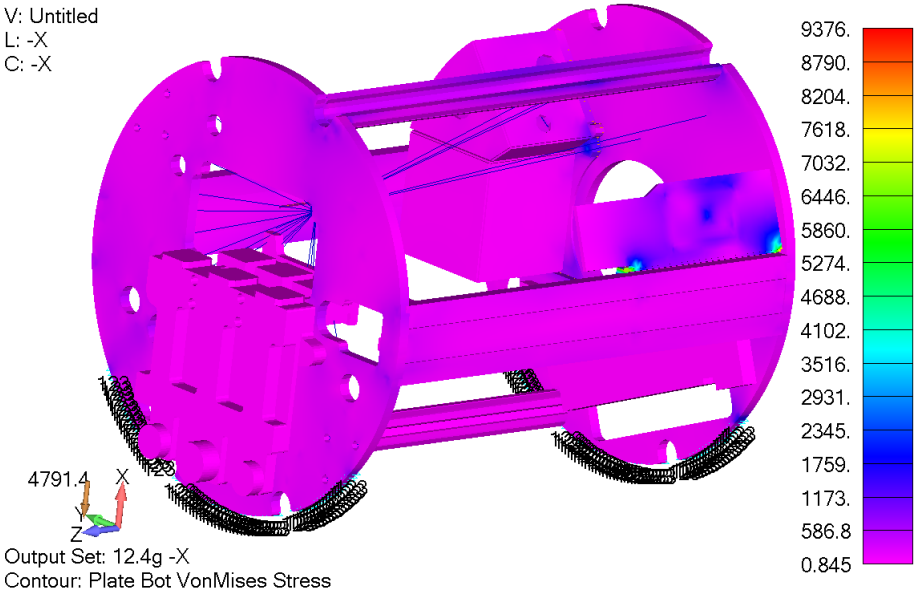
Inertial Loads applied to CIA in the + and – directions for each axis

+X Direction Loading



Max Stress = 9,380 psi
MSu = 1.23
MSy = 1.98

-X Direction Loading

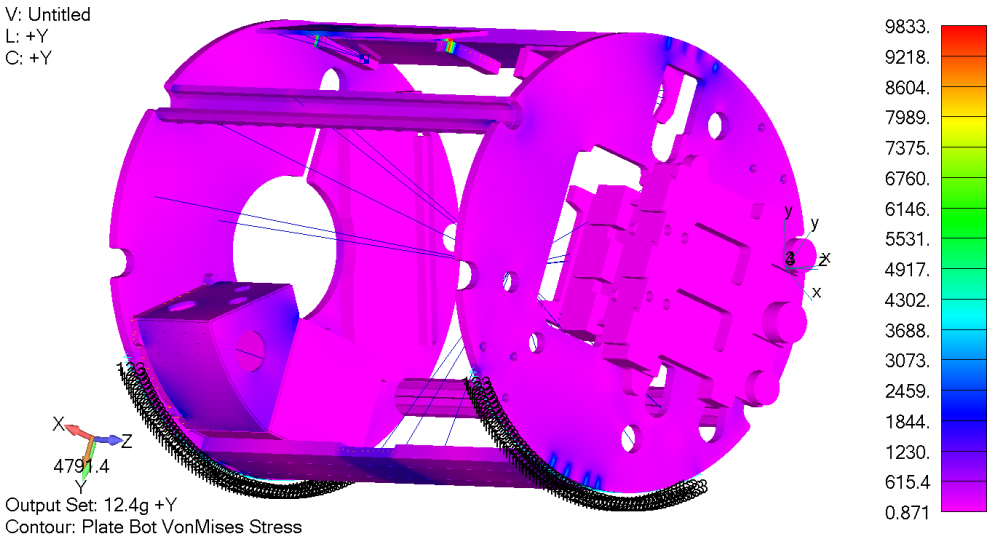


Max Stress = 9,376 psi
MSu = 1.23
MSy = 1.98

Structural Analysis – Launch

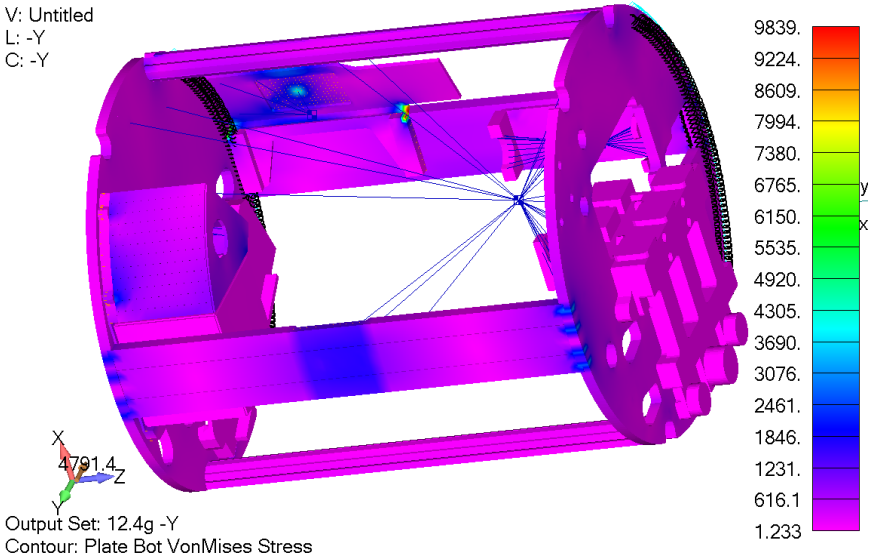
Inertial Loads applied to CIA in the + and – directions for each axis

+Y Direction Loading



Max Stress = 9,833 psi
MSu = 1.13
MSy = 1.84

-Y Direction Loading

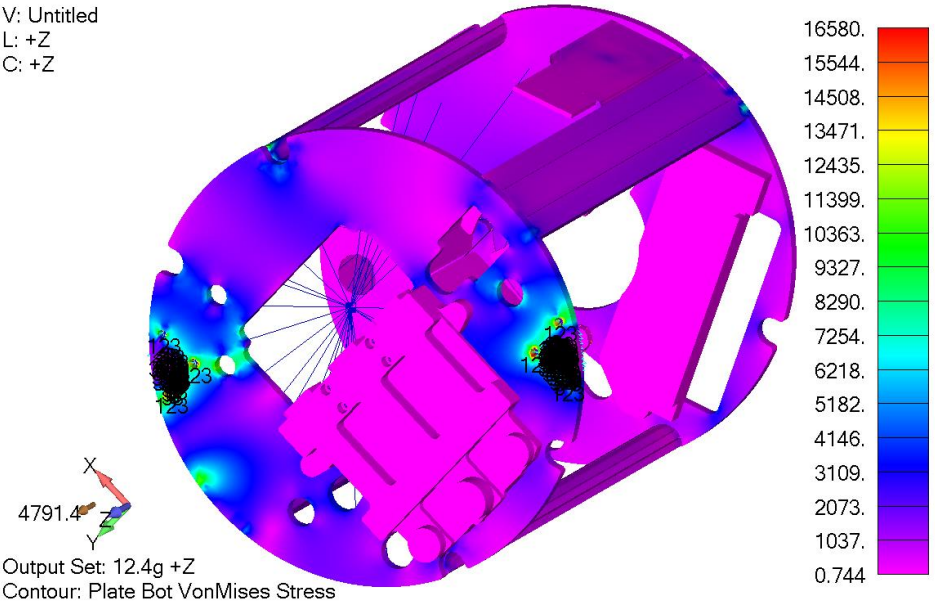


Max Stress = 9,839 psi
MSu = 1.13
MSy = 1.84

Structural Analysis – Launch

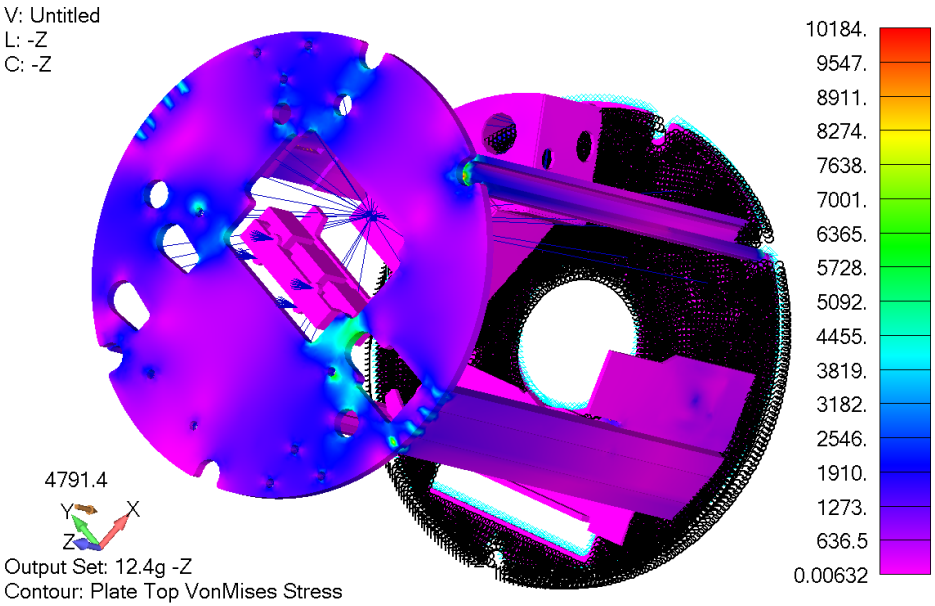
Inertial Loads applied to CIA in the + and – directions for each axis

+Z Direction Loading



Max Stress = 16,580 psi
MSu = 0.26
MSy = 0.68

-Z Direction Loading



Max Stress = 10,184 psi
MSu = 1.06
MSy = 1.74

Structural Analysis – Open Work

- Avionics Package – Crew Loading
  - Similar overall dimensions to MDCA
  - Stiffer Structure should produce positive margins of safety

Depress / Repress

- Unless part of a pressurized system, all volumes are designed to adequately vent due to station / vehicle depressurization
- Vented Containers will be verified by the stored energy method (SDP STD-ACME-1 17.a) where the volume has a stored energy < 3,063 ft-lbs. Volumes with a stored energy > 3,063 ft-lbs will have a depress rate calculated.

- Depress rates are calculated* for each volume to show that the volume's depress rate is greater than maximum station / vehicle depress rate so no pressure load is generated
- Maximum depress rate is defined in section 3.1.1.2.1.5 part B of SSP-50835 Rev C as 13.3kPa/s (1.93psi/sec)

*AIAA Journal, October 1986, Vol. 24, No. 10, pg 1709-1711, "Formulas for Venting or Charging Gas from a Single Volume", H.T. Yang

Rotating Equipment / Fracture Control

- Rotating Machinery consists of linear and rotating motors
- All Rotating Machinery is shown to have low kinetic energy significantly less than the fracture critical limit of 14,240 ft-lbs

- Fracture Control is implemented on all safety critical hardware using the requirements defined in NASA-STD-5019
- All pressurized components will be non-fracture critical using the disposition of Non Hazardous Leak Before Burst NHLBB
- All safety critical fasteners will be non-fracture critical using the failsafe disposition.

Thermal Analysis

Bill Dial – Thermal Analyst

ACME Thermal Analysis

Thermal Analysis Requirements

- Safety Requirements - Touch Temperature less than 45°C
  - SSP 51700, 220, 2d.
  - FCF-IDD-CIR-ACME, 14.2.0
- Facility Requirements
  - FCF-IDD-CIR-ACME, 5.1.1.3.1 The UML/PIL cooling is limited to a flow rate of 1.4m^3/min.
  - FCF-IDD-CIR-ACME, 5.1.1.3.2 The UML/PIL air inlet has a temperature range of 20-30°C.
  - FCF-IDD-CIR-ACME, 5.1.1.3.3 The UML/PIL mounted component shall have a flow impedance that yields 100±12Pa.
  - FCF-IDD-CIR-ACME, 5.6.4 The cooling water return shall be less than 49°C
- Mission Success
  - GLPR 7120.5.30, 3.7.4.

Thermal Analysis Report: ACME-ANA-003 (DRAFT)

ACME Thermal Analysis

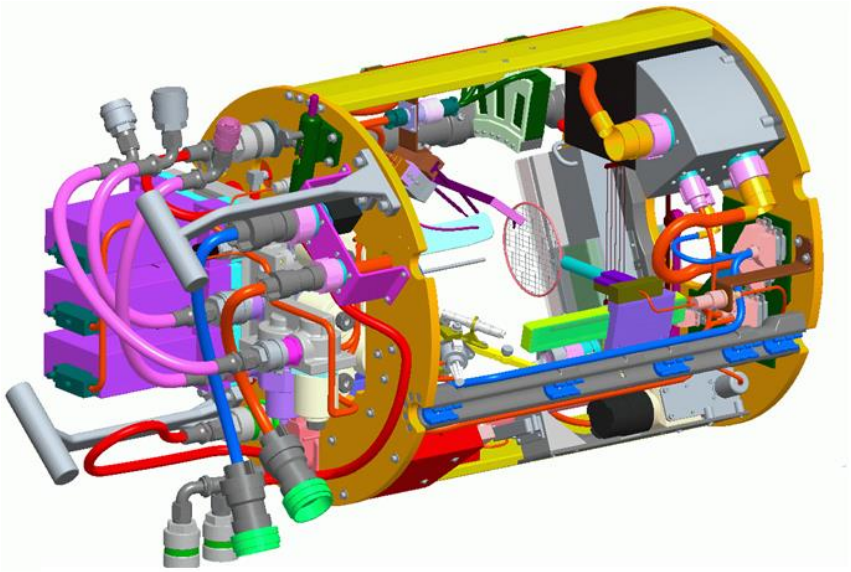
Thermal Analysis Approach

- There are three primary packages analyzed
  - Insert Assembly, S1101MFA1000
  - Camera Package, S1101MFA2000
  - Avionics Package, S1101MFA3000

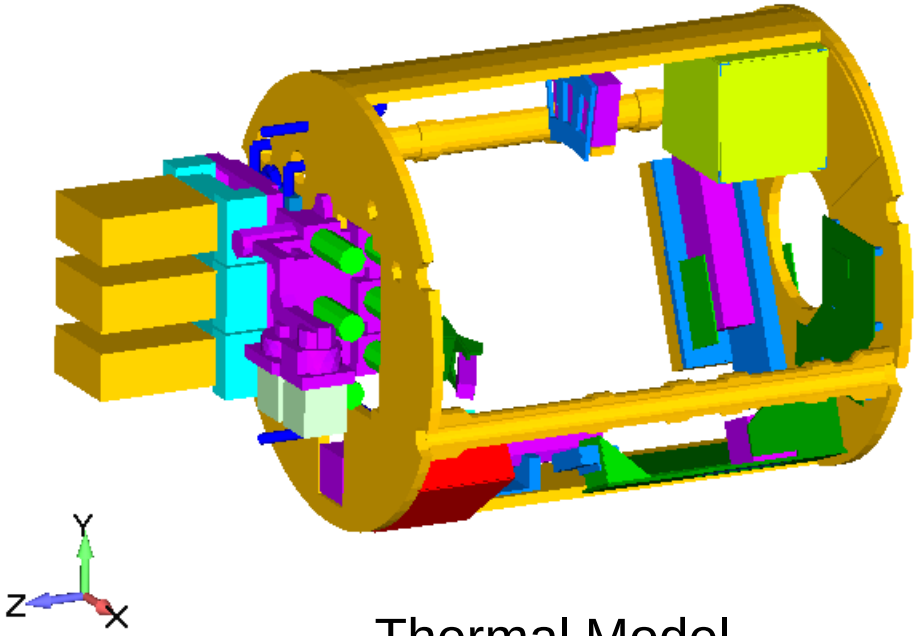
- COTS Parts - PC104 Stack & Other COTS PCBs
  - Operational temperatures are based on ambient environment in 1g. Verify the forced air is equal to or greater than free convection.
- Custom Electronic Components
  - Calculate the specific temperatures on these components to verify operating conditions are acceptable.
- Thermal Uncertainty Margin
  - Use SMC-TR-06-11 (MIL-STD-1540) as a guideline; thermal margin on components is 11°C

ACME Thermal Analysis

Insert Assembly, S1101MFA1000



Pro-E Model



Thermal Model

Primary Heat Dissipaters:

Solenoid Valves (Qty. 10), Mass Flow Controllers (MFC), DNC Cube, Igniter Converter, Upstream and Downstream PCBs.

ACME Thermal Analysis

Insert Assembly, Heat Loads Summary

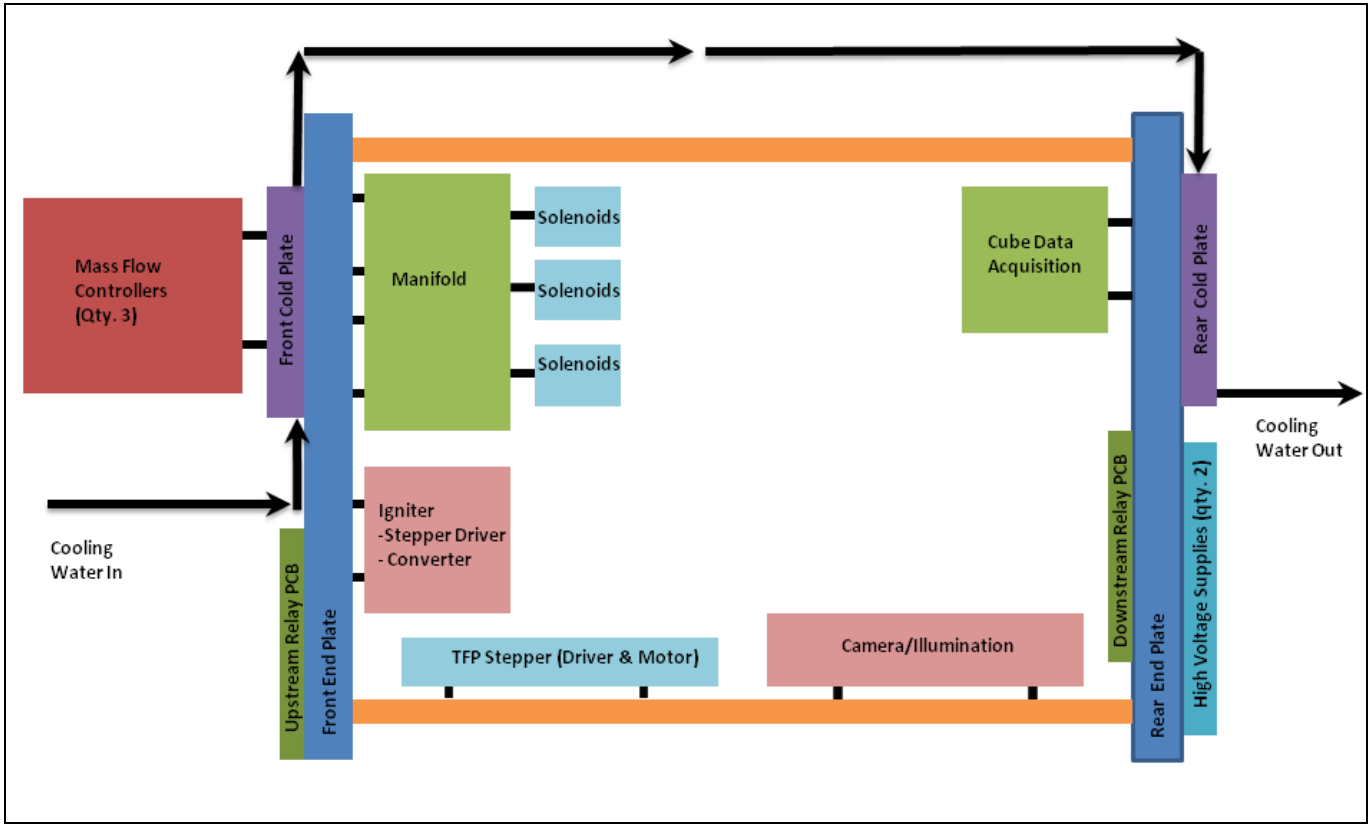
ID	Component	Quantity	Typical Power Per Unit (W)	Typical Power Total (W)	Maximum Operating Temperature (°C)
Insert Assembly [S1101MFA1000]					
1	DNA CUBE – Processor PCB	1	2.2	2.2	103[b]
	DNA CUBE – 'NIC' PCB	1	2.2	2.2	105[b]
	DNA CUBE – Al-225 PCB	1	4.29	4.29	100[b]
	DNA CUBE – AO-333 PCB	1	3.3	3.3	101[b]
	DNA CUBE – Al-207 PCB	1	2.42	2.42	103[b]
2	Igniter Motor	1	0.19	0.19	130
3	Driver (Igniter Stepper)	1	0.1	0.1	70
4	Igniter Converter	1	6.75	6.7	100
5	Solenoid Valves	10	3.0	30	200
6	Mass Flow Controllers	3	7.5	22.5	50
7	High Voltage Supplies	2	7.2	14.4	65
8	Illumination	1	0.3	0.3	100
9	Camera	1	2.64	2.64	50[a]
10	TFP Stepper Motor	1	0.48	0.48	100
11	TFP Stepper Driver	1	0.1	0.1	130
12	PMT	3	0.62	1.86	50
13	Upstream Relay PCBs	1	5.16	5.16	125
14	Downstream Relay PCBs	1	25.27	25.27	85
			Insert Assembly Total:	124.16	

a – Maximum temperature is for component and ambient.

b – Component case temperature based on testing with CUBE fan running.

ACME Thermal Analysis

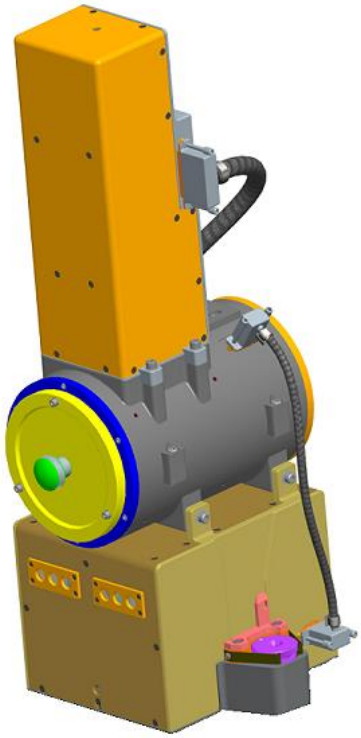
Insert Assembly, Heat Flow Diagram

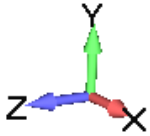


Cooling Water: FCF-ICD-CIR-ACME, 5.6.2 & 5.6.3
Mass flow rate = 25lbs/hr, Inlet water temperature = 16.0-18.3^{o}C

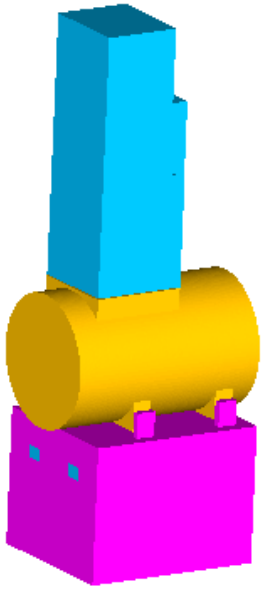
ACME Thermal Analysis

Camera Package, S1101MFA2000





Pro-E Model



Thermal Model

- Heat Dissipation Path: Air cooling from CIR (UML)
- Total Heat Load: 14.61W
- Cooling Air inlet temperature = 30°C
- Pressure Drop Set to 100Pa

ACME Thermal Analysis

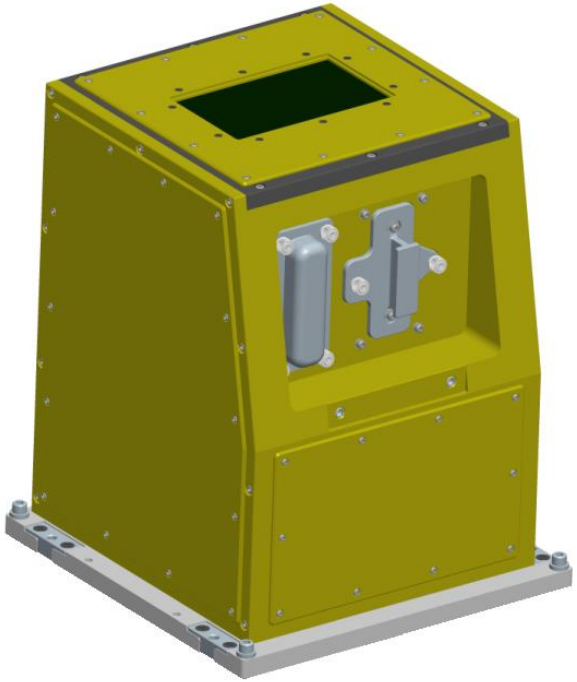
Camera Package, Heat Load

Component	Quantity	Typical Power Per Unit (W)	Typical Power Total (W)	Maximum Operating Temperature (°C)
Camera Package [S1101MFA2000]				
Camera Motor	4	0.067	0.267	70[a]
Camera	1	3.5	3.5	50[a]
Camera Stepper Driver PCB				
Stepper Driver	4	0.74	2.96	150
Voltage Regulator	1	0.6	0.6	125
Misc. Smeared		0.081	0.081	--
Camera Control Power Distribution PCB				
EMI Filter	1	0.31	0.31	100
DC-DC Converter	1	1.00	1.00	100
Misc. Smeared		0.024	0.024	--
Camera Power PCB [COT]	1	2.87	2.87	85
Motion Control PCB [COT]	1	3.0	3.0	70[a]
		Camera Package Total:	14.61	

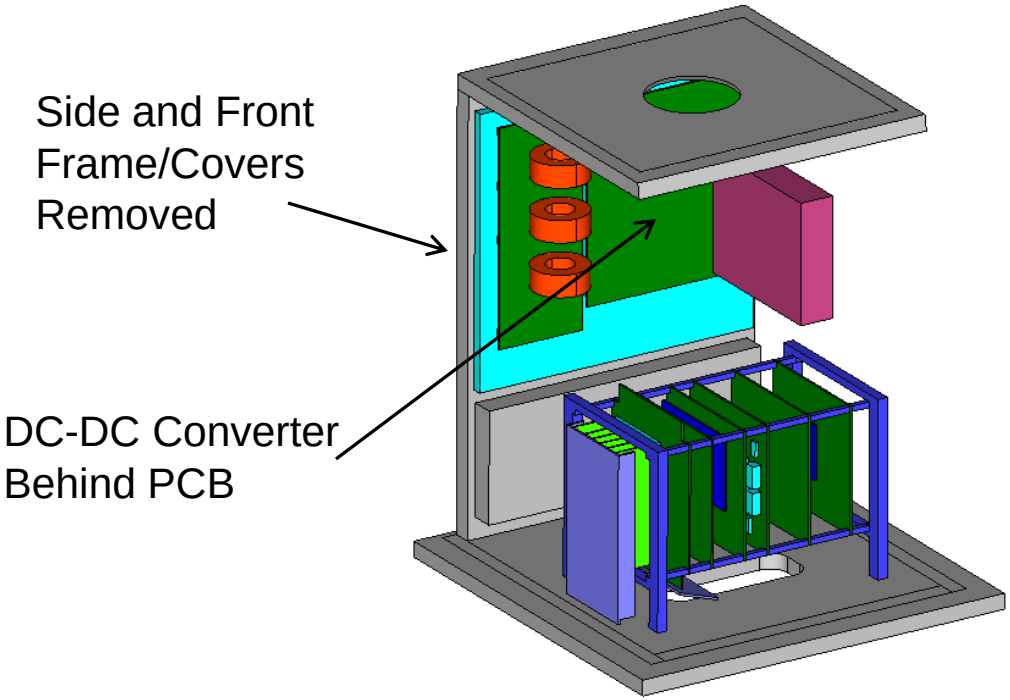
a – Maximum temperature is for component and ambient.

ACME Thermal Analysis

Avionic Package, S1101MFA3000



Pro-E Model



Thermal Model

Heat Dissipation Path: Air cooling from CIR (PIL), T_{inlet}=30°C, ΔP=100Pa
Total Heat Load: 92.25W
Primary Heat Dissipaters: PC104 Stack & DC-DC Converters/Filter

ACME Thermal Analysis

Avionic Package, Heat Loads

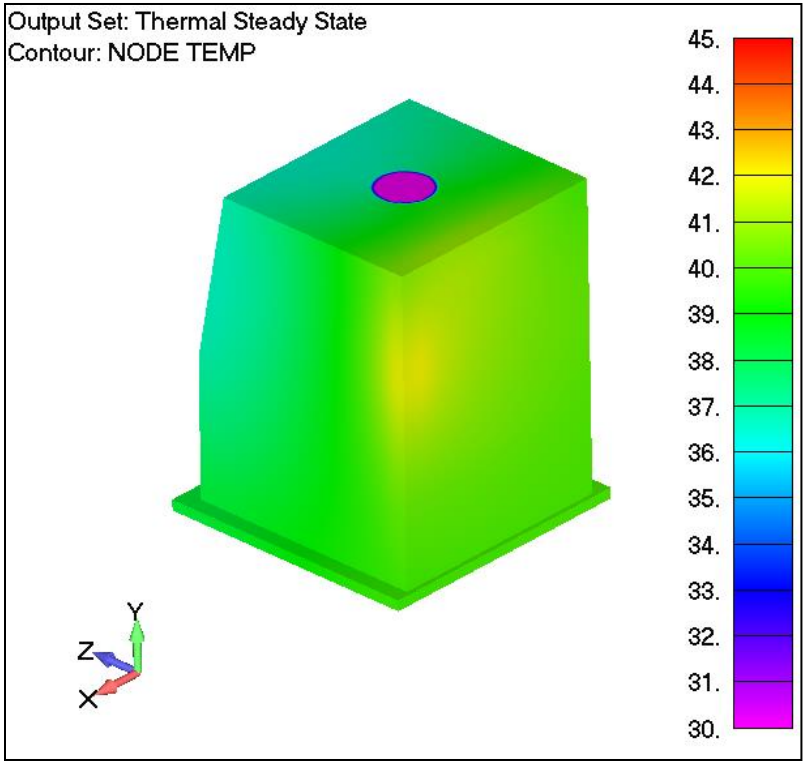
Component	Quantity	Typical Power Per Unit (W)	Typical Power Total (W)	Maximum Operating Temperature (°C)
Avionic Package [S1101MFA3000]				
Power Distribution Assembly				
EMI Filters (Toroids)	3	1.293	3.88	130
DC-DC Converter, U1	1	1.56	1.56	100
DC-DC Converter, U2	1	1.56	1.56	100
DC-DC Converter, U3	1	14.32	14.32	100
EMI Filter	1	4.67	4.67	100
Solid State Hard Drive	1	0.16	0.16	70[a]
PC104 Stack – CPU	1	32.5	32.5	100
PC104 Stack – Ethernet	1	3.0	3.0	85[a]
PC104 Stack – Power	1	20.6	20.6	85[a]
PC104 Stack – Serial	1	3.0	3.0	70[a]
PC104 Stack – Motion Control	1	3.0	3.0	70[a]
PC104 Stack – VGA	1	4.0	4.0	85[a]
		Camera Package Total:	92.25	

a – Maximum temperature is for component and ambient.

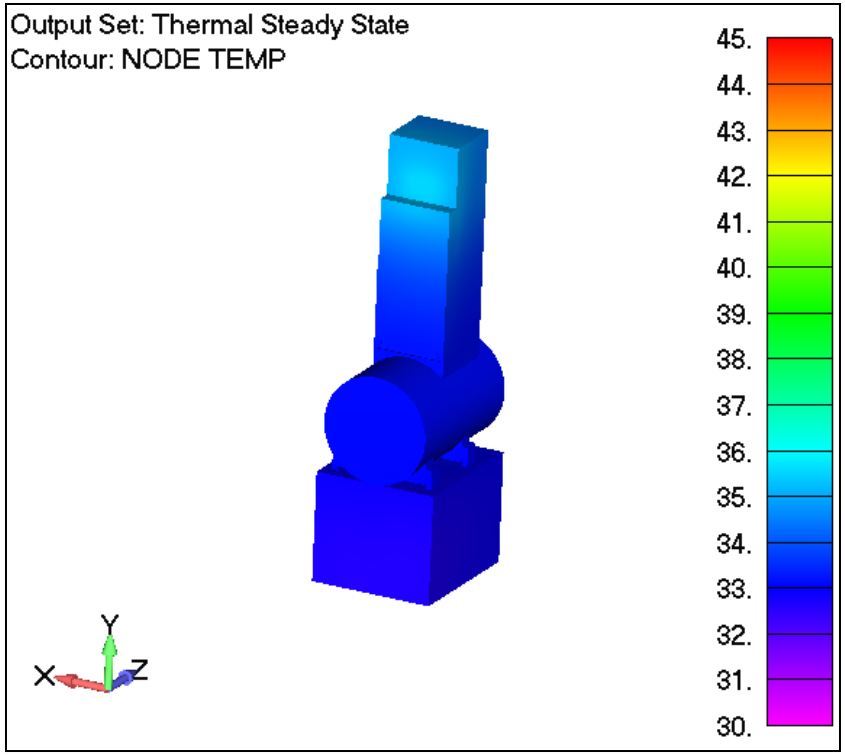
ACME Thermal Analysis

Glenn Research Center

Touch Temperature



Avionic Package: 42.0°C
Air Flow Rate: 1.05 m^3/min



Camera Package: 35.5°C
Air Flow Rate: 0.14m^3/min

- Insert Assembly is not applicable to crew loading.

ACME Thermal Analysis

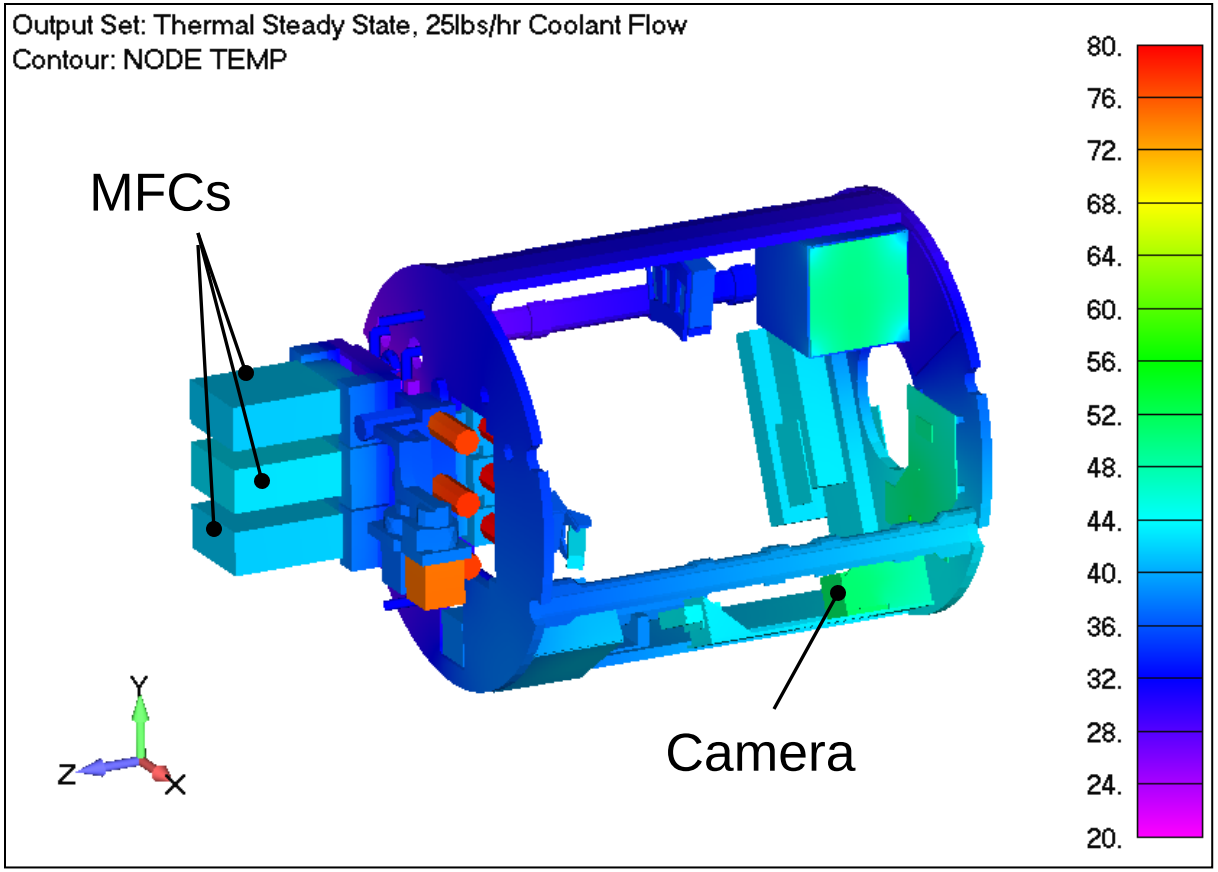
COTS PCBs – Thermal Results

- Compare Air Velocities: Forced versus Free Convection
  - Using Grashof number and Reynolds number ratio to estimate equivalent air velocities. Velocity = 0.33m/sec for PCBs
- Assess Air Flow in Camera Package
  - Velocity assessed at 0.66m/sec
  - All COTS components have case temperature with more than 11°C margin on the PCBs maximum ambient temperature.
- Assess Air Flow in Avionic Package
  - Velocity assessed at 0.66m/sec
  - All COTS components have velocities greater than 0.66m/sec

- DNA CUBE PCBs
  - No air flow
  - CUBE PCBs sub-modeled and tuned to test data.
  - Minimum thermal margin: 21.0^{o}C on the Al-225 board.

ACME Thermal Analysis

Other COTS Components – Thermal Results



Insert Assembly

- Two COTS components fail to meet the 11°C margin guideline from SMC-TR-06-11
  - Camera: -1.1°C
  - MFCs: 7.1°C
- All other COTS components have margins greater than 11°C.
- Possible Mitigation
  - Increase water flow rate
  - Refine thermal loads
  - Revise thermal allowable based on ambient conditions in 1g.

ACME Thermal Analysis

Custom Components – Thermal Results

- All Custom Components have large thermal margins
  - Insert Assembly
    - Upstream Relay PCBs: 28.2°C
    - Downstream Relay PCBs: 44.8°C
  - Camera Package
    - Camera Stepper Driver PCB: 64.4°C
    - Camera Control Power Distribution PCB: 61.2°C
  - Avionic Package
    - Power Distribution Assembly: 56.3°C

ACME Thermal Analysis

Conclusion

- All other components with the exception of two have more than 11°C thermal margin.
  - Exceptions:
    - Insert Assembly, Camera
    - Insert Assembly, Mass Flow Controller (MFCs)
- The ACME hardware meets the touch temperature requirement of SSP 51700.
- The air flow rates through the camera package and the avionic package were less than 1.4m^{3}/min with a 100Pa pressure drop across the package.

Actions

- Addition work is needed to resolve the thermal margins on the mass flow controllers (MFCs) and the Camera in the Insert Assembly.
  - Follow the mitigation approached defined earlier.

CIR Capabilities Used by ACME

Chris Rogers – CIR/ACME Integration Lead

Combustion Integrated Rack (CIR) Overview

Goal

Use the space environment as a laboratory for strategic research & technology experiments in physics, chemistry and biology to support Agency goals & the Nation's vision new for space exploration.

Objectives

- Develop the Combustion Integrated Rack (CIR) and the Fluids Integrated Rack (FIR) to support physical and biological research on board the ISS.
- Develop the Ground Infrastructure, Integration, Operations, Logistics & Maintenance plans required for FCF sustaining engineering and operations.
- CIR provides the only large environmental chamber (100 liters) on ISS in which the space environment, Lunar, Mars or other desired environment may be simulated (i.e., pressure, temperature, gaseous atmospheres, particulates, etc.) for long duration, on-orbit investigations and technology validations.
- CIR provides easily reconfigurable diagnostics, customizable software, passive rack-level vibration isolation, and other subsystems that will support a wide range of investigations



Combustion Integrated Rack

Combustion Integrated Rack (CIR) Overview

Glenn Research Center

Fuel/Oxidizer Management Assembly (FOMA)
- Gas Distribution
- Exhaust Vent

Environmental Control (ECS)
- Air Thermal Control
- Fire Detection & Suppression
- Water Thermal Control
- Gas Interfaces (GN2, VES)

International Standard Payload Rack (ISPR)

Combustion Chamber

Rack Door

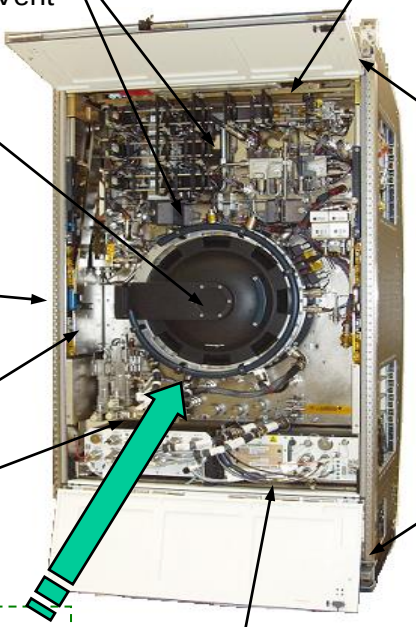
Image Processing and Storage

SAMS RTS

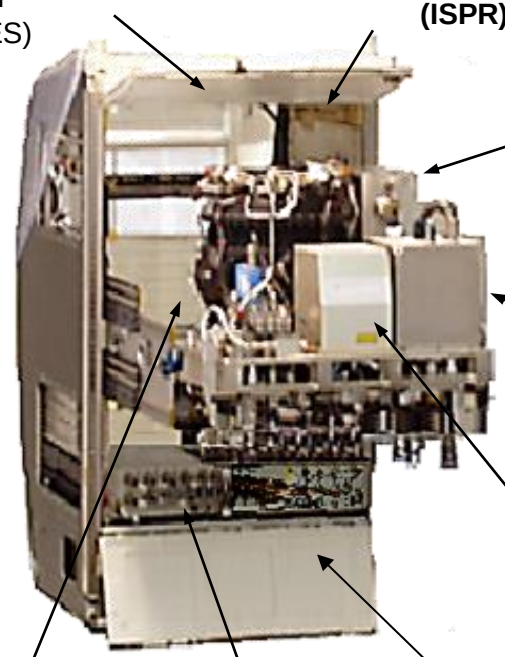
FOMA Control Unit (FCU)

Optics Bench Slides

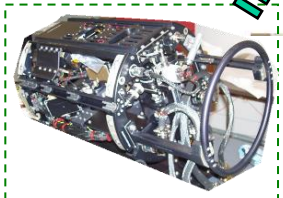
Optics Bench



Passive Rack Isolation Subsystem (PaRIS)



PI Avionics



ACME CIA

Science Diagnostics
- Illumination Package
- Low Light Level (2 Units)
- High Bit Depth Multi-Spectral
- High Frame Rate/High Resolution

OR

Experiment Specific Diagnostics

Electrical Power Control Unit (EPCU)



Laptop Computer

Input/Output Processor (IOP)

FCF Common Element
CIR Element
PI Hardware Element

ZIN Technologies

CIR Sub-rack Payload Relationship

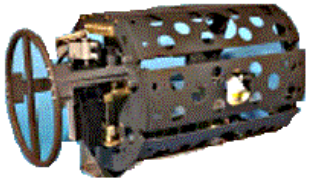
Glenn Research Center

The Combustion Integrated Rack (CIR) includes support subsystems and combustion diagnostics commonly needed for combustion investigations and supplements the laboratory with unique science hardware developed for each Principal Investigator (PI).

Payload specific and multi-user hardware customizes the CIR in a unique laboratory configuration to perform research effectively.



Advanced Combustion via Microgravity Experiments Insert Assembly





Payload Specific Hardware

- Unique hardware components
- Specific Diagnostics
- Unique PI fuels

ACME Chamber Insert

- Platform for exchangeable burner:
  - Gas Delivery
  - Retracting ignitor
  - Electric field generator
- Infrastructure that uniquely meets the needs of PI experiments
- Unique science requirements

FCF Combustion Integrated Rack

- Power Supply
- Avionics/Control
- Structural Support & Power
- Common Diagnostics and Illumination
- Environmental Control
- Data Processing/Distribution
- Fluids Control/Distribution
- Combustion Containment

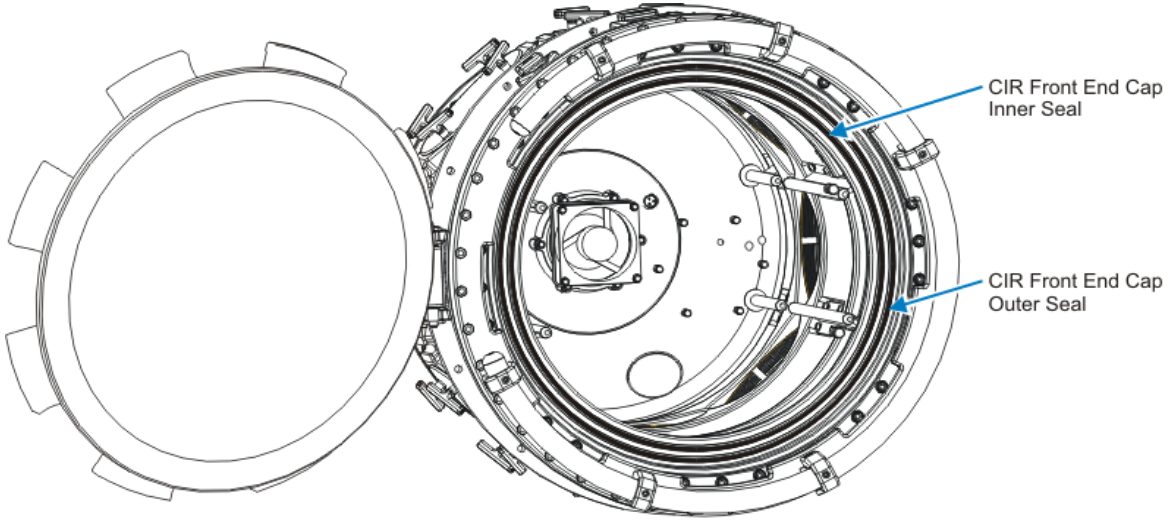
CIR Combustion Chamber

Combustion Chamber General Info:

- PIs may operate from .02 to 3.0 ATM
  - CIR Combustion Chamber rated 0 to 9 ATM
- CIA useable: 396 mm (W) x 600 mm (L)
- Internal free volume: 105 liters
- Maximum Oxygen concentration of 30%
- 6 Fluid Connections include (High Pressure Gas (1) GC sample port (1), SS Coolant inlet and outlet (2), Fuel (1) and Static Mixer (1).

Combustion Chamber Power and Data Interface includes:

- JCTC1– Provides 40 pairs of #22 gauge wire between IRR and PI Avionics connector through the Optics Bench
- JCTC2 – Provides 27 pairs of #20 wire pairs between IRR and PI Avionics connector through the Optics Bench
- JCTC3– Provides 2 Fiber Optic, single-mode 9.0 core/125 Cladding, 2 RG-179 coaxial cables and 5 pairs of #16 gauge wires to the PI Avionics location through the Optics Bench.
- JCTC4 – Provides 49 pairs of #22 gauge wire pairs between IRR and PI Avionics connector through the Optics Bench



Combustion Chamber Atmosphere Mixing

The oxidant MFC can provide up to 30 slpm (based on nitrogen (N2)) flow. The oxygen gas concentration is a function of the gas bottle size, as listed below.

- 2.25 liter bottle provides a maximum of 50% O2, balance diluent at 2000 psia.
- 3.8 liter bottle provides a maximum of 30% O2, balance diluent at 2000 psia.

A Nitrogen manifold can supply inert gas types such as N2, He, and CO2, as well as oxidant blends below 30% oxygen, to the chamber. In addition, this manifold provides the interface for the utilization of the ISS supplied N_2. This manifold also can directly route the pressurized gas to the chamber, by-passing the regulator. The N2 MFC can provide up to 30 slpm (based on nitrogen (N_2)) flow.

CIR Provided Equipment

FCF/CIR diagnostics, equipment and resources to be provided to ACME are documented in the CIR to ACME Integration Agreement (FCF-IA-MV-ACME)

TABLE 1 CIR EQUIPMENT REQUIRED BY THE PAYLOAD

FCF Provided Equipment	CIR Quantity Available	CIR Quantity Required by ACME	ACME Provided
CIR HiBMS Imaging Package	2	2	0
CIR HFR/HR Imaging Package	0	0	0
CIR LLL-IR Imaging Package	0	0	0
CIR LLL-UV Imaging Package	1	1	0
CIR Window Coated	7	7	0
CIR Window Uncoated	1	1	0
CIR Illumination Package	1	1	0
FCF IPSU	4	3	0
FCF IPSU-Analog	2	0	0
CIR IPSU Adapter	2	2	0
CIR IPSU Corner Adapter	1	0	0
PaRIS	1	1	0
90mm Diameter Lens (HiBMs)	1	1	0
80mm Diameter Lens (UV)	1	1	0
CIR Gas Chromatograph	1	1	0
CIR N2 Bypass Hose*	0	1	1
CIR N2 Bypass Pressure Regulator*	0	1	1

*The CIR N2 Bypass Hose and the CIR N2 Bypass Pressure Regulator will be funded by ACME but procured, tested and verified by CIR per line item F in Section 3.2.3

TABLE 2 CIR CONSUMABLES REQUIRED BY THE PAYLOAD

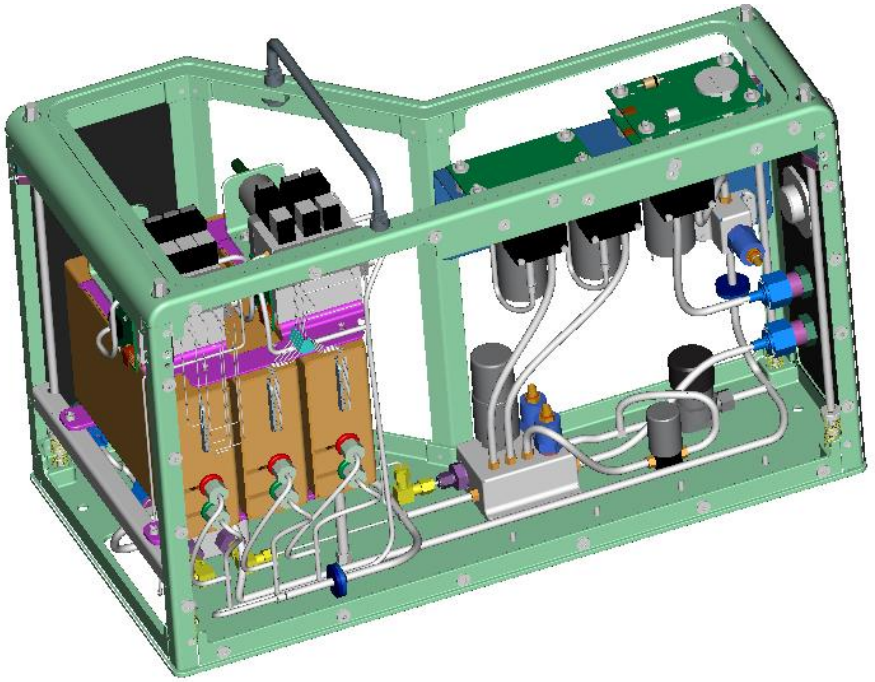
FCF Provided Equipment	CIR Quantity Available	CIR Quantity Required by ACME	ACME Provided
CIR Adsorber Cartridge	2	1	0
CIR Manifold # 4 Bottle 1.00 L	TBD*	14	0
CIR Manifold # 2 Bottle 2.25 L	TBD*	20	0
CIR Manifold # 1 Bottle 2.25 L	TBD*	2	0
CIR Manifold # 2 Bottle 3.80 L	TBD*	10	0
CIR High Percentage Oxygen Bottle (1.00 L)	TBD*	0	0
CIR GC Argon Bottle	5	3	0
CIR GC Helium Bottle	5	3	0
CIR GC Check Gas Bottle	5	3	0
FCF I/O Processor Hard Disk Drive	4	2	0
CIR 150cc	TBD	7**	0

*Number of bottles available will be dependent upon how many bottles can be returned from on orbit.

**Based upon assumption that 5 test points can be conducted per bottle. There are 35 inverse flame test points for ACME.

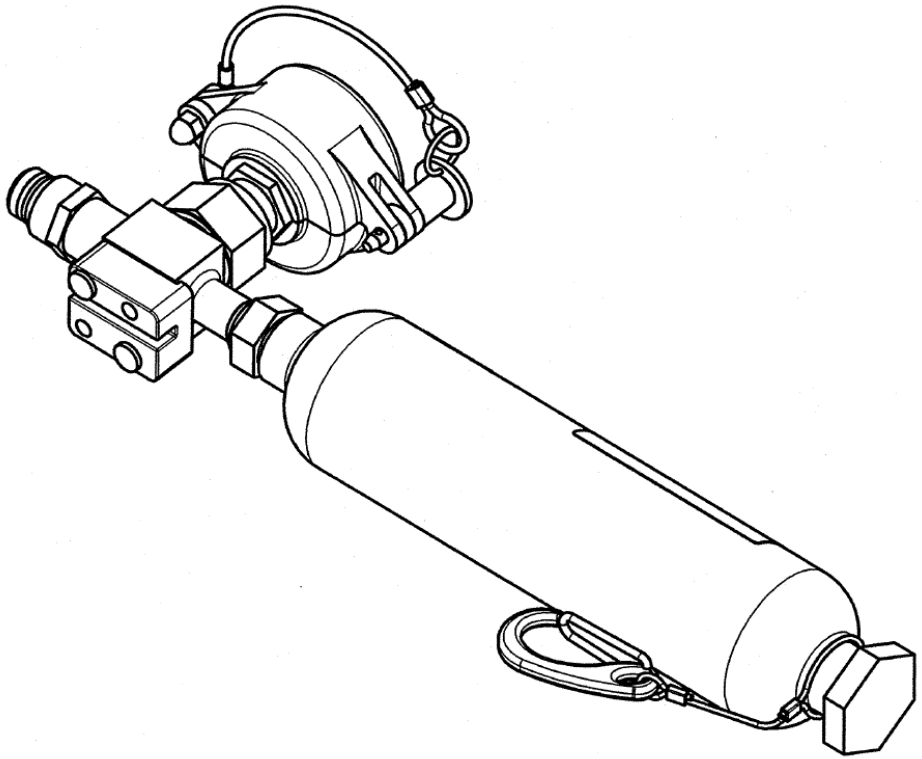
CIR GCIP

- Design based on re-packaged commercial Agilent Micro GC instrument
  - Un-modified vendor supplied software for analysis
  - 2 PCB controllers
  - 3 detector columns
    - Column A – Molecular Sieve with Backflush
    - Column B – PoraPlot Q
    - Column C – OV1701
- Structural mounting, air-cooling, power and data interfaces through Optics Bench
- Sample inlet and vent I/F's through FOMA



Small FOMA Gas Bottle

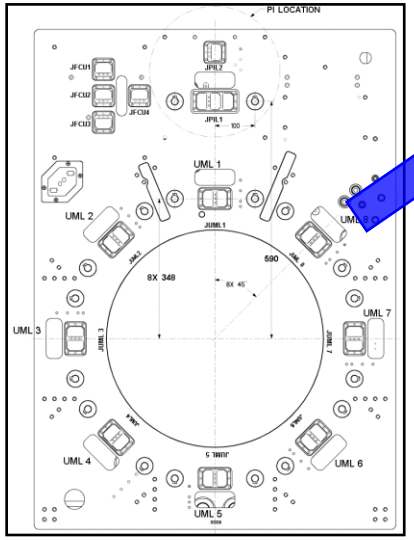
- Necessary as a control to hazard FCF-ACME-05, hazard control 1.1
- Based on analysis driven by hazard control the bottle properties are:
  - 150cc
  - 1815 psia
  - 1.5 x MDP
  - DOT 3E
  - oxygen mixtures only
  - Based on current FOMA bottle design

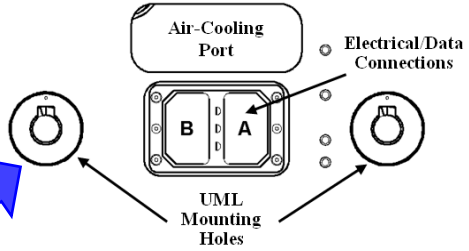


CIR Diagnostics – Optics Bench/UML

UML - *Universal Mounting Location*

- 8 locations on rear of CIR Optics Bench
- Correspond to Chamber Windows

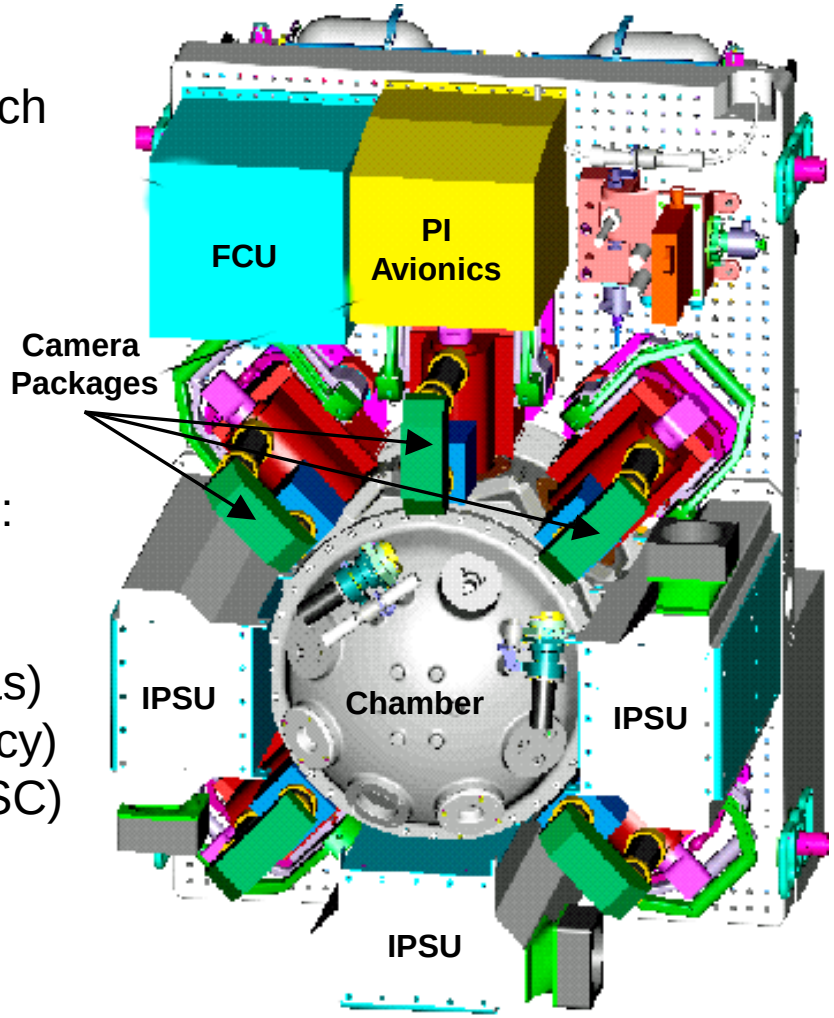




UML provides interfaces for:

- Mechanical w/ Air Cooling
- 28 VDC power
- Ethernet (cmd, control, h&s)
- Sync Bus (± 1 Hz accuracy)
- Analog Video (RS170/NTSC)

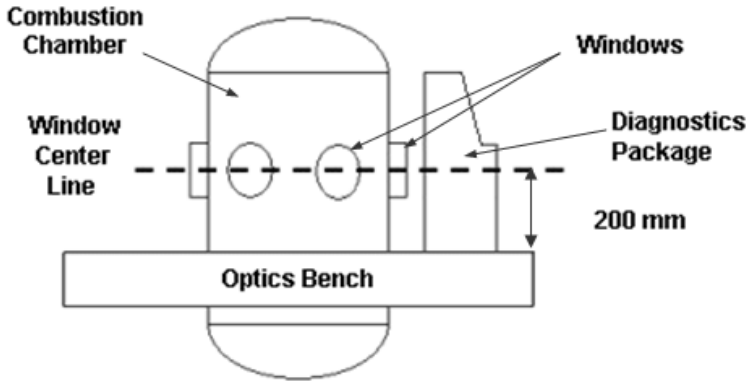
- PI Mounting Location – Additional unique UML for PI Avionics Package



CIR Diagnostics – Package Form

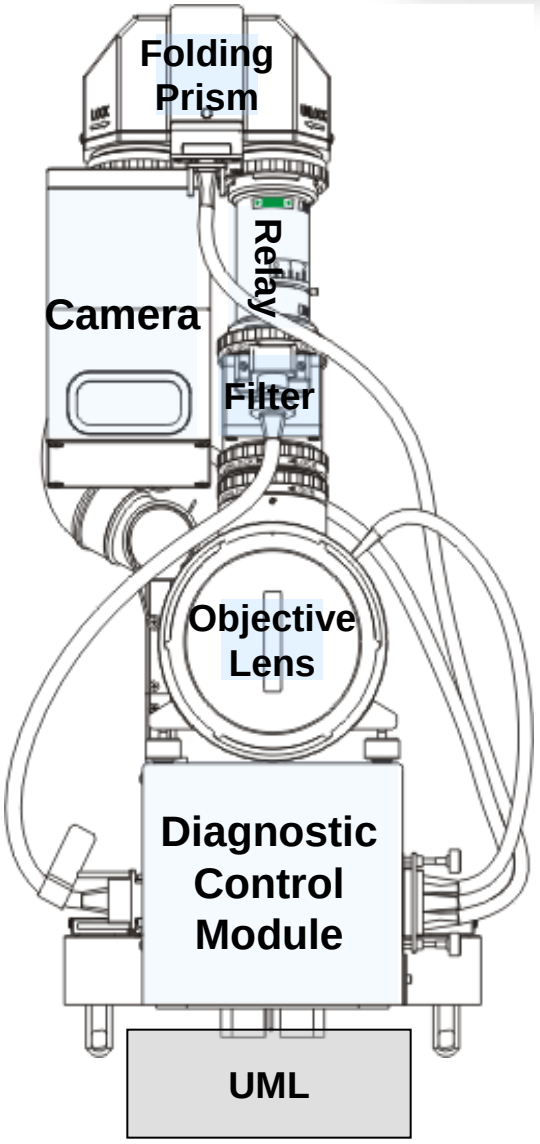
CIR Diagnostics Packages are mounted on the Optics Bench at one of the 8 UML locations around the Chamber.

Diagnostics Packages are designed to align with the centerline of the Chamber Windows.



Modular build format allows for on-orbit reconfiguration of components.

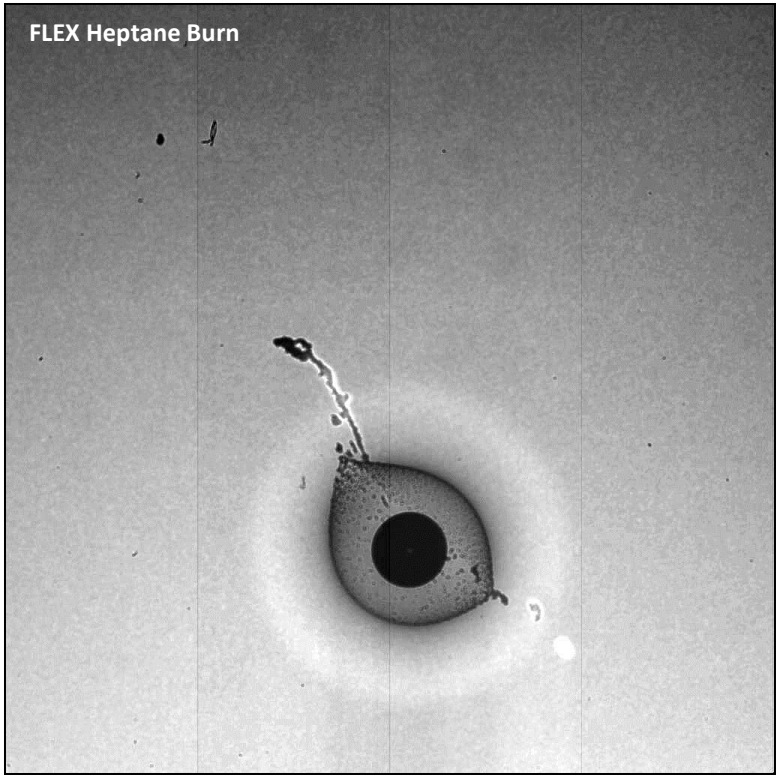
Changes require Crew Time!



CIR Diagnostics – HiBMs/Illumination

HiBMs Package

- Used by ACME for Soot Volume Fraction View
- Based on 1024 x 1024 CCD, 12 bit output
- Frame Rate: 30 fps (60 fps via Binning)
- Modular Relay Lens determines FOV
  - 30 mm, 50 mm, *90 mm (not on orbit)*
- Narrow Band Filter @ 653 nm to isolate background illumination



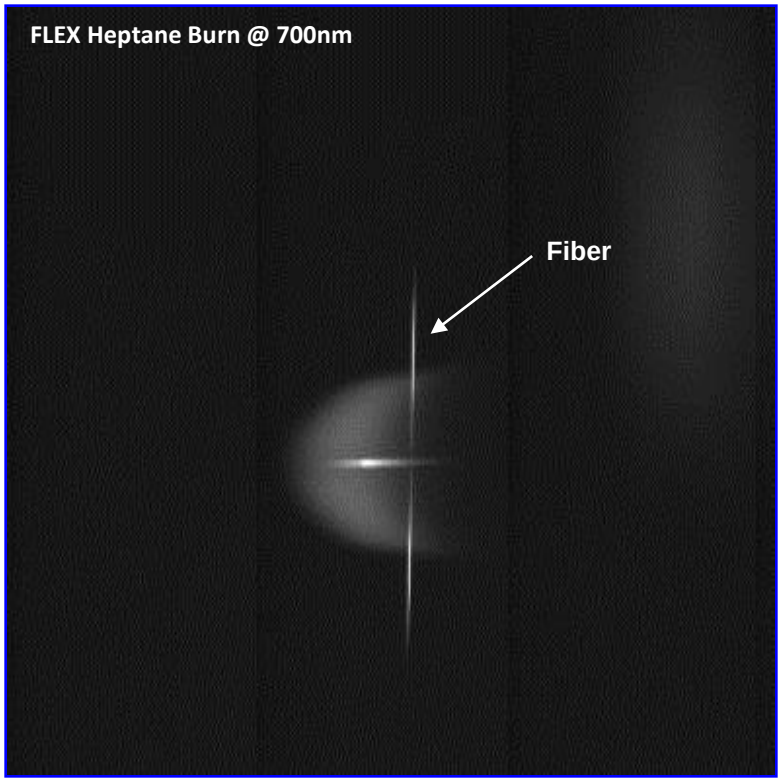
Illumination Package

- Used as backlight for HiBMs Package
- Laser Diode @ 653 nm provides monochromatic, uniform background

CIR Diagnostics – HiBMs-2

HiBMs-2 Package

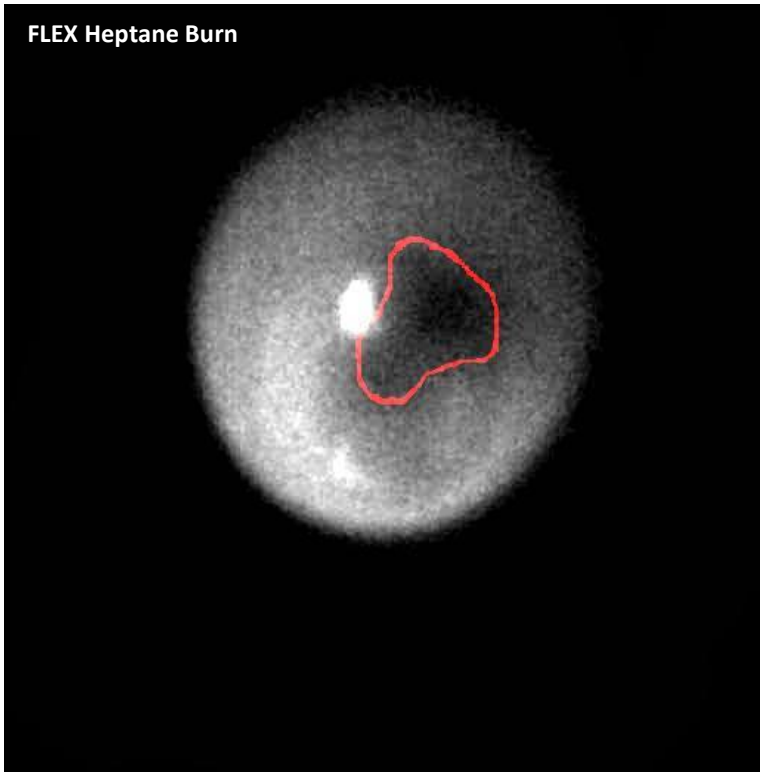
- Used by ACME for Soot Temperature Imaging & Thin Filament Pyrometry View
- Based on 1024 x 1024 CCD, 12 bit output
- Frame Rate: 30 fps (60 fps via Binning)
- Modular Relay Lens determines FOV
  - 30 mm, 50 mm, *90 mm (not on orbit)*
- Liquid Crystal Tunable Filter (LCTF) is switchable - Wavelengths: 650 – 1050 nm
  - Allows for 2 Wavelength Pyrometry
- Broadband Filter (BK7) also available (in stowage) for higher light throughput.



CIR Diagnostics – LLL-UV

LLL-UV Package

- Used by ACME for Ultraviolet Imaging View
- Based on 1024 x 1024 CCD, 12 bit output
- Frame Rate: 30 fps (60 fps via Binning)
- Modular Relay Lens determines FOV
  - 50 mm, *80 mm (not on orbit)*
- Used w/ integrated Gen-II UV Intensifier
- Flame Imaging Filter Options (on orbit):
  - OH* @ 310 nm
  - CH* @ 431 nm



CIR Diagnostics – IPSU

IPSU (Image Processing & Storage Unit)

- Currently 1 IPSU is required for each Camera Package
- Mounted on the Optics Bench at one of the 8 UML locations around the Chamber.
- Camera data interface based on custom fiber-optic link: Serial Data Link (SDL)
  - SDL interface is not compatible w/ any off-the-shelf Cameras
  - Requires Fiber-optic cable from Camera to IPSU
- Images to the ground is 2 step process:
  - Transfer: IPSU to IOP (CIR main computer) ~ 10 mbps
  - Downlink: IOP to Ground via HRDL (high-rate data link) ~ 20 mbps



CIR Diagnostics – IPSU-G

IPSU-G (GigE upgrade to IPSU)

- Currently in development by CIR & scheduled to fly in early 2016.
- Supports both GigE (GigE Vision Std) & USB-3 Camera interfaces.
  - Supports up to 4 cameras
  - Not compatible w/ current CIR Cameras
- Mounts on Optics Bench at UML 1, 3 or 7.
- Includes a direct HRDL interface to ISS.
- ACME Color Camera compatible w/ IPSU-G (Images can be Downlinked via HRDL directly)
- Images from existing IPSUs to the ground is still a 2 step process, but faster:
  - Transfer: IPSU to IPSU-G ~ 32 mbps
  - Downlink: IPSU-G to Ground via HRDL (high-rate data link) ≥ 20 mbps

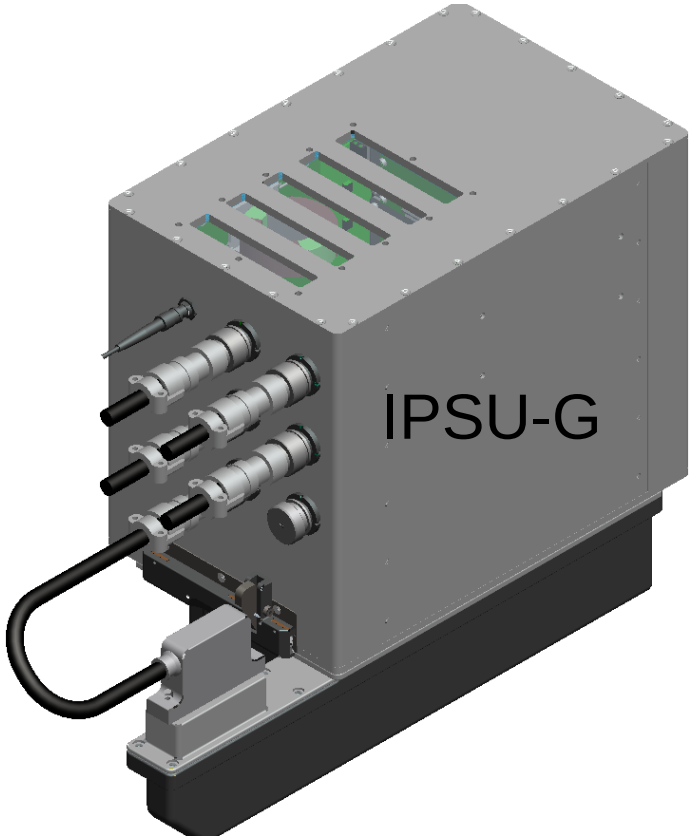


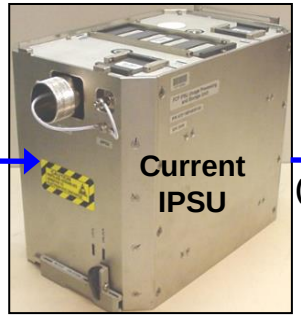
Image Transfer/Downlink – Current

Glenn Research Center



Serial Data Link (SDL)

Storage



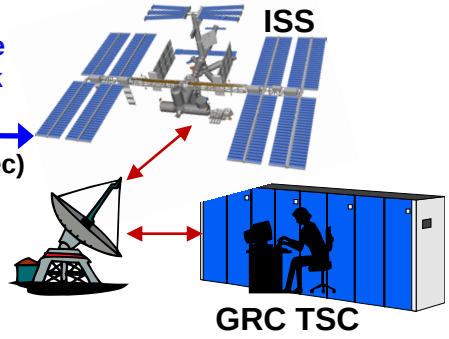
Ethernet (10 mbit/sec)

Transfer



High Rate Data Link (HRDL) (20 mbit/sec)

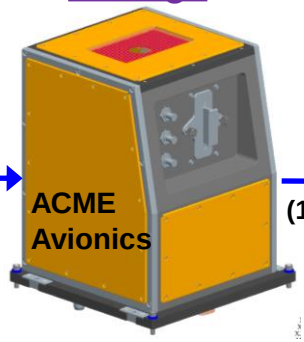
Downlink





Gigabit Ethernet (GigE)

Storage



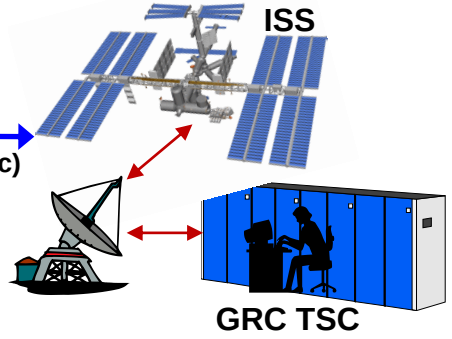
Ethernet (10 mbit/sec)

Transfer



High Rate Data Link (HRDL) (20 mbit/sec)

Downlink



Data Transfer & Downlink - 1 Test Point (60 sec. duration)				
Camera Package	Image Data (GB)	Transfer (min)	Downlink (min)	*Total Time (hrs)*
HiBMs	3.8	50.3	25.2	*1.3*
HiBMs-2	0.9	12.6	6.3	*0.3*
LLL-UV	0.9	12.6	6.3	*0.3*
ACME Color	2.8	37.6	18.8	*0.9*
Total	8.5	113.1	56.6	*2.8*

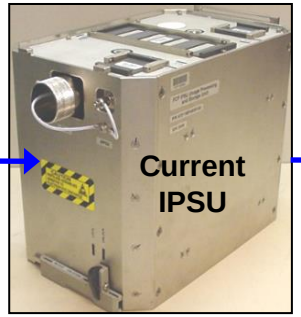
Image Transfer/Downlink – IPSU-G

Glenn Research Center



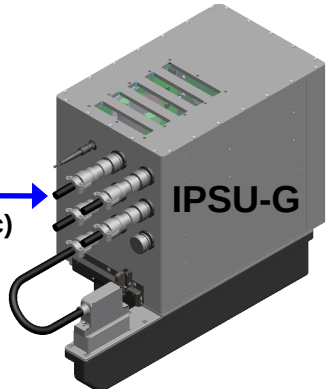
Serial Data Link (SDL)

Storage



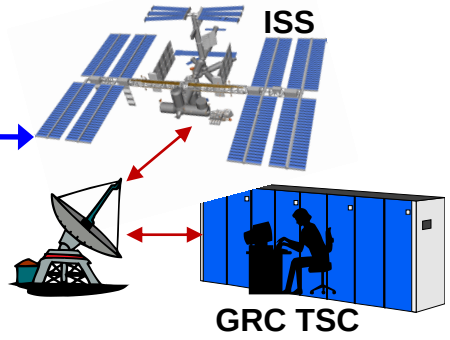
Ethernet (24 mbit/sec)

Transfer



High Rate Data Link (HRDL) (20 mbit/sec)

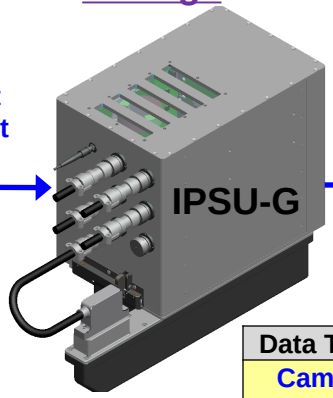
Downlink





Gigabit Ethernet (GigE)

Storage

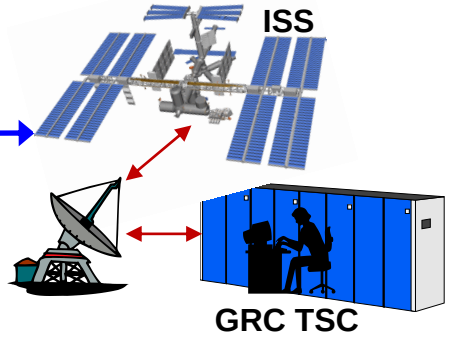


(No Transfer Required)

Transfer

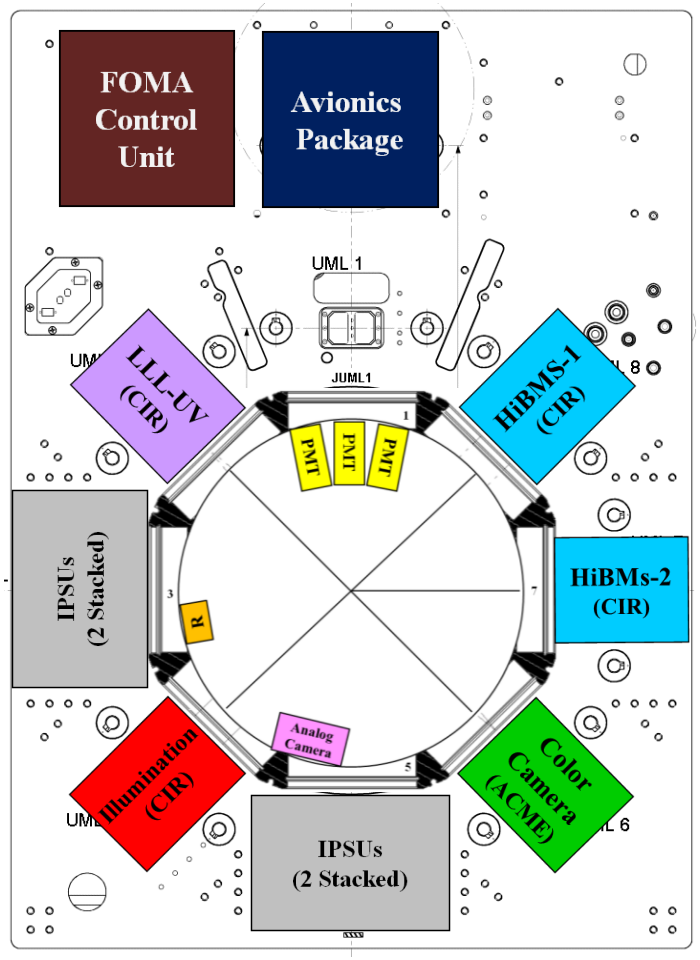
High Rate Data Link (HRDL) (20 mbit/sec)

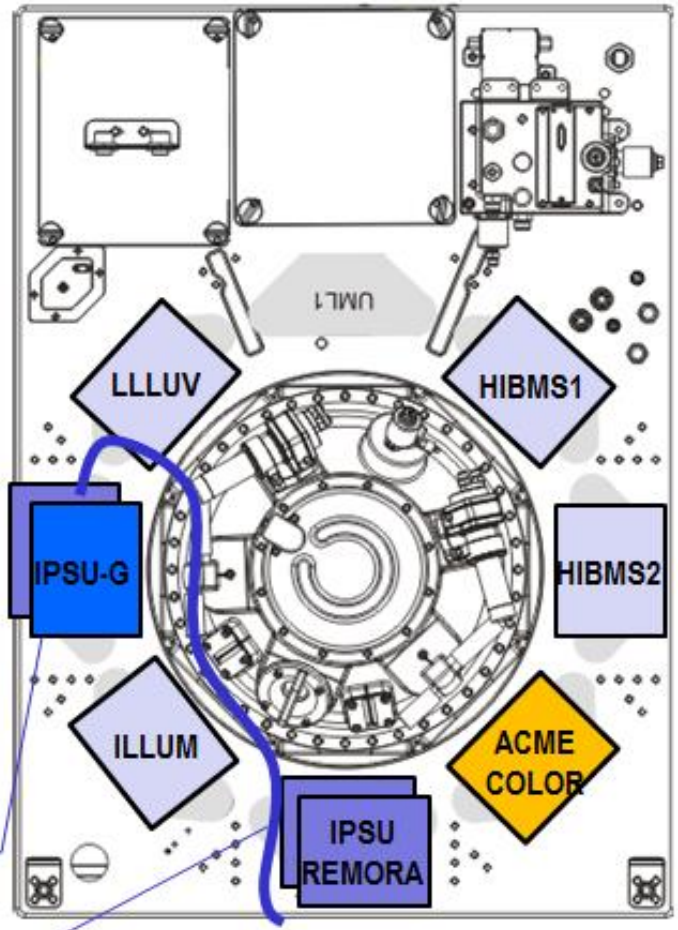
Downlink



Data Transfer & Downlink - 1 Test Point (60 sec. duration)				
Camera Package	Image Data (GB)	Transfer (min)	Downlink (min)	*Total Time (hrs)*
HiBMs	3.8	21.0	25.2	*0.8*
HiBMs-2	0.9	5.2	6.3	*0.2*
LLL-UV	0.9	5.2	6.3	*0.2*
ACME Color	2.8	0.0	18.8	*0.3*
Total	8.5	31.4	56.6	*1.5*

CIR Diagnostics – ACME Layout





IPSU-G stacked on top of IPSU Remora at UML 3

HRDL Cable

CIR Optics Bench (rear view)

ZIN Technologies

ACME to CIR Integration

Chris Rogers – CIR/ACME Integration Lead

FCF Utilization Process

- The FCF Utilization Process is defined in FCF-PLN-0875
  - Primary functions of the Integration process are:
    - Payload Planning
    - Mission Integration
    - Engineering Integration
    - Operations Planning
  - FCF/CIR provides the following resources
    - An integration liaison to assist with the Utilization Process
      - Integrating Sub-rack Payload into CIR
      - Integrating CIR/Sub-rack Payload to ISS
    - Services in support of Sub-rack Payload
      - CIR Provided Hardware (IPSUs, HiBMs, UV Imaging, IOP hard drives, etc)
      - Gas filling services
      - Launch Site Processing Support
      - TADs updates
        - Further resources provided by CIR and agreements between CIR and payload provider are documented in the CIR/Payload Unique Integration Agreement

FCF Utilization Process

- Payload Planning
  - Payload Kick-Off Meeting
    - Documents and Templates
  - Develop payload to CIR Integration Agreement (IA)
  - Develop payload to CIR Interface Control Document (ICD)
  - Collect payload Integration Agreement (IA) Addenda data inputs and combine inputs for FCF submittal per increment
  - Preliminary Payload Data Collection and Analysis
    - Utilization Tools
    - Payload Data Inputs
  - Payload-Unique Integration Documentation
    - IA Main Volume (MV)
      - Utilization Schedule
    - External Preliminary Interface Revision Notices (PIRNs)/ Change Requests (CRs)/Exceptions

FCF Utilization Process

- Engineering
  - Engineering Integration
    - FCF to Payload Compatibility Assessment
    - Ground Processing
    - Payload Testing, Reviews and Certification
    - Ground Operations
  - Sustaining Engineering
    - Logistics and Maintenance
  - Develop and maintain ACME to CIR integration schedule

ISS Payload Integration

- Payload Engineering Integration
  - Assist in requirement identification, verification and verification submittal
    - Includes:
      - identification of unique requirements for multiple launch vehicles
      - on-orbit requirements from ISS
      - Requirements unique to the payload
  - Assistance with Safety Reviews and Safety Data Packages
  - Issue Resolution
    - Identification of problems or issues at the ISS Program Level
    - Work directly with ISS Program Offices to mitigate and resolve issues.
  - Manifesting
    - Submit required manifest requests and associated data
    - Identify launch restrictions or packaging constraints unique to payload hardware

FCF Utilization Process

- Mission Integration
  - Integration Documentation
    - FCF IA with ISS
      - Payload Data Library Data Set
    - Interface Control Document (ICD)
    - Payload Verification Plan (PVPs)
    - Payload Tactical Plan (PTP)
    - Payload Increment 2-Pagers
    - Experiment Science Summary
    - Internal PIRNs/CRs/Exceptions
    - Collect payload Integration Agreement (IA) Data Set inputs from payload and combine inputs for FCF submittal per increment
  - Payload Launch Processing
    - Manifesting
    - Launch Vehicle Identification and Requirement Definition
    - Hardware Processing and Turnover

ISS Payload Integration

- Operations Integration Products
  - Training
    - Crew Training products and services to properly familiarize Astronauts with payload hardware, use and operation
    - Train ground support personnel for on-orbit operations
  - ISS Resource planning
    - Submit request for ISS Resource requirements necessary for payload operations
- Ground & On-orbit Operations
  - Launch/Landing Site Processing
    - Provides real time support in preparation for launch
    - Retrieval of time sensitive samples from landing site
  - Bench Reviews
    - Act on behalf of payload during final crew review and inspections prior to final packing
    - Verify required packing requirements met
  - Continued on-orbit support of operations
    - Perform real time operations
    - Resolve on orbit issues or problems as they arise

ISS Payload Integration

- ISS Payload Integration facilitates support from concept through on-orbit operations. Also known as the End-to-end Integration Process
  - Payload Planning
    - Perform initial planning and evaluate mission criteria
    - Define roles and responsibilities
  - Mission Integration
    - Planning and ISS Data Set Submittals
    - ISS Program Level schedules
      - Based on projected launch or hardware completion date
      - Identifies major milestones leading up to launch and operations
    - ISS Documentation
      - Provide access to ISS Documentation
      - Write and maintain payload unique documentation for the ISS Program

ISS Requirement Flow Down to ACME

ISS requirements applicable to CIR and CIR Sub-rack payloads from SSP 57000 and SSP 57008 (Common Transport) are documented in the CIR to Station ICD SSP 57217 Rev. C

SSP 57217	Section	Requirements
Table 4.2-1	Applicability Verification Matrix	293/366*
Table 4.2-8	Common Transport Requirements	32

*103 of which are HFIT

FCF Top-Level Documents

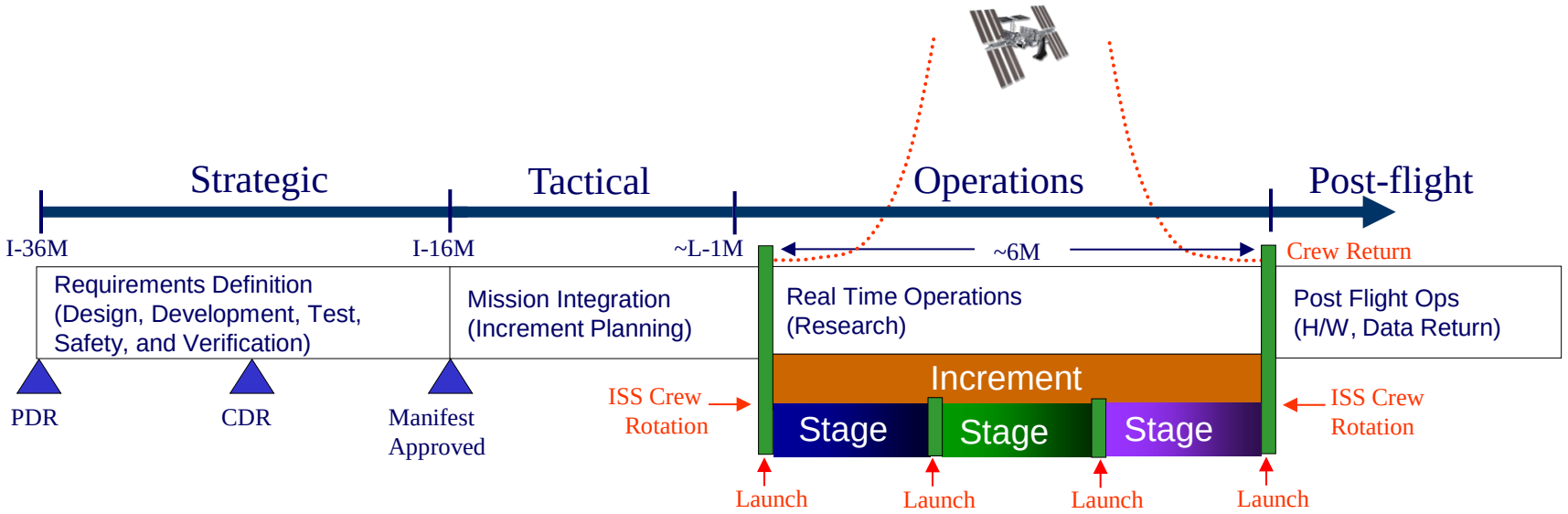
ISS Documents

- SSP 57000: Pressurized Payloads Interface Requirements Document
- SSP 51700: Payload Safety Policy and Requirements for the International Space Station
- SSP 30599: Safety Review Process
- SSP 57008, Appendix P: Common Carrier Transport Requirements
- SSP 57057: Payload Integration Template (PIT)
- SSP 57217: Combustion Integrated Rack (CIR) Hardware Interface Control Document
- SSP 57218: Fluids Integrated Rack (FIR) Hardware Interface Control Document
- SSP 57117: Payload Integration Agreement for Fluids & Combustion Facility

Integration Documents

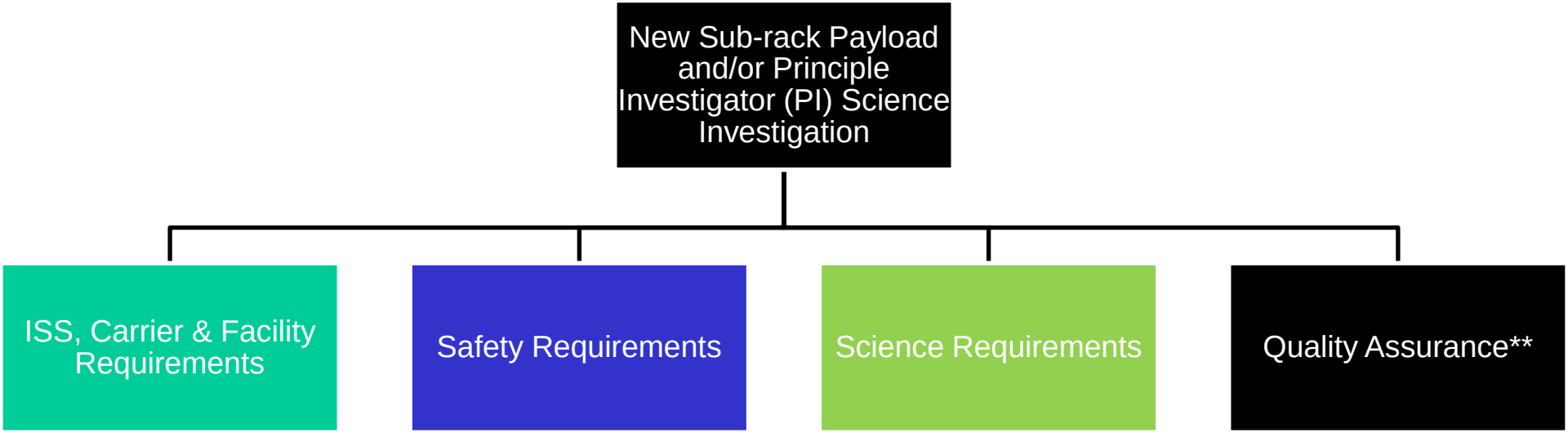
- FCF-PLN-0875: FCF Utilization Process
- FCF-IDD-CIR: Combustion Integrated Rack Payload Interface Definition Document
  - ❖ FCF-ICD-CIR-MDCA
  - ❖ FCF-ICD-CIR-ACME
- FCF-IDD-FIR : Fluids Integrated Rack Payload Interface Definition Document
  - ❖ FCF-ICD-FIR-LMM
- FCF-IA-MV-BB: Integration Agreement Main Volume Blank Book
  - ❖ FCF-IA-MV-MDCA
    - – FLEX
    - – FLEX-2
    - – FLEX-2J
  - ❖ FCF-IA-MV-LMM
    - – CVB
    - – PACE
    - – PACE-2
    - – LMM-Bio
    - – ACE
  - ❖ FCF-IA-MV-ACME
    - – CLD Flame
    - – E-Field Flame
    - – Flame Design
    - – S-Flame
- FCF-ICD-0076: Software Interface Control Document
- CIR-DOC-4064: CIR Payload Accommodations Handbook
- FIR-DOC-1863: FIR Payload Accommodations Handbook

CIR/Sub-rack Payload to ISS Integration



- NASA PIMs provide integration leadership during all phases of the payload's life cycle
  - Strategic – ISS integration requirements, products, and schedule development to ensure that an ISS compatible payload is built; support manifest process (payload data collection and feasibility assessments)
  - Tactical – represent PD interests to Increment and Flight-specific teams to ensure that integration and operations requirements are addressed; provide oversight for payload CoFR and verification submittals
  - Operations – assist with operations issue resolution between the PD and the Increment Payload Manager; maintains payload insight; and coordinates payload resupply or return requirements; assure payload CoFR and verification submittals during payload lifetime on-orbit
  - Post-flight – coordinate vehicle deintegration requirements; return of payload material from the landing site to the PD; and Lessons Learned submittals

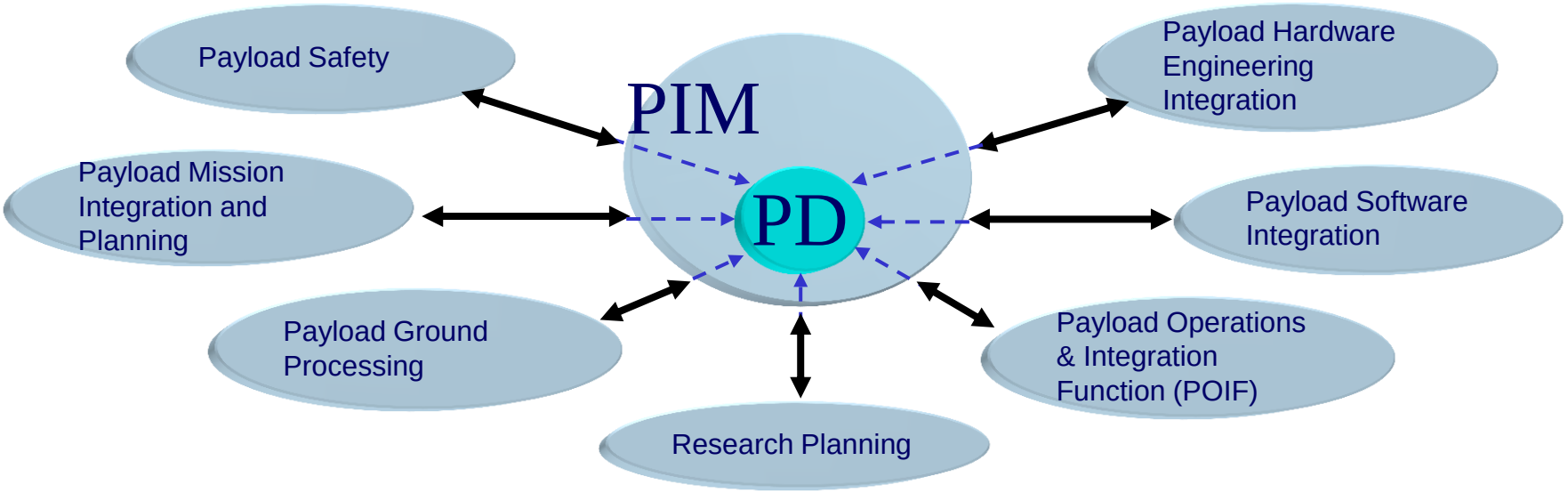
Sub-rack Payload Requirement Categories



** In addition to any quality assurance requirements imposed in the other requirement categories, the Payload Organization for each sub-rack payload (eg GRC/Zin) imposes additional Quality Assurance (QA) requirements (eg. GLPR 7120.5.30 /Zin Quality Management System (QMS), Procedure No. P30012).

Payload Integration Manager (PIM)

- NASA Payload Integration Manager (PIM)
  - Functions as the Payload Developer's primary interface to the ISS Program
  - Serves as payload advocate – but also protects ISS Program Requirements



- Ensures payload requirements are accurately defined and documented
- Facilitates payload integration product development, delivery schedules, and communications with the ISS Program

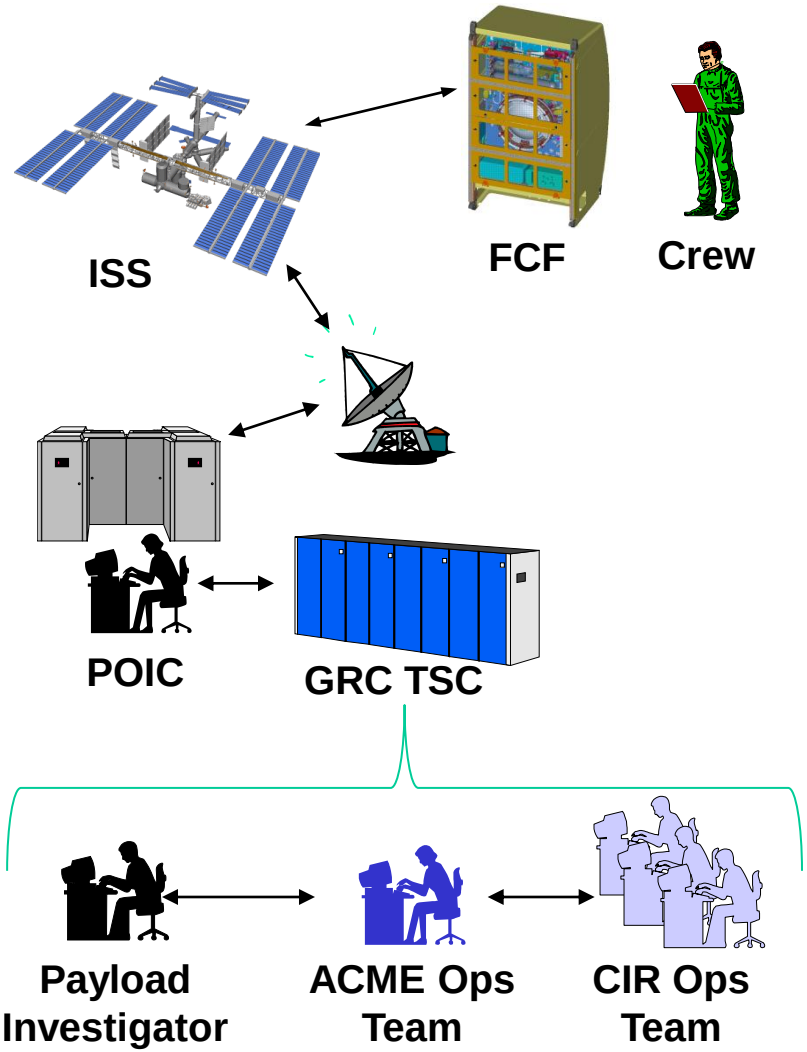
ACME Operations

Angie Adams – ACME Operations Lead

Concept of Operations

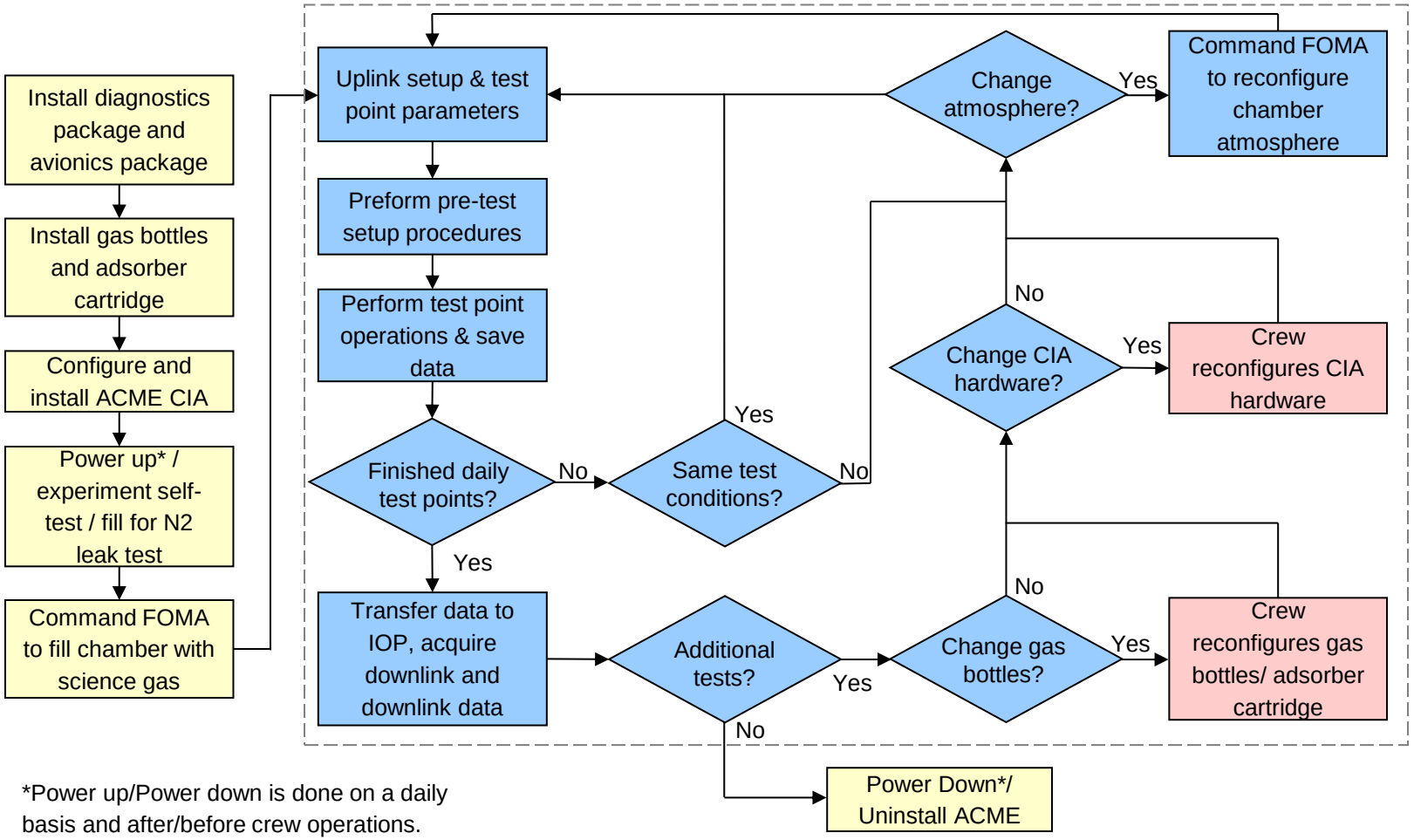
ACME Concept of Operations

- Designed for autonomous, ground tended operations
- Experiment is configured/reconfigured by the crew, then operated from the ground
- Experiment progress monitored by the operations team and PI's
- Ops team will uplink commands to perform experiment operations and collect data
- ACME's primary ground operations site is the GRC TSC (Telescience Support Center)



Operations Flow

Hardware configure, power up for checkout, chamber fill, test point operations, reconfigurations, power down

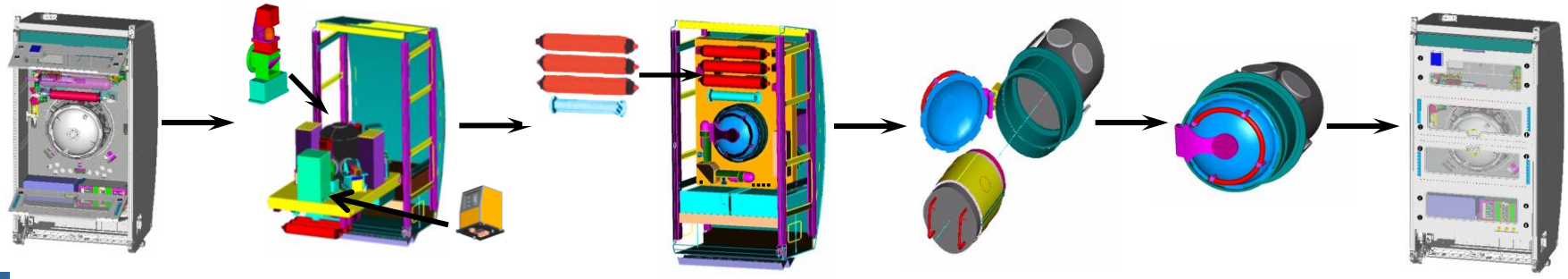


*Power up/Power down is done on a daily basis and after/before crew operations.

ACME Configure

ACME Configure – Crew Activities

- Optics Bench Back (4 hr. 25 min. Crew time)
  - Install ACME Camera Package
  - Install ACME Avionics Package
- Optics Bench Front (2 hr. 50 min. Crew time)
  - Install gas bottles
  - Install adsorber cartridge
  - Install N2 pressure relief
  - Install GC bottles
- ACME CIA (7 hr. 50 min. Crew time)
  - Install hardware components
  - Insert into CIR Combustion Chamber



Experiment Hardware

Experiment	Burner Type	Mass Flow Controllers (SLPM)	E-Field Mesh	Long or Short Thermocouple Array	Fiber	Radiometer
CLD Flame	Coflow	Fuel: 0.1, 0.2, 0.5 Inert: 0.05, 0.1, 0.2 Oxygen: 20	No	No	Yes	Yes
E-Field	Gas Jet: 1.3mm & 2.1mm	Fuel: 0.05, 0.1 Inert: 0.01 Oxygen: n/a	Yes	No	No	Yes
Flame Design	Spherical	Fuel: 0.2 Inert: 0.5, 2.0, 5.0 Oxygen: 0.5	No	Yes: Short and Long	Yes	Yes
s-Flame	Spherical	Fuel: 0.05, 0.5, 1.0 Inert: 0.1, 0.5 Oxygen: n/a	No	Yes: Short	Yes	Yes
BRE	Heat Flux: 25mm & 50mm	Fuel: 0.5, 1.0, 2.0 Inert: 0.1, 0.2, 0.5 Oxygen: n/a	No	No	Yes	Yes – 2nd type

Experiment Resources

CLD Flame – 82 required tests

- 5 – 3.8L 21/79 O2/N2 bottles
- 1 – 1L 100% CH4 bottle*
- 1 – 1L 40/60 CH4/N2 bottle
- 1 – 1L 50/50 C2H4/N2 bottle
- Station N2 – 90L

Flame Design – 76 required tests

- 4 – 2.25L 40/60 O2/N2 bottles
- 7 – 150cc 85/15 O2/N2 bottles
- 3 – 1L 100% C2H4 bottle*
- Station N2 – 1244.623L

E-Field – 58 required tests

- 1 – 3.8L 30/70 O2/N2 bottle
- 1 – 2.25L 30/70 O2/N2 bottle
- 1 – 1L 100% CH4 bottle*
- 1 – 1L 30/70 CH4/N2 bottle
- 1 – 1L 100% C2H4 bottle*
- 1 – 1L 30/70 C2H4/N2 bottle
- Station N2 – 264.093L

Experiment Resources (cont.)

s-Flame – 46 required tests

- 2 – 3.8L 30/70 O2/N2 bottles
- 2 – 3.8L 30/70 O2/He bottles
- 2 – 2.25L 100% He bottle
- 1 – 1L 100% H2 bottle
- 1 – 1L 66.7/33.3 H2/CH4 bottle
- 1 – 1L 55.6/44.4 H2/CH4 bottle
- 1 – 1L 44.4/55.6 H2/CH4 bottle
- 1 – 1L 45/55 CH4/N2 bottle
- 1 – 1L 45/55 CH4/He bottle
- Station N2 – 346.375L

BRE – 73 required tests

- 15 – 2.25L 40/60 O2/N2 bottles
- 1 – 1L CH4 bottle*
- 1 – 1L C2H4 bottle*
- Station N2 – 1229.952L

All 5 Experiments

- 3 – CIR GC Argon bottles
- 3 – CIR GC Helium bottles
- 3 – CIR GC Check Gas bottles

*Fuel Bottles with same contents can be combined into 1 bottle (makes total 1L fuel bottles = 14)

Summary of Operations

Experiment	# Test Points	# OPS Days Required	# Weeks	Crew time
CLD Flame	82	54	22	53.08*
E-Field	58	38	18	36.58
Flame Design	76	64	27	56.75
s-Flame	46	37	19	37.92
BRE	73	60	23	41.33
TOTAL	**335**	**253**	**109**	**225.66**

*Includes initial configure

Assumptions:

- OPS Day = test point, transfer/downlink or crew ops
- 3 powered ops days per week
- 1 powered ops day with 1 crew ops day per week
- 2 – 5 test points per day
- 30 – 70 second test point durations
- Transfer/downlink after every test point day

Sparing

Sparing Philosophy
•Consumables
•Fragile hardware
•Flight vs. Ground

Flight Spares

- Hard Drives (2)
- Igniter Tips (20)
- E-Field Mesh (1)
- Burners (1-3 each)
- Fiber 1 (10)
- Fiber 2 (3)
- Thermocouples (5 each)

Ground Spares

- Hard Drives (5+)
- Cameras (1each + parts)
- Mass Flow Controllers (1 each)
- Data Acquisition Unit (1)
- Gas and Water Hoses (1 each)
- Radiometers (parts)
- Photomultiplier Tubes (parts)

Crew Procedures & Training

Crew Procedures

- Crew procedures will be written and baselined for all initial configure activities as well as nominal replacement activities (Initial July 2014, Baselined May 2015)

Crew Training

- Training Strategy Team (TST) meeting to be held Fall 2013 to develop training plan
- Crew training will consist of an ACME class and the CIR class (Start Fall 2014)
- ACME will provide a CIA trainer with working interfaces for crew training (Summer 2014)

HFIT

- Initial meeting with Crew Office and HFIT held September 2013
  - Developed Operations Nomenclature acceptable by crew office (Final due June 2014)
  - HFIT looked at drawings and found no concerns
  - Developed initial packing strategy for shipping hardware to orbit
  - Developed initial labeling strategy
- IPLAT review of crew interface label drawings:
  Initial ~ May 2014, Final ~ May 2015
- Final HFIT Hardware evaluation, COC Issued ~August 2015

Quality Assurance

Wayne Borrelli – ACME Safety/Quality Engineer

ACME SAR Checklist

- The ACME SAR Checklist is ACME-PLN-011 and customizes the requirements from GLPR 7120.5.30, Space Assurance Requirements, to the scale and needs of the ACME project.

- This document was created with consultation from ZIN Quality, the NASA Safety and Mission Assurance Office and with help from the Chief Engineer's Office and either accepts requirements as applicable or provides justification as to why they are not.

- Requirements deemed Not Applicable include those that require long term trend analysis, requirements for payloads exposed directly to space and those relating to reflight hardware.

- Refer to ACME-PLN-011 for more information. These requirements are also included in the ACME Verification Tracking Log.

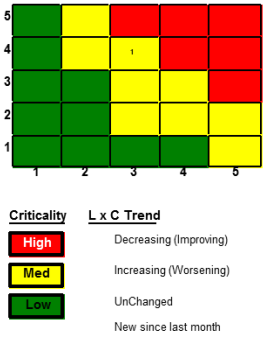
Risk Management

- SpaceDOC Risk Management Plan P30015
- GRC RMIT Risk Management Database
  - Utilized to assist in the continuous risk management process.
  - Online input/processing/reporting of risk information.
- Risk Mgmt Working Group meets monthly to perform risk management activities.
  - ACME Systems Lead supports this meeting.
- 26 ACME Risks have been identified to this point in time (24 closed)
  - 2 Active Technical Risks
  - 14 Closed Technical Risks
  - 5 Closed Schedule Risks
  - 1 Closed Cost Risk
  - 3 Closed Management Risks
  - 1 Closed Safety Risk



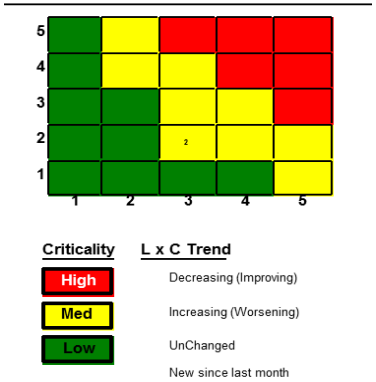
ACME Risk Summary

- Risk ID: ACME-014
- Risk Title: IPSU to IOP image transfer rates take too long
- Risk Statement: Given that the current data transfer rates from the IPSU to the IOP is severely limited, transfer of ACME data may take an unacceptable amount of time and may reduce obtainable science for the allotted operational time on board ISS.
- Likelihood: 4
- Consequence: 3
- Approach: Watch. Need to keep an eye on this and follow up with the CIR team to keep updated on transfer improvements.
- Latest Status: 09/17/13 - The IPSU-G is going to have their PDR September 26th 2013. The IPSU-G will eliminate the need to transfer image data from an IPSU to the IOP for downlink because the IPSU-G will be able to directly downlink image data to the ground. A trigger for this risk would be the cancellation of the IPSU-G project that would require ACME to mitigate, accept, etc...



ACME Risk Summary

- Risk ID: ACME-026
- Risk Title: ACME E-Filed EMI exceedences impact science
- Risk Statement: Given that the current ACME design exceeds the carrier EMI requirement; then there is the possibility that design changes made to meet this EMI requirement causes a loss of science.
- Likelihood: 2
- Consequence: 3
- Approach & Latest Status: Watch. 10/22/13 Report received and the team is waiting to perform EMI testing on the flight system.



Product Assurance

ZIN Product Assurance will be involved extensively with the ACME Investigation verification process.

Product Assurance overview will consist of:

- Inspections
- Closing MWO's
- Witnessing tests
- Performing Mandatory Inspection Points (MIPs)
- Signing drawings and documents

Problem reports, failures, non-conformances, and corrective actions are captured through conformance with the ZIN QMS using the non-conformance and corrective action processes.

ZIN Technologies, Inc maintains compliance to the AS9100 Aerospace Standard.

Software Product Assurance

ACME Software Product Assurance Plan, ACME-PLN-005

- Review of software documentation.
- Peer reviews as needed for ACME S/W life cycle activities.
- S/W audits.
- Oversight on ACME S/W verifications.
- Nonconformance Reporting & Corrective Action Process – S/W issues recorded in SpaceDOC NCR system.
- Support of software development lifecycle milestone activities (PDR, CDR, SAR, etc.).

Reliability Analysis

Results documented in ACME-ANA-001

Failures are mitigated using ORU's and components that are known to have a high reliability.

There are no critical components based on a redundancy screen assessment.

- Failures would be immediately apparent.
- Mitigations outlined.

Reliability assessed to be sufficient to obtain required success criteria.

- Approximately 61 hours of required test point operation (not including startup/warm up times)
- Risk of non-ORU component failure within this period of time is minimal

ACME Safety

Wayne Borrelli – ACME Safety/Quality Engineer

ACME Flight Safety

•ACME Flight Phase 0/1 Flight Safety Data Package reviewed and approved by PSRP. All hazard reports signed.

- Phase II FSDP Submittal to PSRP planned for 12/29/2013.
- ACME Phase II Flight Safety Review is scheduled for February 12 & 13, 2014.
- ACME PSE is currently David Cho

ACME Phase II Hazard Reports

STD-ACME-1: JSC Form ISS_OE_1298 Payload Standard Hazard Control Report

ACME-2: Leakage of Water Cooling System

ACME-3: Rupture of Pressurized Systems

ACME-4: Crew Exposure to Broken Materials

ACME-5: High Oxygen Concentration Results in Fire

ACME-6: Electrical Shock

ACME Phase II Hazard Reports

JSC Form ISS_OE_1298

- **Flammable Materials** – Review & approval of MIUL/MUAs utilizing MAPTIS, and flammability assessment.
- **Materials Offgassing** – Review & approval of MIUL/MUAs utilizing MAPTIS, and offgas assessment.
- **Mechanical Hazards (Sharp Edges, Corners, and/or Protrusions)** – Verify ACME hardware conforms to the approved drawings, including requirements to remove sharp corners and protrusions.
- **Touch Temperature** – Perform Thermal analysis and testing of the ACME hardware to prove the maximum temperature ACME hardware reaches (worst case experiment), and cool down time needed to reach below 45 degrees C.
- **Shatterable Material Release** – All optical glass components, radiometer lenses and photomultiplier tube optics will be vibration tested. All optical glass components will be packaged in clear plastic bags prior to flight, and inspected before removal.

ACME Phase II Hazard Reports

JSC Form ISS_OE_1298

- **Nonionizing Radiation** - Perform EMI/EMC testing using the test methods/procedures defined in SSP 30237 and SSP 30243, and prepare a report confirming requirements have been met.
- **Lasers** – N/A: ACME hardware contains no lasers.
- **Noise Exposure** – Perform acoustic testing per SSP 50021 to prove ACME hardware will not create noise exposure hazard causing damage to hardware, injury to crew, or interfere with communications or caution/alarm detection.
- **Battery Failure** – Perform battery loaded / open circuit voltage measurements and leakage check under vacuum to DNA battery. Perform functional check of Lippert Board Assembly (with soldered button battery) and visual inspection of Lippert Board battery.
- **Capacitors used as energy storage devices -** ACME does not utilize asymmetric, lithium-based or free electrolyte capacitors as energy storage devices.

ACME Phase II Hazard Reports

JSC Form ISS_OE_1298

- **Electrical Power Distribution -** Perform analysis of the ACME electrical design to verify that proper circuit protective devices and wire sizing are implemented, proper bonding and proper grounding and prepare a report confirming that ISS requirements from SSP 30240 and 30245 have been met. Perform inspection to verify that the as-built electrical circuitry is in accordance with approved drawings and parts lists.
- **Mating/demating Powered Connectors** – Inspect hardware for design features that completely enclose or shroud pins. Inspect connectors for powered side terminating in sockets rather than pins. Inspect crew procedure inputs to ensure that a step is place to remove power from the CIR experiment power connector prior to mate/demate of ACME connectors.

ACME Phase II Hazard Reports

JSC Form ISS_OE_1298

- **Rotating Equipment** – Inspect drawings and hardware to ensure rotating equipment is contained. Certify that rotating parts do not exceed 8 inches in diameter and 8000 revolutions per minute.
- **Interference with Translation Paths** – Perform analysis to verify that ACME hardware does not impede emergency IVA egress.
- **Structural Failure** – Perform a structural analysis, and generate a Fracture Control Plan and a Structural Verification Plan, including analysis of kick loads and 0.2 g reboost. Inspect final flight stowage configuration.
- **Structural Failure of Sealed Containers** – Perform an analysis of all sealed containers to insure that they do not contain hazardous substances. Analyze all sealed containers to ensure they contain less than 14,240 foot pounds of stored pressurized energy and can withstand a maximum delta pressure of 1.5 atm.

ACME Phase II Hazard Reports

JSC Form ISS_OE_1298

•**Structural Failure of Vented Containers** – Perform an analysis that demonstrates that the vented containers provide an internal to effective vent area ratio that results in a differential pressure loading of no greater than 0.01 psid. Perform inspection to insure that the volume and vent area are in accordance with the above analysis.

ACME Phase II Hazard Reports

Unique Hazard Report ACME-2

Title: Leakage of Water Cooling System

Description: Leaks in the ACME Chamber Insert Assembly water lines results in damage to hardware or inadvertently contaminates the vacuum exhaust system. Excessive cooling water leakage can cause an automatic shut down of the ISS ITCS, a must work system.

Controls:

1. Design to appropriate factors of safety.
2. Build in accordance with approved drawings and assemble per approved procedures.
3. Install accumulators, where necessary.
4. Utilize crew procedures to ensure proper assembly.
5. Utilize redundant seals in all quick disconnects.

ACME Phase II Hazard Reports

Unique Hazard Report ACME-3

Title: Rupture of Pressurized Systems

Description: Rupture of pressurized system hardware results in injury to crewmembers or damage to lab module.

Controls:

1.Design to appropriate factors of safety.

2.Build in accordance with approved drawings and assemble per approved procedures.

3.Install accumulators, where necessary.

4.Materials selected to avoid susceptibility to stress corrosion cracking in accordance with MAPTIS database.

ACME Phase II Hazard Reports

Unique Hazard Report ACME-4

Title: Crew Exposure to Broken Materials

Description: Crew exposed to fragments of ceramic and silicon-carbide materials.

Controls:

1. Igniter tips, fiber arrays, burner insulators, E-Field Mesh Insulator, and E-Field Power Supply Insulator will be designed to a Factor of Safety of 2.0.
2. Igniter tips, fiber arrays, burner insulators, E-Field Mesh Insulator, and E-Field Power Supply Insulator will be bagged for stowage. Crew procedures will include steps for inspection of hardware prior to removal from clear plastic bags for installation.

ACME Phase II Hazard Reports

Unique Hazard Report ACME-5

Title: High Oxygen Concentration Results in Fire

Description: Fire resulting from material incompatibility with high oxygen concentrations. **Note**: Current Science Requirements specify an 85% oxygen bottle for Inverse Flame experiments.

Controls:

1.All CIA materials will be analyzed and selected for compatibility with up to 40% oxygen.

2.To compensate for the lack of flammability test data of some materials over 30% oxygen levels, non-metallics used inside the chamber will be assessed in the final configuration per JSC 29353 (to minimize ignition sources and flame propagation paths).

ACME Phase II Hazard Reports

Unique Hazard Report ACME-5

3. CIR solenoid valve SV5 is nominally closed and prevents flow of oxygen to ACME. CIR Ground Procedure will verify that, when 85% oxygen bottle is in use, the CIR chamber pressure is above 14.0 psia before a command is issued to open solenoid valve SV5.

4. ACME solenoid valves (SV A1, SV A4 and SV A5) are nominally closed and prevent flow of oxygen into CIR chamber. ACME Ground Procedure will verify that, when 85% oxygen bottle is in use, the CIR chamber pressure is above 14.0 psia before a command is issued to open any of solenoid valves SV A1, SV A4, or SV A5.

5.ACME Pressure Switch (PSA1) inhibits ignition in the chamber until a chamber pressure of 14.0 psia is reached.

6.Materials in high O2 areas of the gas system shall be selected for high O2 environment per the MAPTIS database, or if required, an MUA will be written and approved by GRC PPAD per M&P Intercenter agreement.

ACME Phase II Hazard Reports

Unique Hazard Report ACME-6

Title: Electrical Shock

Description: Incidental contact by the crew with high voltage of +/- 10KVDC can lead to crew injury.

Controls:

1.Proper design of the ACME power distribution system, including compliance with bonding/grounding requirements.

2.Provide two bond paths to ground as required by letter MA2-99-142.

3.Provide electrical isolation as required by SSP 57000.

ACME Phase II Hazard Reports

Unique Hazard Report ACME-6

4. Utilize connectors of scoop-proof design that completely encloses the pin/socket contacts. Powered side of all ACME cable connectors above 32 VDC are terminated in sockets to provide protection against debris/inadvertent shorting. To prevent mis-mating, all connectors are of different sizes and/or keying. All connectors will be labeled.

5. At least one verifiable upstream inhibit exists. ACME will utilize the CIR Rack Power Switch (RPS) as its upstream inhibit.

6. Manufacturing, assembly processes, and fabrication shall be performed In accordance with approved drawings and procedures.

7. High Voltage circuitry will be designed with resistor divider networks that will discharge the high voltage signals within one second (both positive and negative voltages) to chassis ground.

8. Wire size derating analysis will be performed per Interpretation Letter TA-92-038.

CIR/ACME Integrated Safety

Clarise Shinn – FCF Safety Engineer

CIR/ACME Integrated Flight Safety

- CIR/ACME Integrated Phase 0/1 Flight Safety Data Package submitted to the PSRP, December 2011

- CIR/ACME Integrated Phase 0/I Flight Safety Review, February 2012
  - All Phase 0/I integrated hazard reports are approved by the PSRP
  - Safety Data Package was approved by the PSRP
  - No Action Items or Agreements from Phase 0/I Review

- Phase II Safety Data Submittal to PSRP planned for 12/29/2013.

- CIR/ACME Integrated Phase II Flight Safety Review is scheduled with ACME Phase II Review on February 12 & 13, 2014.

Integrated Phase II Hazard Reports

FCF-ACME-01	Release of Adsorber Cartridge Contents into Habitable Volume
FCF-ACME-02	Exposure of Crew to Toxic Byproducts During Chamber Operations
FCF-ACME-03	Overpressurization of CIR Pressurized System by ISS Nitrogen Supply
FCF-ACME-04	Adiabatic Combustion Ruptures CIR Combustion Chamber or CIR Exhaust Vent Package during ACME experiments (except for the Inverse Flame Experiment)
FCF-ACME-05	Adiabatic Combustion Ruptures CIR Combustion Chamber or CIR Exhaust Vent Package during ACME Inverse Flame Experiment

Integrated Phase II Hazard Reports

FCF-ACME-01

Title: Release of Adsorber Cartridge Contents into Habitable Volume

Description: Generation of (toxicity hazard level = 0, see HMST) particles due to launch loads and subsequent release into habitable volume during Adsorber Cartridge replacement results in injury to crew

Controls:

1. The Adsorber Cartridge is designed with a 40 micron sintered metal filter, which is smaller than the grain size (1680 microns) of chemicals, located at both the inlet and outlet ports, to keep particles from escaping.
2. The Adsorber Cartridge is designed so that each end of the Adsorber Cartridge has redundant seals QDs and a manual valve which seals the Adsorber Cartridge for removal.
3. Constituents of Adsorber Cartridge are limited to Silica Gel or Molecular Sieve for ACME.

Integrated Phase II Hazard Reports

FCF-ACME-02

Title: Exposure of Crew to Toxic Byproducts During Chamber Operations

Description: Carcinogenic soot particles from the combustion chamber contaminate the habitable atmosphere causing crew illness if inhaled or floating particles could cause injury to the eyes.

Controls:

1. The soot production for ACME experiments will be below the SMAC limit and does not exceed Tox 0.
2. The chamber air will be circulated through a tested Adsorber Cartridge for 15 minutes until free floating soot is removed, prior to venting.
3. The chemicals selected for the Adsorber Cartridge will remove the soot and combustion by-products.
4. Each Adsorber Cartridge shall have a content label and serial number.
5. Approved crew procedures shall be used for installation and removal of an Adsorber Cartridge. (TBR) *(This control is also covered in CIR Hazard Report, FCF-C05 Inadvertent Release of Toxic Fuels and Combustion By-products and CIR Hazard Report, FCF-C28 Inadvertent Venting of High Oxygen, Pre-mixed Gases or Fuel through ISS Vent)*

Integrated Phase II Hazard Reports

FCF-ACME-03

Title: Overpressurization of CIR pressurized system by ISS Nitrogen Supply

Description: Overpressurization of the CIR pressurized system by the ISS nitrogen supply via the CIR nitrogen bypass line results in rupture of CIR hardware, damage to ISS hardware or injury to the crew.

Controls:

1. The ACME nitrogen pressure system design will include a pressure relief valve for the CIR nitrogen bypass line, and will be set at 135 psia.

Note: CIR provides the additional hazard controls for overpressurization due to the ISS nitrogen gas supply. See CIR Hazard Report FCF-C10, hazard controls, 3.4, 3.5, 3.6 and 3.8.

Integrated Phase II Hazard Reports

FCF-ACME-04

Title: Adiabatic Combustion Ruptures CIR Combustion Chamber or CIR Exhaust Vent Package during ACME experiments (except for the Inverse Flame Experiment)

Description: Rapid overpressurization during the combustion of ACME gaseous fuel and oxidizer/diluent exceeds the MDP of the CIR Combustion chamber and/or exhaust vent package resulting in damage to CIR or ISS hardware or injury to the crew.

Note *(on hazard report)***:**

1. This hazard report works in conjunction with CIR hazard report FCF-C07, “Adiabatic Combustion ruptures the chamber walls, causing damage to station and/or injury to crew.“ Overpressurization due to combustion will be controlled by limiting the fuel volume in each gas bottle. The CIR Valve Timers will be placed in the fuel mode. The chamber will be pre-filled with gaseous oxidizer and gaseous fuel will flow through the ACME burner.
2. CIR is responsible for filling the experiment gas bottles. The ACME detonation analyses determine the limits on fuel volumes in each bottle. The fuel gas volumes will be chosen such that even if the entire contents of a bottle were to be released into the CIR Combustion Chamber, any adiabatic combustion would result in a pressure less than the MDP of the CIR Combustion chamber. The MDP of the CIR chamber is 135 psia.
3. The CIR bottle leakage hazard controls and verifications are covered in CIR hazard report FCF-C08 Leakage of Pressurized System.
4. The CIR system rupture hazard controls and verifications are covered in CIR hazard report FCF-C10 Rupture of Pressurized Systems.

Integrated Phase II Hazard Reports

FCF-ACME-04

Controls:

1. Fuel volume in each gas bottle will be limited so that adiabatic combustion of the total fuel bottle volume at the worst-case chamber environment cannot exceed the MDP of the chamber.
2. The chamber atmosphere will be vented between fuel gas bottle changeouts.
3. Procedures for gas bottle ground handling and filling will prevent filling a bottle with the incorrect constituent, or the incorrect volume or pressure.
4. The crew procedure steps will require the crew to confirm the fuel bottle serial number and contents with POIC after installation.
5. An ACME pressure sensitive switch will inhibit the ACME igniter if the ACME maximum operating pressure is exceeded.

Integrated Phase II Hazard Reports

FCF-ACME-05

Title: Adiabatic Combustion Ruptures CIR Combustion Chamber or CIR Exhaust Vent Package during ACME Inverse Flame Experiment

Description: Rapid overpressurization during the combustion of ACME gaseous fuel and oxidizer/diluent exceeds the MDP of the CIR Combustion chamber and/or exhaust vent package resulting in damage to CIR or ISS hardware or injury to the crew.

Note (on hazard report) :

1. This hazard report works in conjunction with CIR hazard report FCF-C07, “Adiabatic Combustion ruptures the chamber walls, causing damage to station and/or injury to crew.“ Overpressurizaton due to combustion will be controlled by limiting the oxygen volume in each gas bottle. The CIR Valve Timers will be placed in the oxygen mode. The chamber will be pre-filled with gaseous fuel and gaseous oxygen will flow through the ACME burner.
2. CIR is responsible for filling the experiment gas bottles. The ACME detonation analyses determine the limits on oxygen volumes in each bottle. The oxygen gas volumes will be chosen such that even if the entire contents of a bottle were to be released into the CIR Combustion Chamber, any adiabatic combustion would result in a pressure less than the MDP of the CIR Combustion chamber. The MDP of the CIR chamber is 135 psia.
3. The CIR bottle leakage hazard controls and verifications are covered in CIR hazard report FCF-C08 Leakage of Pressurized System.
4. The CIR system rupture hazard controls and verifications are covered in CIR hazard report FCF-C10 Rupture of Pressurized Systems.

Integrated Phase II Hazard Reports

FCF-ACME-05

Controls:

1. Oxygen volume in each gas bottle will be limited so that adiabatic combustion of the total fuel bottle volume at the worst-case chamber environment cannot exceed the MDP of the chamber.
2. The chamber atmosphere will be vented between oxygen gas bottle changeouts.
3. Procedures for gas bottle ground handling and filling will prevent filling a bottle with the incorrect constituent, or the incorrect volume or pressure.
4. The crew procedure steps will require the crew to confirm the oxygen bottle serial number and contents with POIC after installation.
5. An ACME pressure sensitive switch will inhibit the ACME igniter if the ACME maximum operating pressure is exceeded.

ACME Readiness

- All project requirements are documented and agreed to
- ACME design reflects complete compliance with the project requirements (detailed in ACME-DOC-002D)
- Drawing status (complete/total):
  - CIA – 295/375
  - AVP – 92/92
  - Data Camera – 93/109
  - Total – 480/576 83%
  - 222 Flight drawing ECOs have been processed as a result of EM build and testing
- Engineering Model Build: AVP complete, CIA and Data Camera underway

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ACME Readiness

- Flight Model Build: Build on all 3 major packages has started for low risk items
- Build documents and test plans have been generated
- Functional, validation and verification test planning is complete spanning all levels of the build
- All analyses (structural, thermal, reliability, materials, etc.) show no major issues with ACME design
- Risks are identified and managed
- Carrier integration has already begun and will continue per the integration schedule
- All hazards have sufficient controls
- Project planning shows adequate technical margins and resources exist to complete the project within budget and schedule

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